Trainingsdaten

Im folgenden sind alle Trainingsdaten, welche in der Facharbeit anlysiert werden. Dargestellt. Die dargestellten Daten von Training 1-16 sind die Hauptdaten. Jedes dieser Trainings war ein full_ training(...). Für diese Trainings sind folgende Parameter immer gleich:

epsilon = 1,0, epsilon_min = 0,01, epsilon_decay = 0,99995, win_reward = 1,0, draw_reward = 0,1, lose_reward = -1,0

*Das Basistraining startet mit 100.000 Episoden, die 20 folgenden League-play-Sessions sind n*10.000 Episoden lang. Dadurch ist ein vollständiges Training 2.200.000 Episoden lang.*

Die anderen Werte unterscheiden sich wie folgt:

	hidden_size	gamma	batch_size	survive_reward
Trainer 1	256	0,95	64	-0,005
Trainer 2	256	0,95	64	0,001
Trainer 3	256	0,95	128	-0,005
Trainer 4	256	0,95	128	0,001
Trainer 5	256	0,99	64	-0,005
Trainer 6	256	0,99	64	0,001
Trainer 7	256	0,99	128	-0,005
Trainer 8	256	0,99	128	0,001
Trainer 9	128	0,95	64	-0,005
Trainer 10	128	0,95	64	0,001
Trainer 11	128	0,95	128	-0,005
Trainer 12	128	0,95	128	0,001
Trainer 13	128	0,99	64	-0,005
Trainer 14	128	0,99	64	0,001
Trainer 15	128	0,99	128	-0,005
Trainer 16	128	0,99	128	0,001

Die Werte wurden gewählt, um zu vergleichen, inwiefern es einen Unterschied macht: das neuronale Netz zu vergrößern oder zu verkleinern, einen größeren Wert auf eine langfristige Strategie zu legen, mehr Erinnerungen in jedem Lernschritt miteinzubeziehen, das bloße Überleben zu bestrafen oder zu belohnen.

Das Training in allen Konstellationen hat insgesamt 35.200.000 Episoden simuliert und 26 Stunden gedauert. Die Größe der gesammelten Daten (Checkpoints, Grafiken etc. beträgt 38,96 Gigabyte.

Die Grafiken werden pro Trainer nach folgendem Schema präsentiert: erst ein Überblick über das volle Training, anschließend in chronologischer Reihenfolge ausgewählte Grafiken der einzelnen Trainingsschritte (base_training, league_play_1, league_play_2, ..., league_play_20). Dargestellt sind die Win-Rate des Modells gegen einen Zufallsgegner, der berechnete Loss pro Episode und die finale Belohung nach einer Episode. Als Referenz für die bessere Einschätzung des zeitlichen Ablaufs führt die Übersicht auch die Entwicklung von Epsilon auf.

Grundtraining

1.1 Trainer 1 3-5

1.2 Trainer 2 6-8

1.3 Trainer 3 9-11

1.4 Trainer 4 12-14

1.5 Trainer 5 15-17

1.6 Trainer 6 18-20

1.7 Trainer 7 21-23

1.8 Trainer 8 24-26

1.9 Trainer 9 27-29

1.10 Trainer 10 30-32

1.11 Trainer 11 33-35

1.12 Trainer 12 36-38

1.13 Trainer 13 39-41

1.14 Trainer 14 42-44

1.15 Trainer 15 45-47

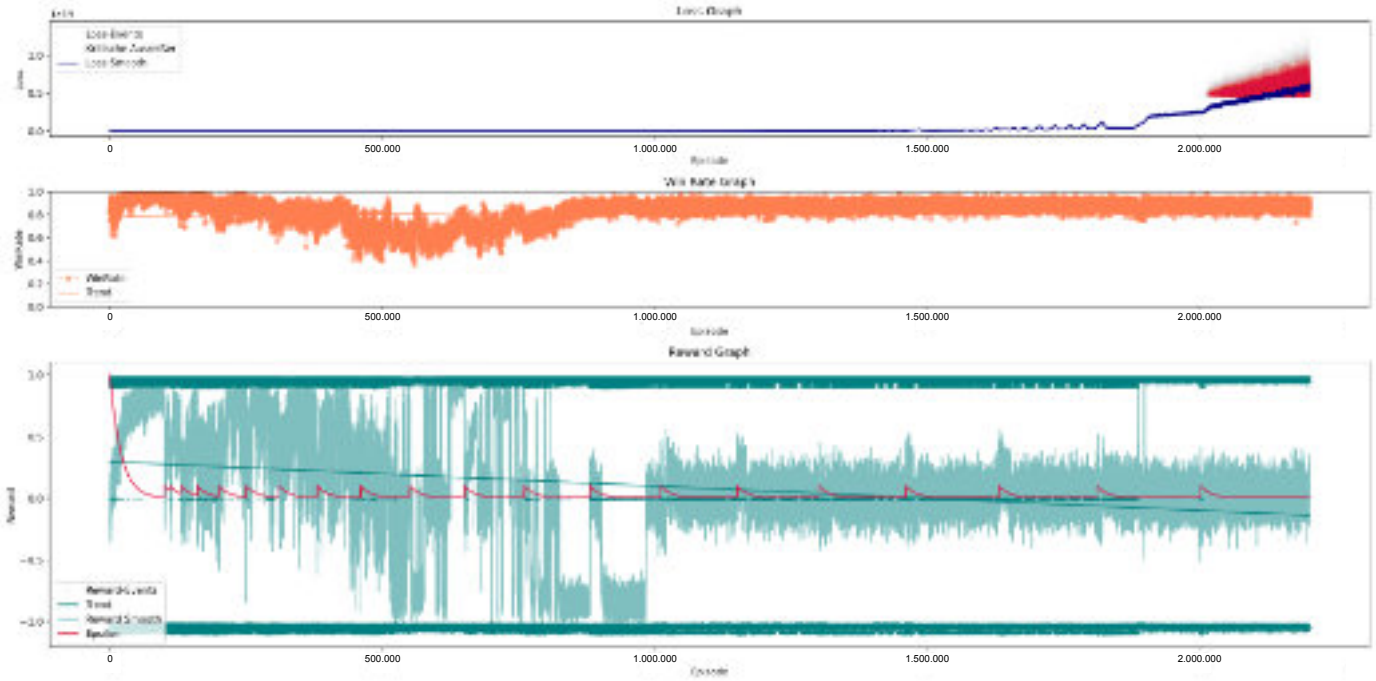
1.16 Trainer 16 48-50

Trainer 1

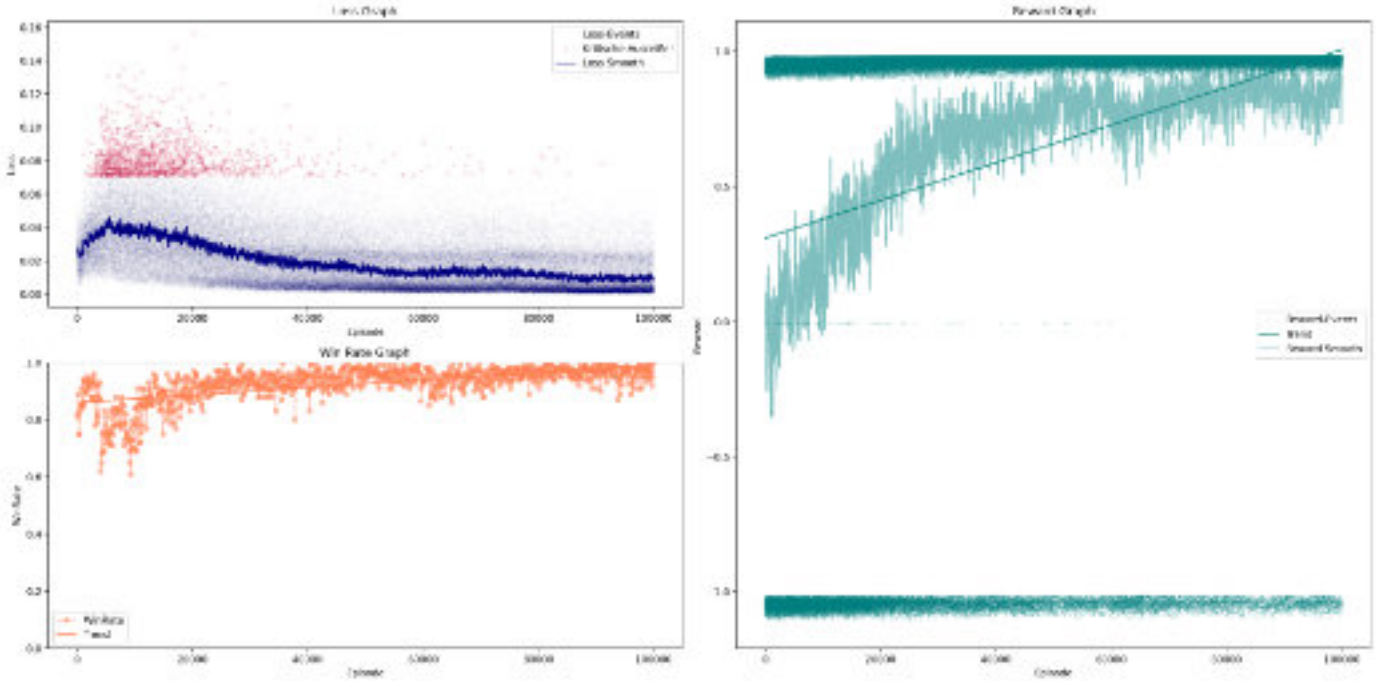
Haperparameter: hidden_size = 256, gamma = 0,95, batch_size = 64, survive_reward = -0,005

Erwartung: Durch gamma = 0,95 und einen negativen survive_reward, sollte der Agent auf ein kurzes Spiel setzen und schnellstmöglich versuchen zu gewinnen.

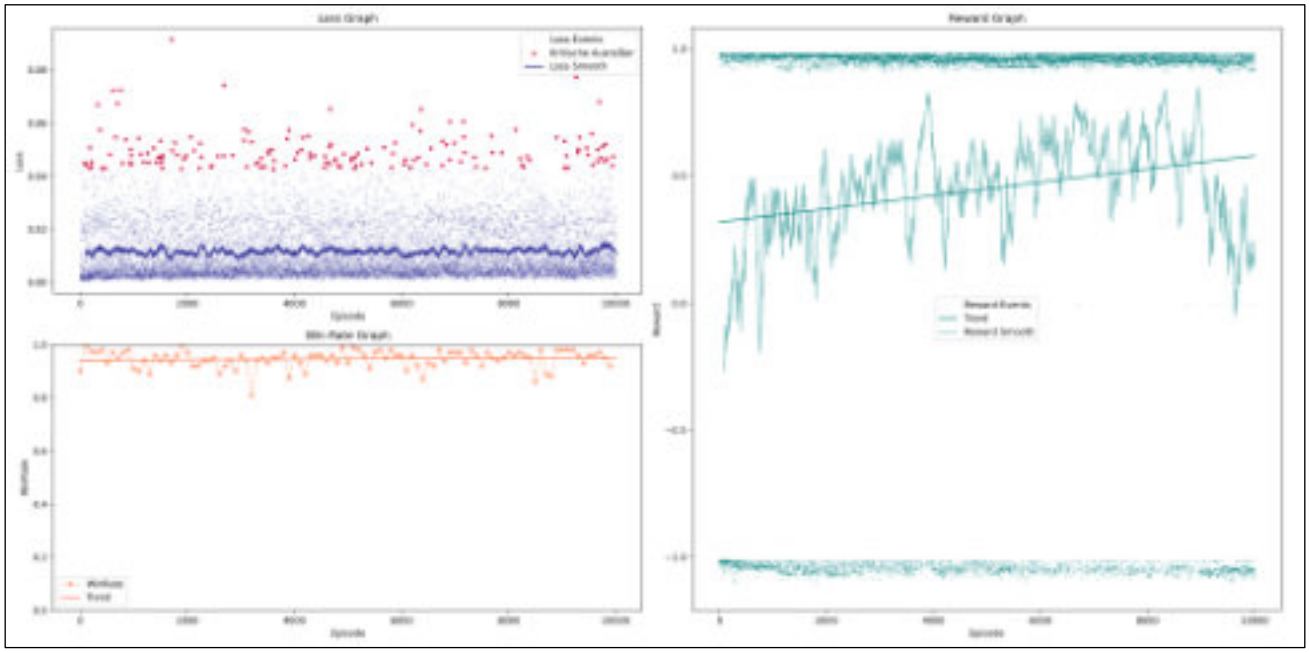
Überblick



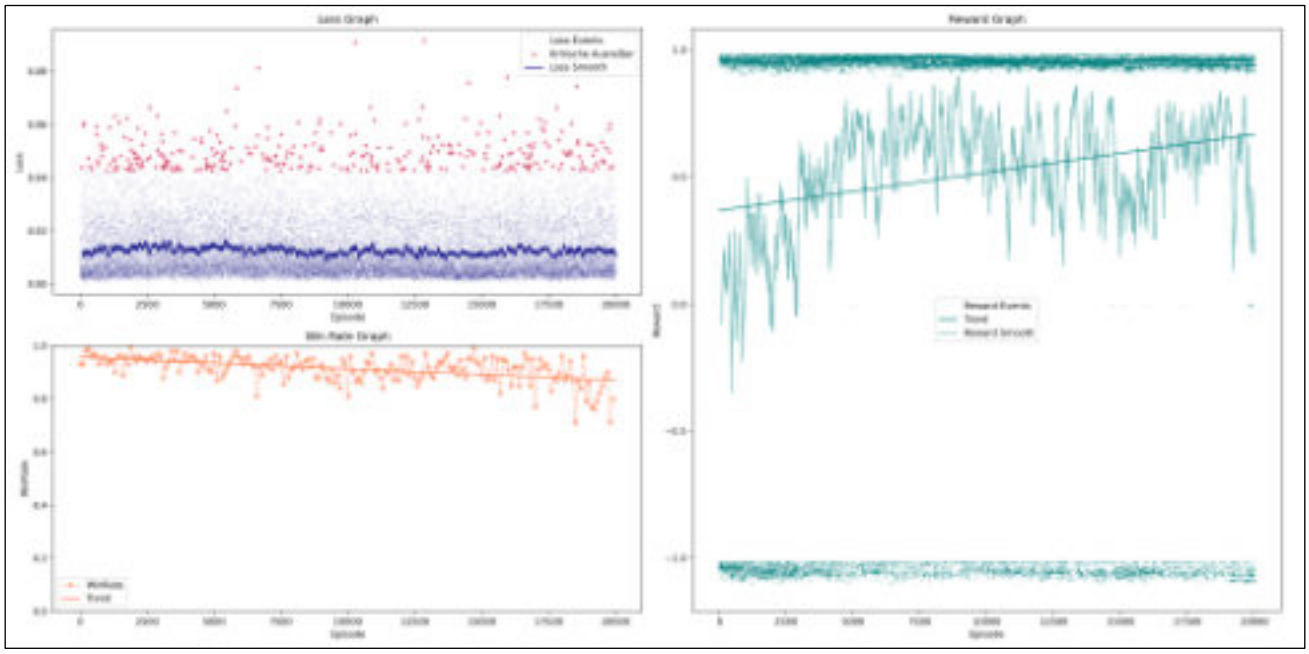
base_model



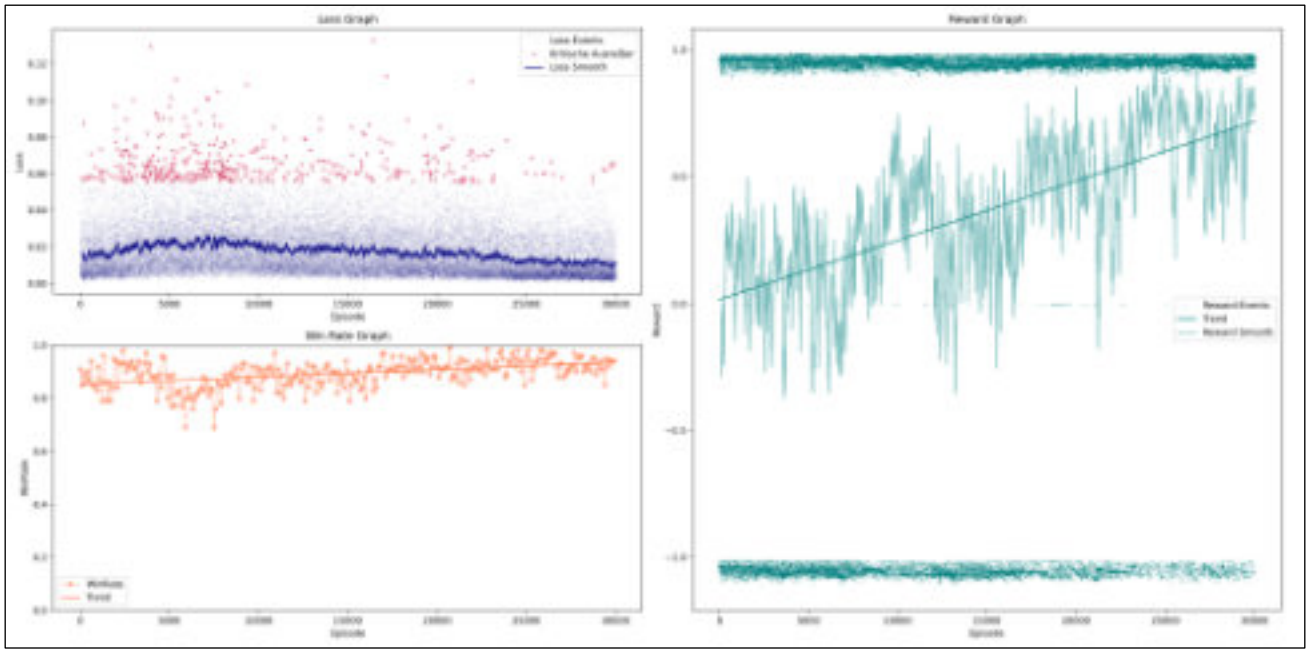
league_play_model_1



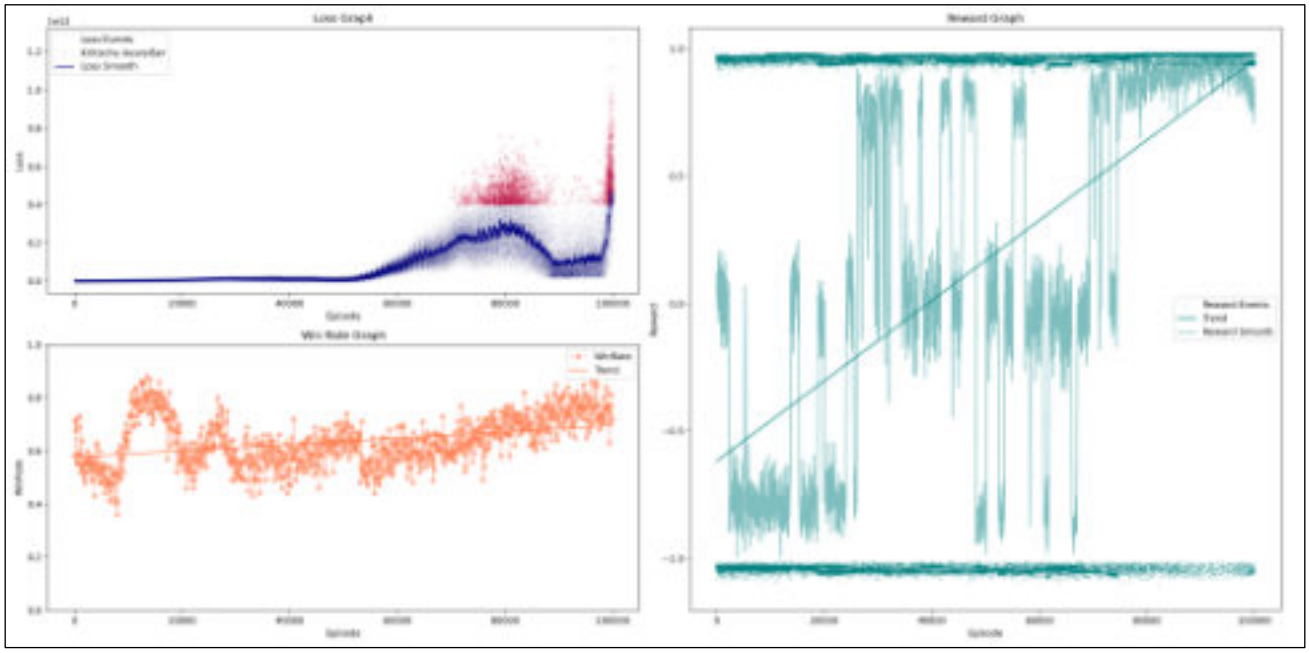
league_play_model_2



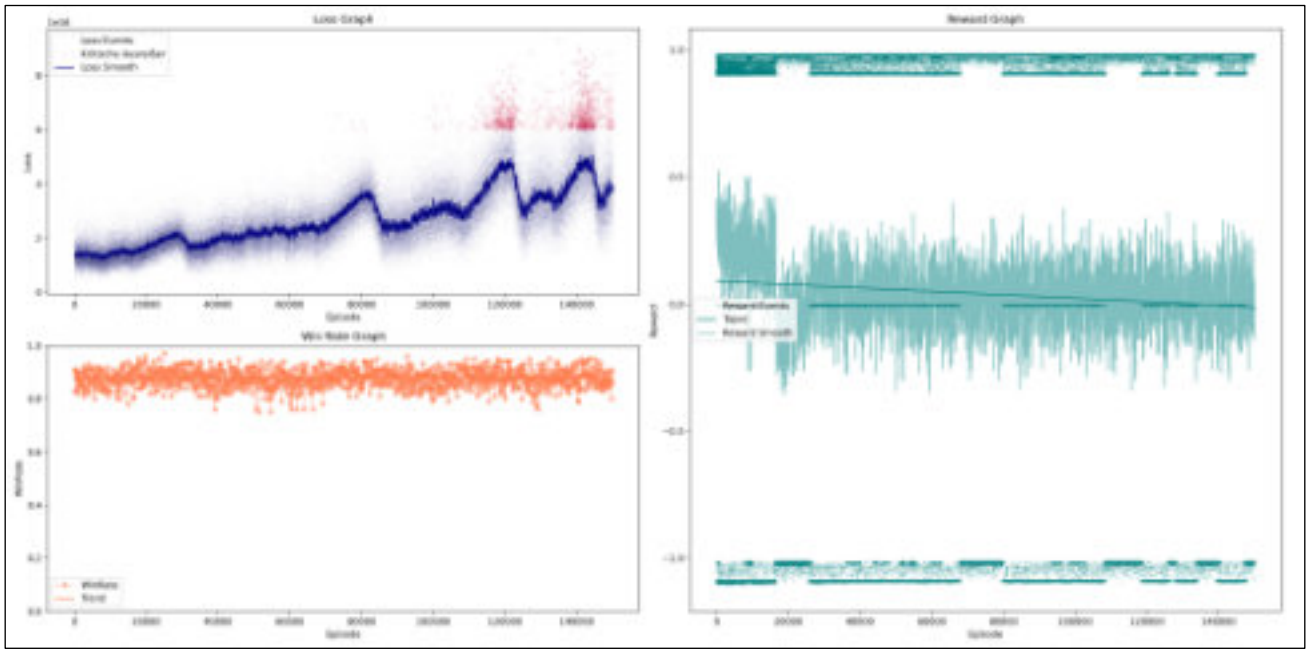
league_play_model_3



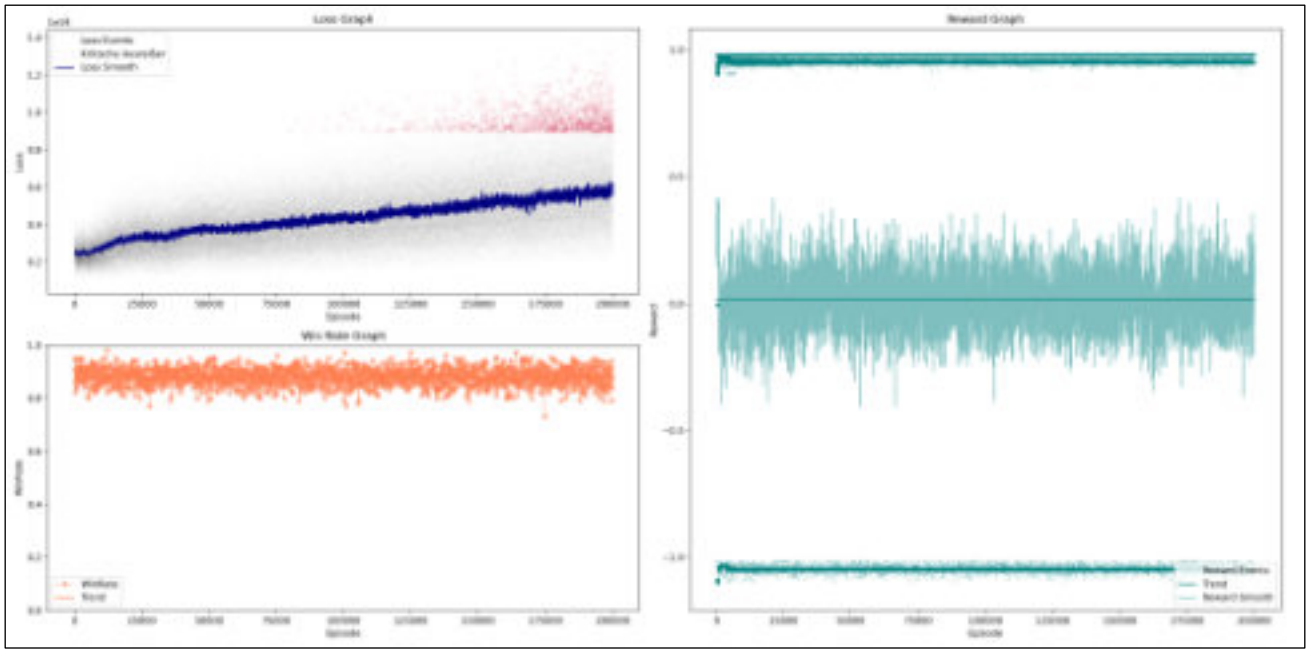
league_play_model_10



league_play_model_15



league_play_model_20

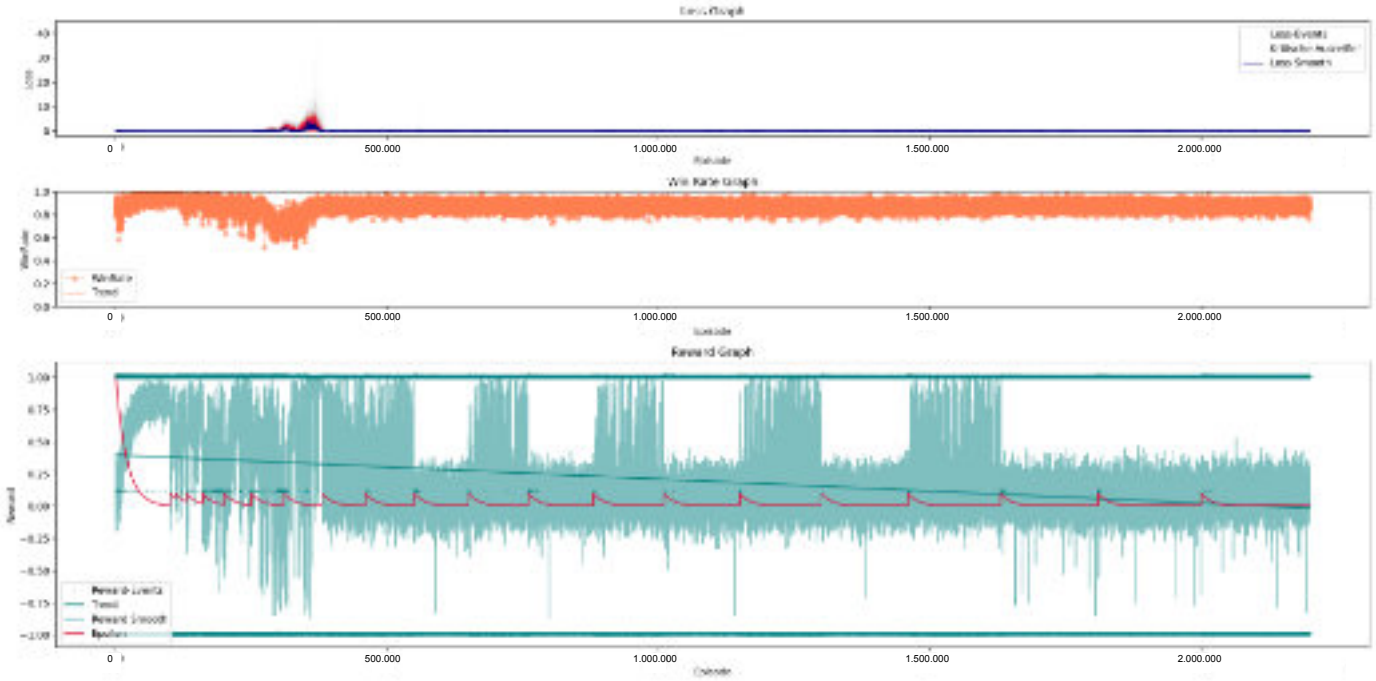


Trainer 2

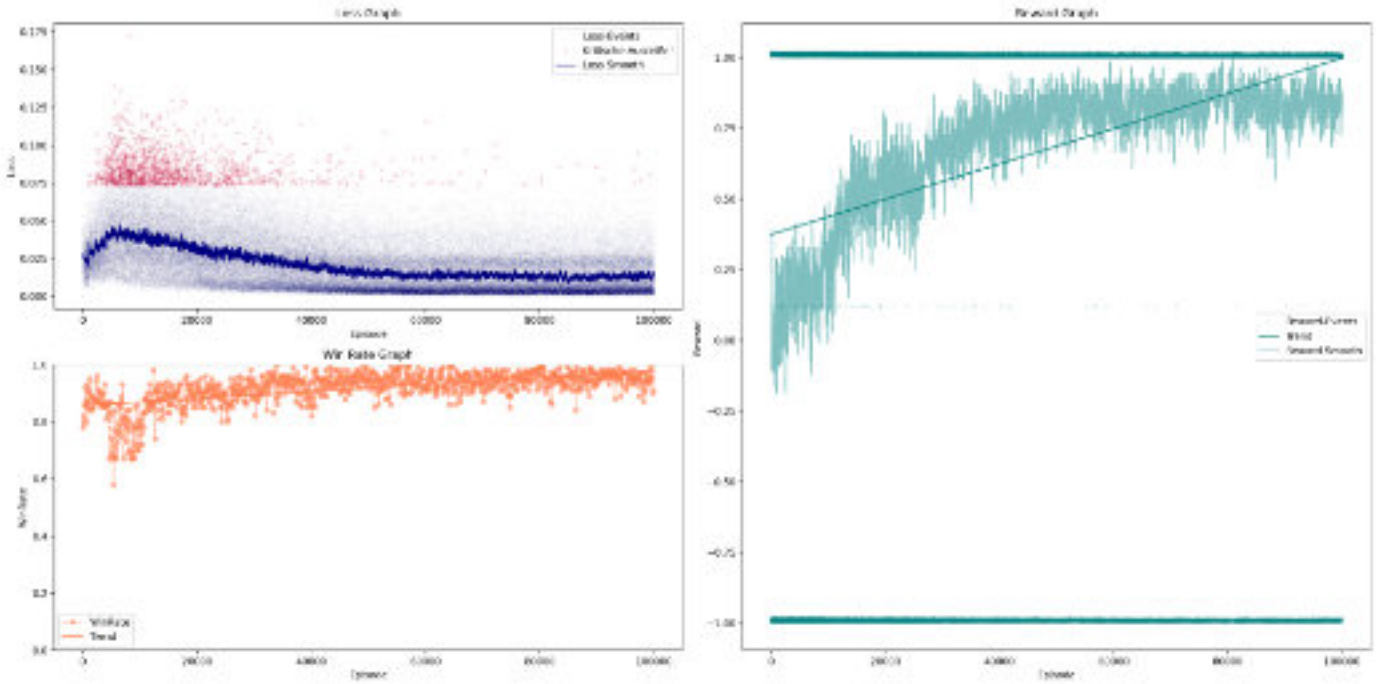
Haperparameter: hidden_size = 256, gamma = 0,95, batch_size = 64, survive_reward = 0,001

Erwartung: Durch gamma = 0,9, aber den positiven survive_reward sollte eine balance entstehen aus möglichst schnell gewinnen, aber Spiel nicht in die Länge ziehen.

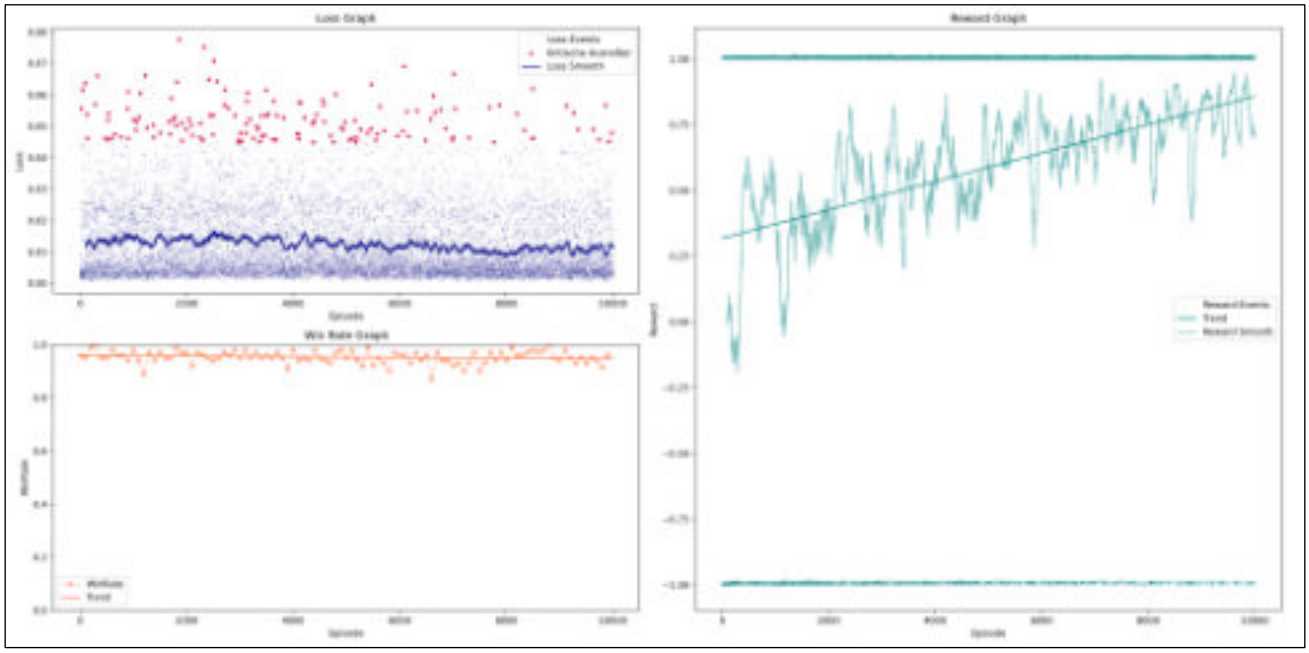
Überblick



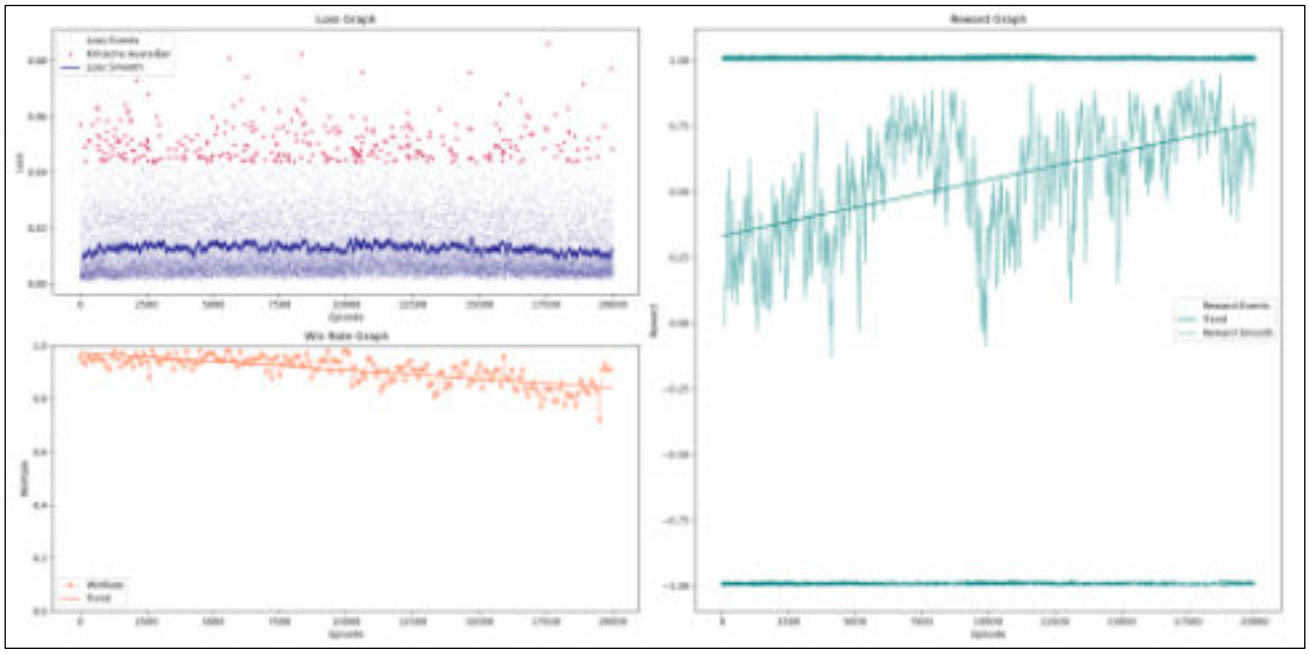
base_model



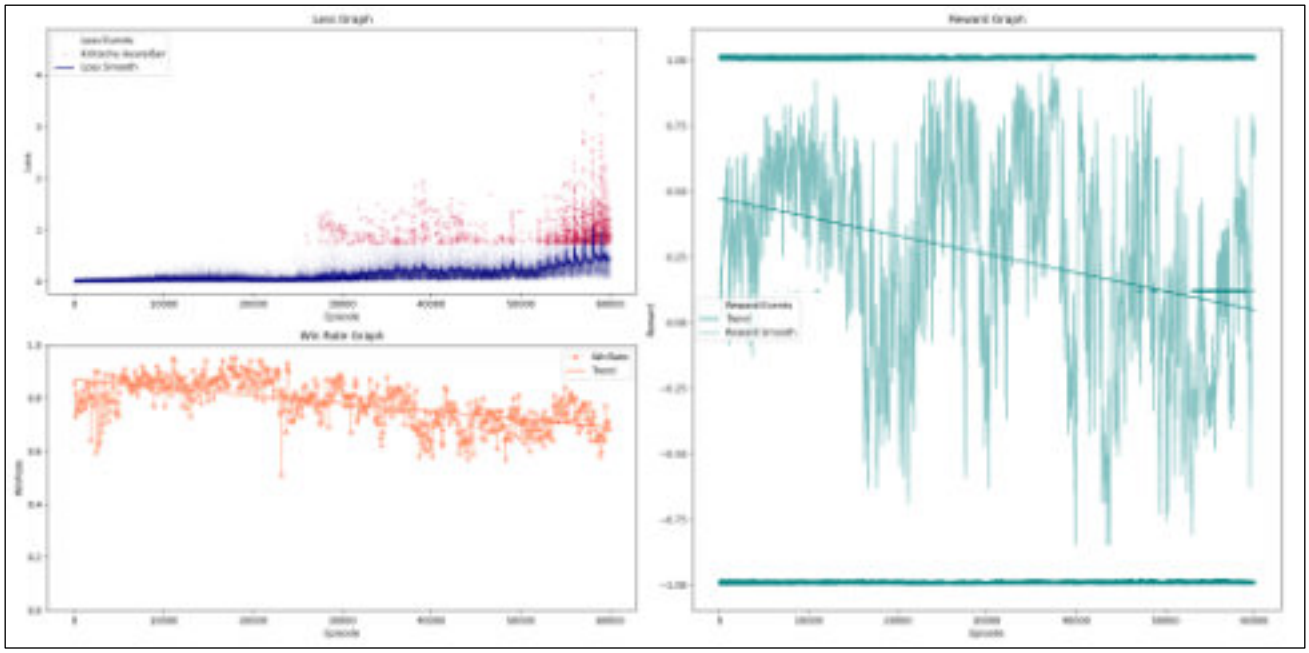
league_play_model_1



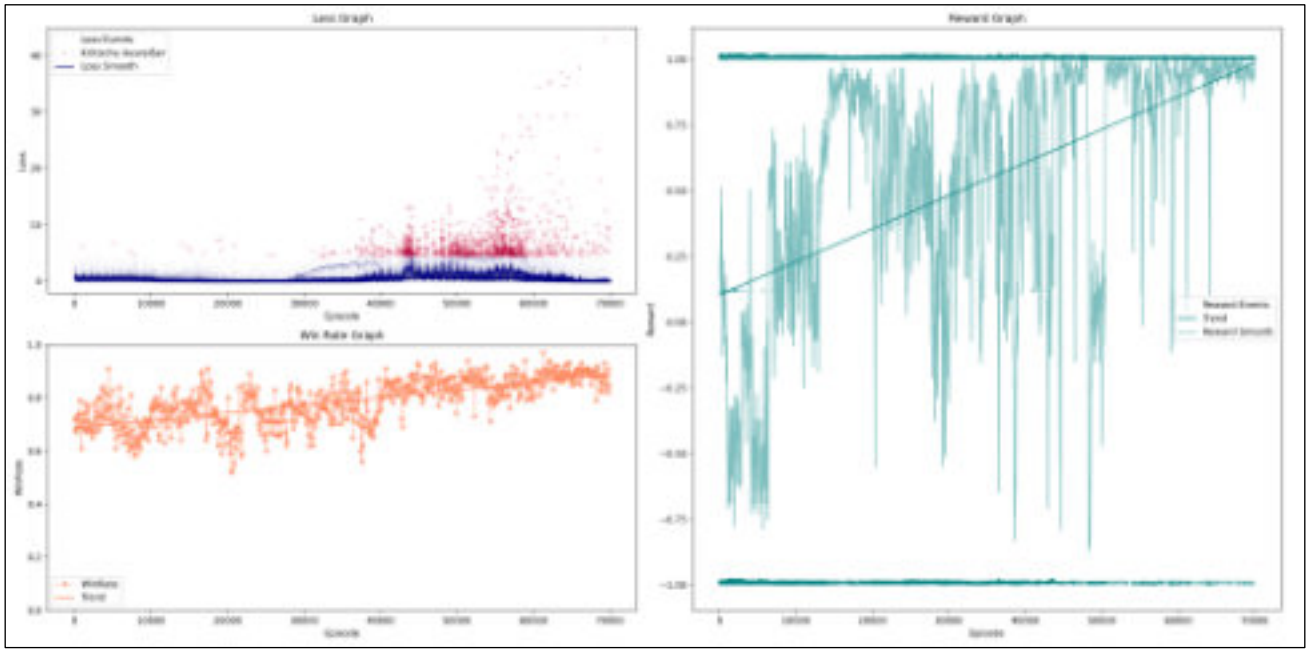
league_play_model_2



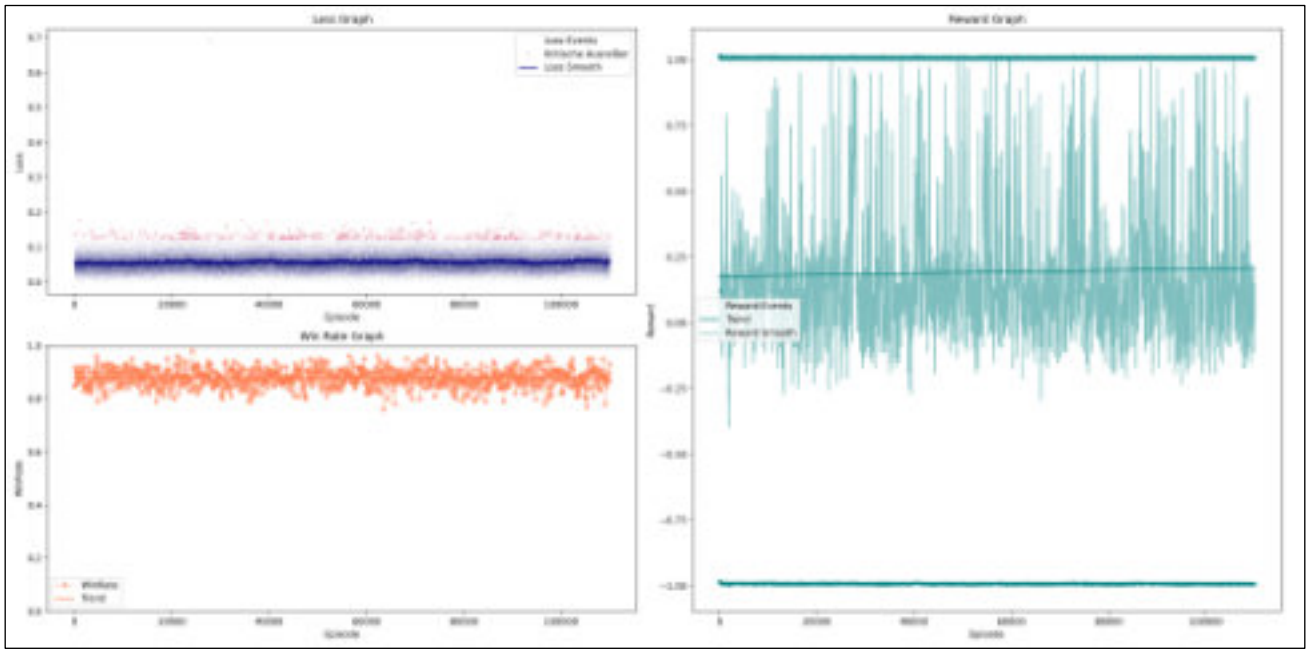
league_play_model_6



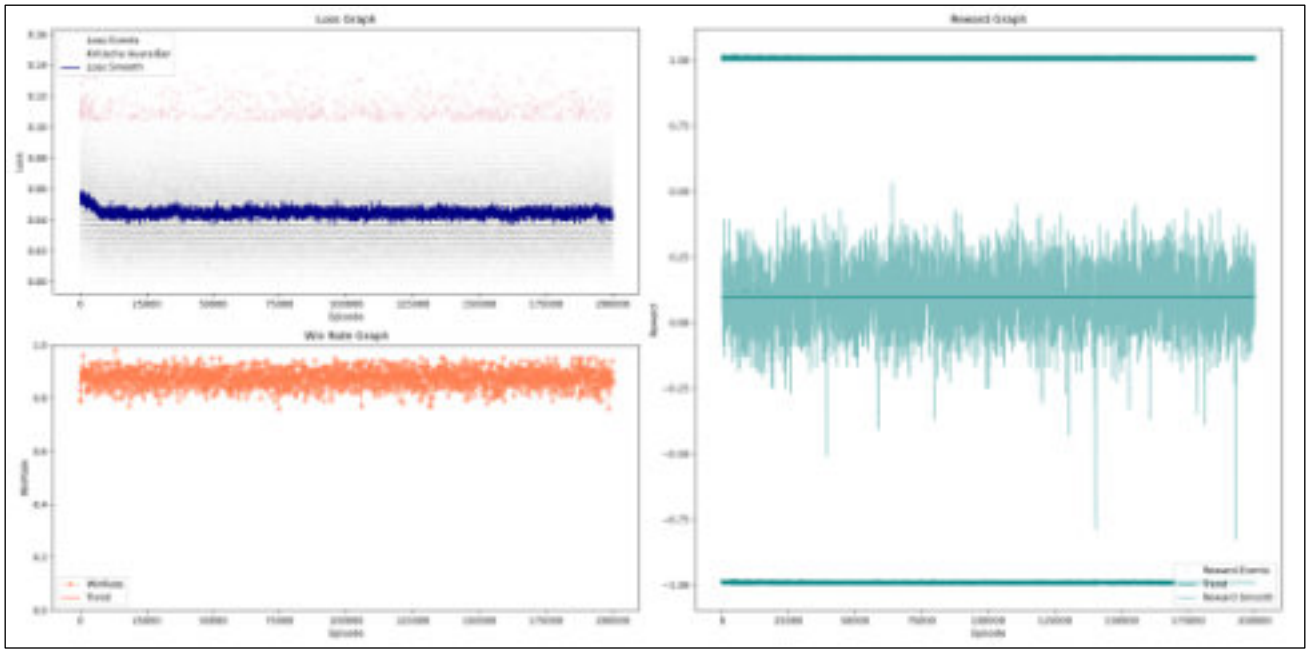
league_play_model_7



league_play_model_10



league_play_model_20

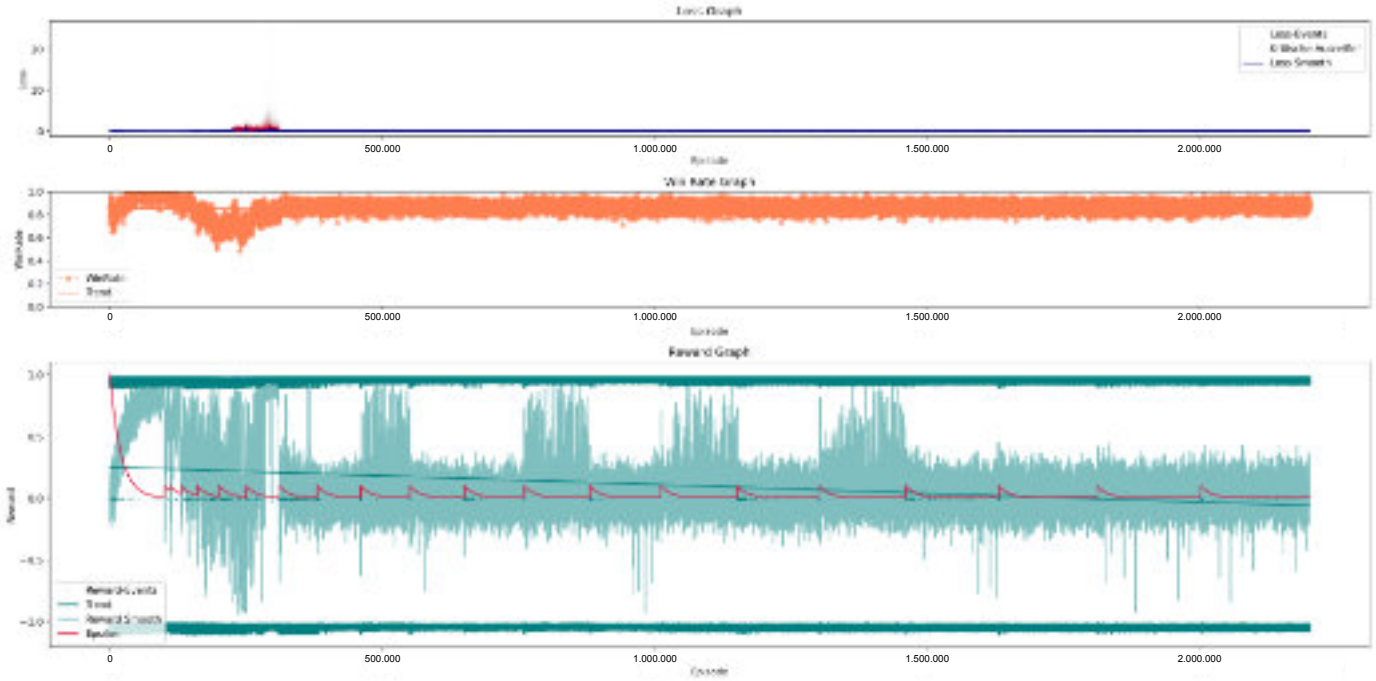


Trainer 3

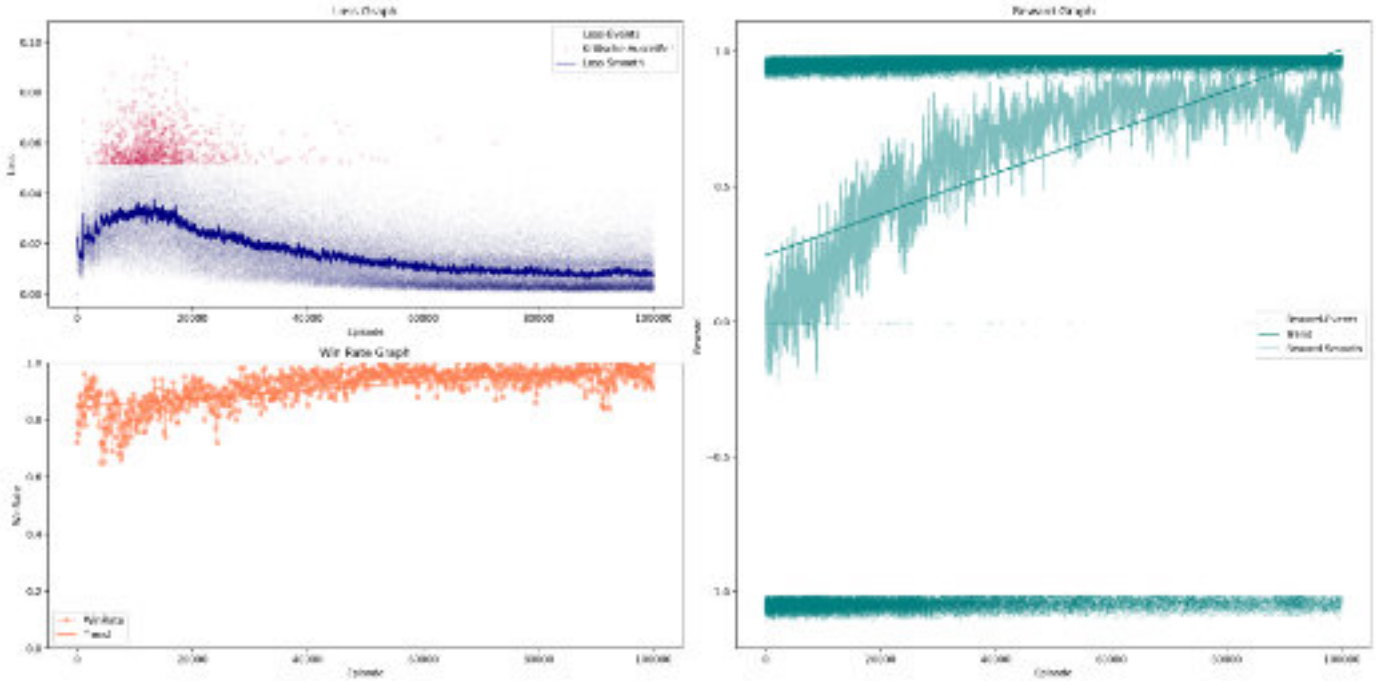
Haperparameter: hidden_size = 256, gamma = 0,95, batch_size = 128, survive_reward = -0,005

Erwartung: Der Agent sollte sich ähnlich zu Trainer 1 verhalten, es werden jedoch mehr Daten im Lernprozess einbezogen, wodurch potentiell weniger relevante Daten miteibezogen werden.

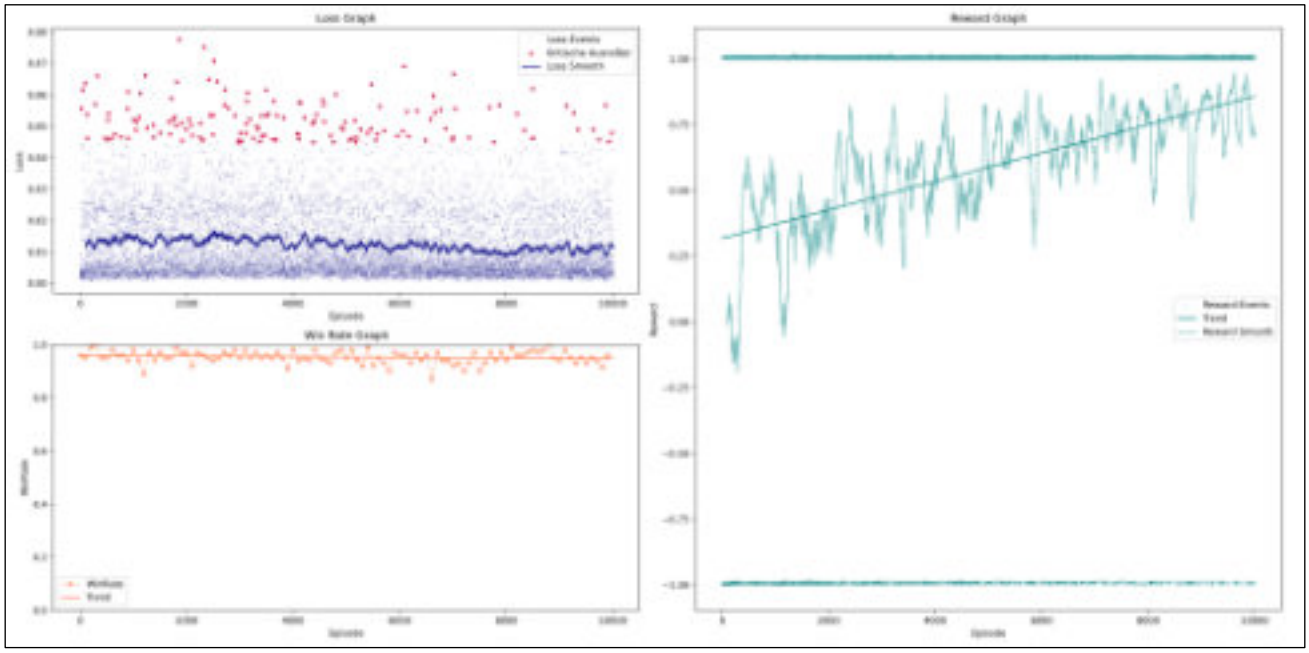
Überblick



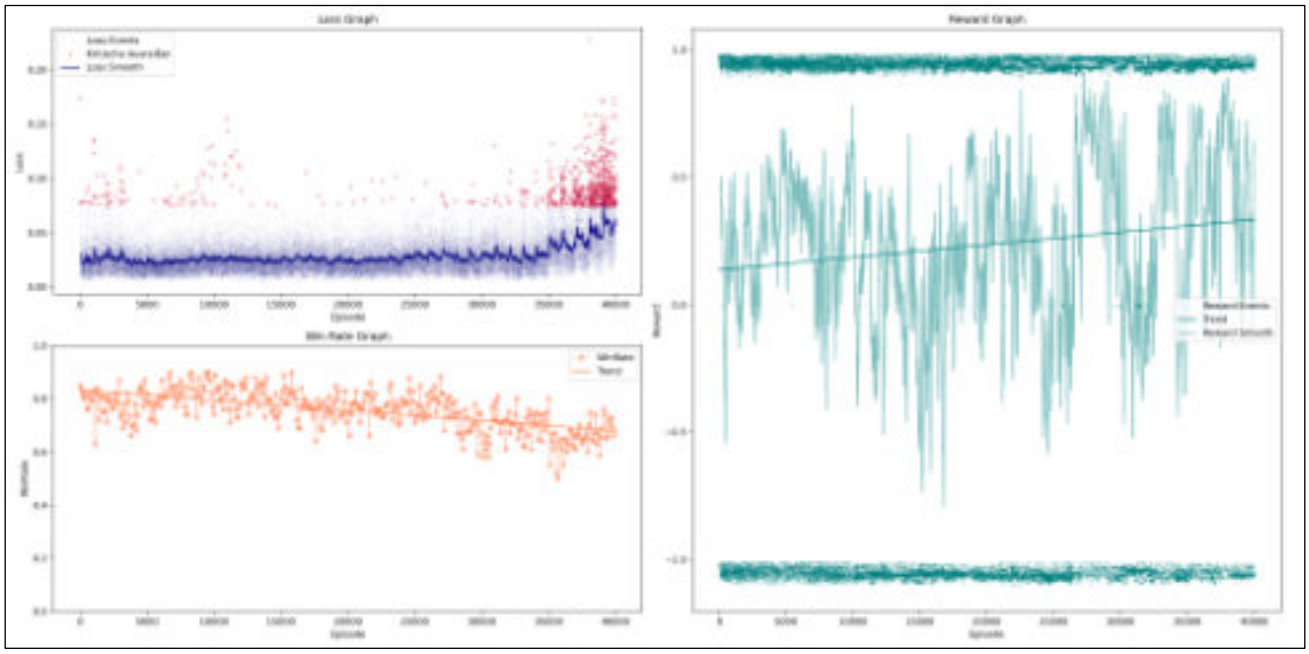
base_model



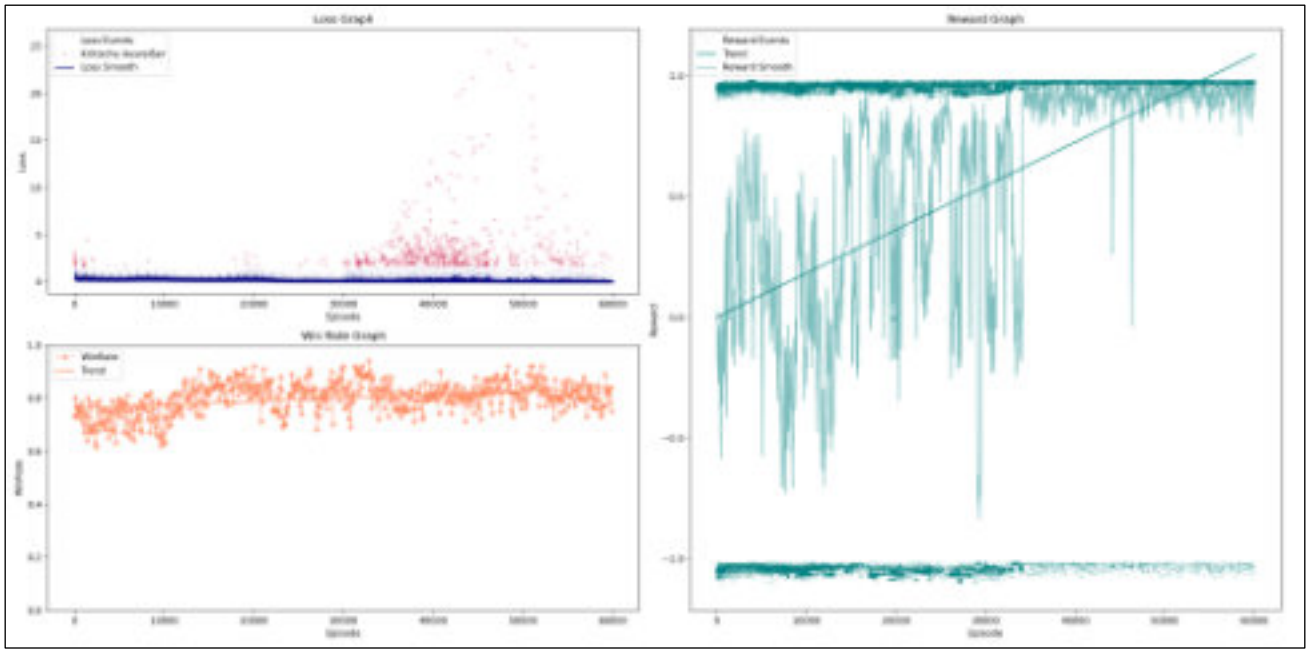
league_play_model_1



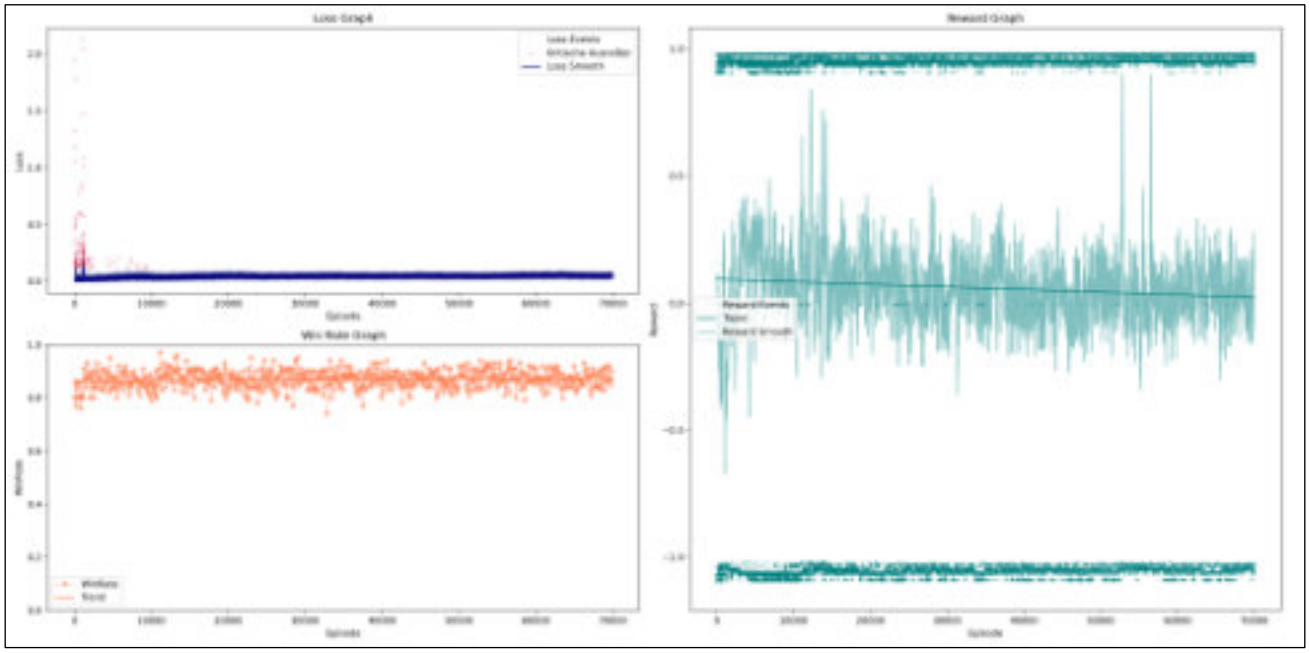
league_play_model_4



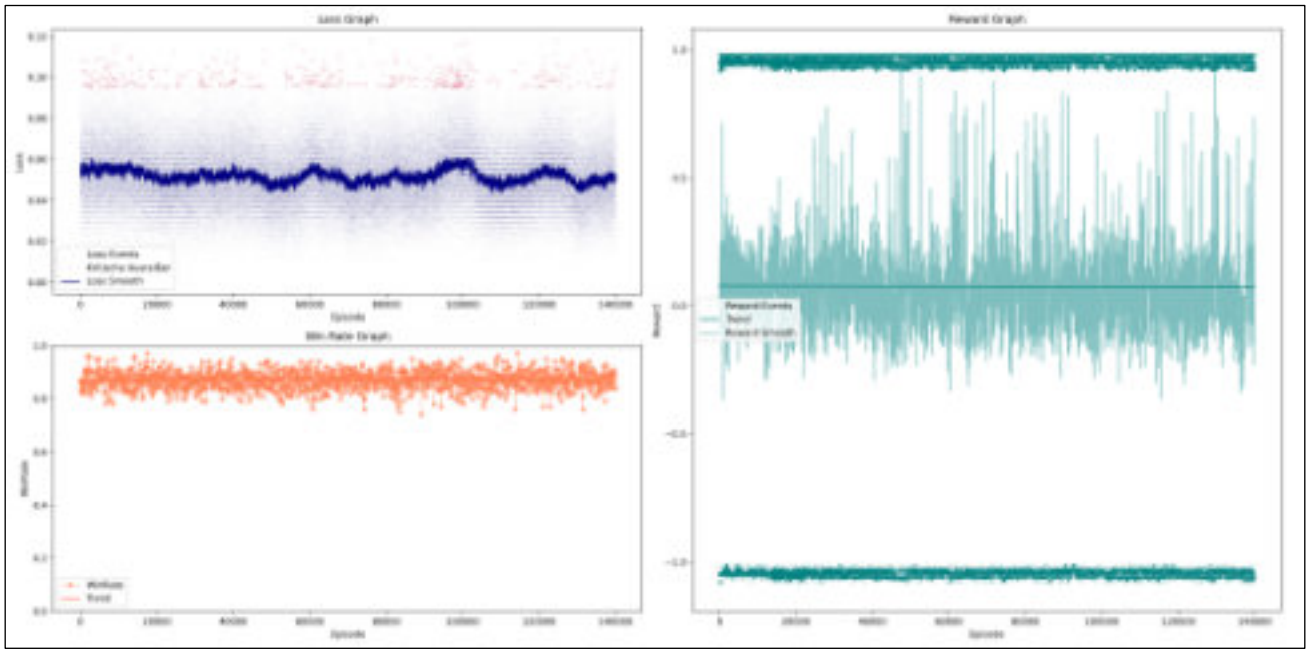
league_play_model_6



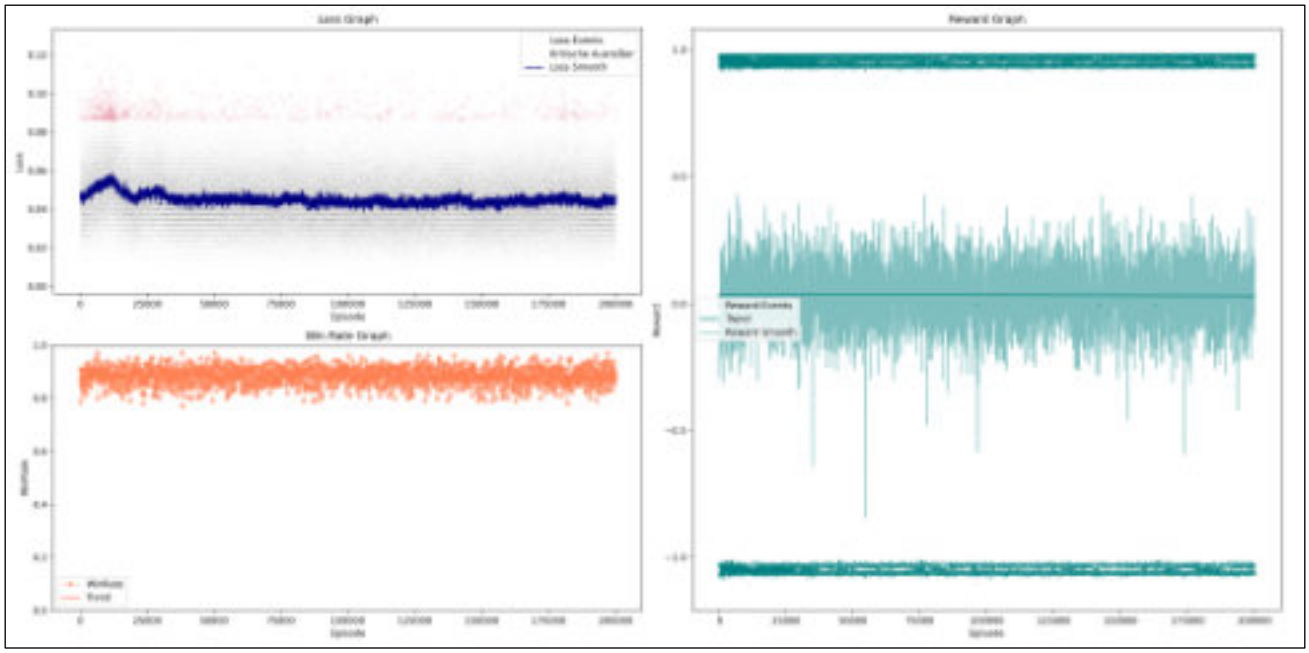
league_play_model_7



league_play_model_14



league_play_model_20

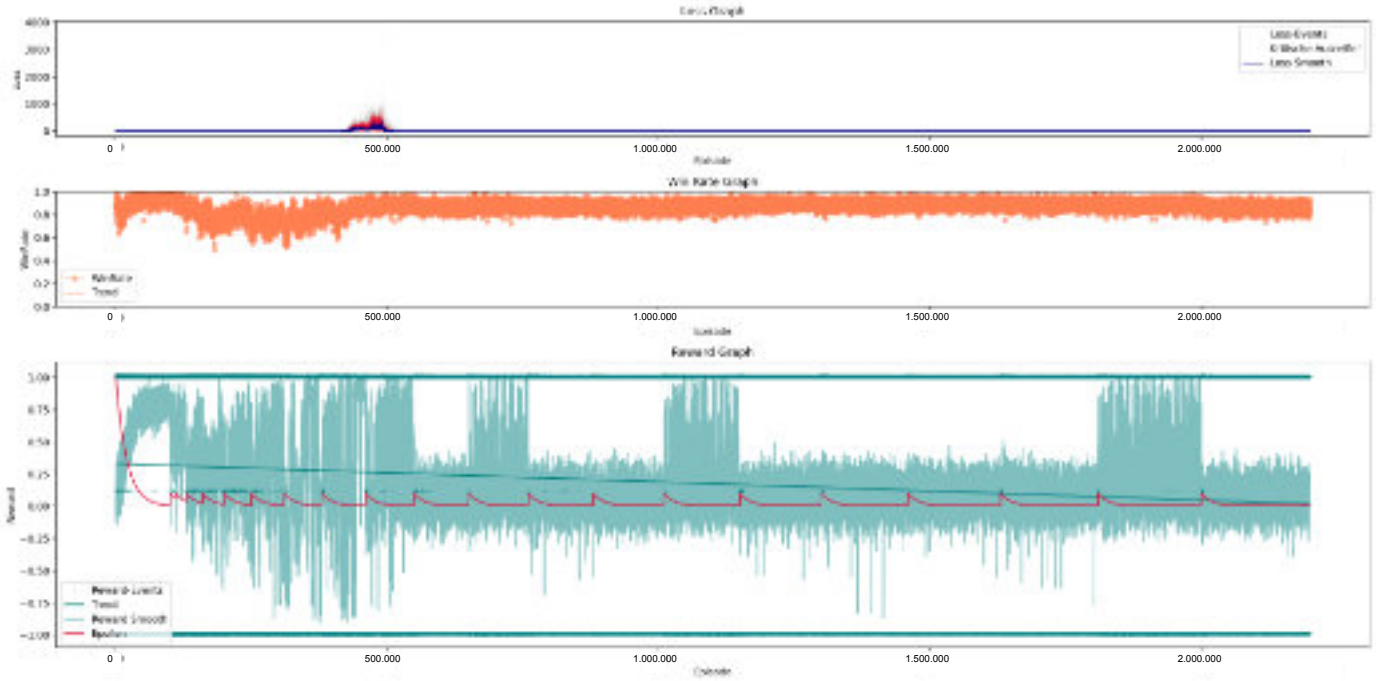


Trainer 4

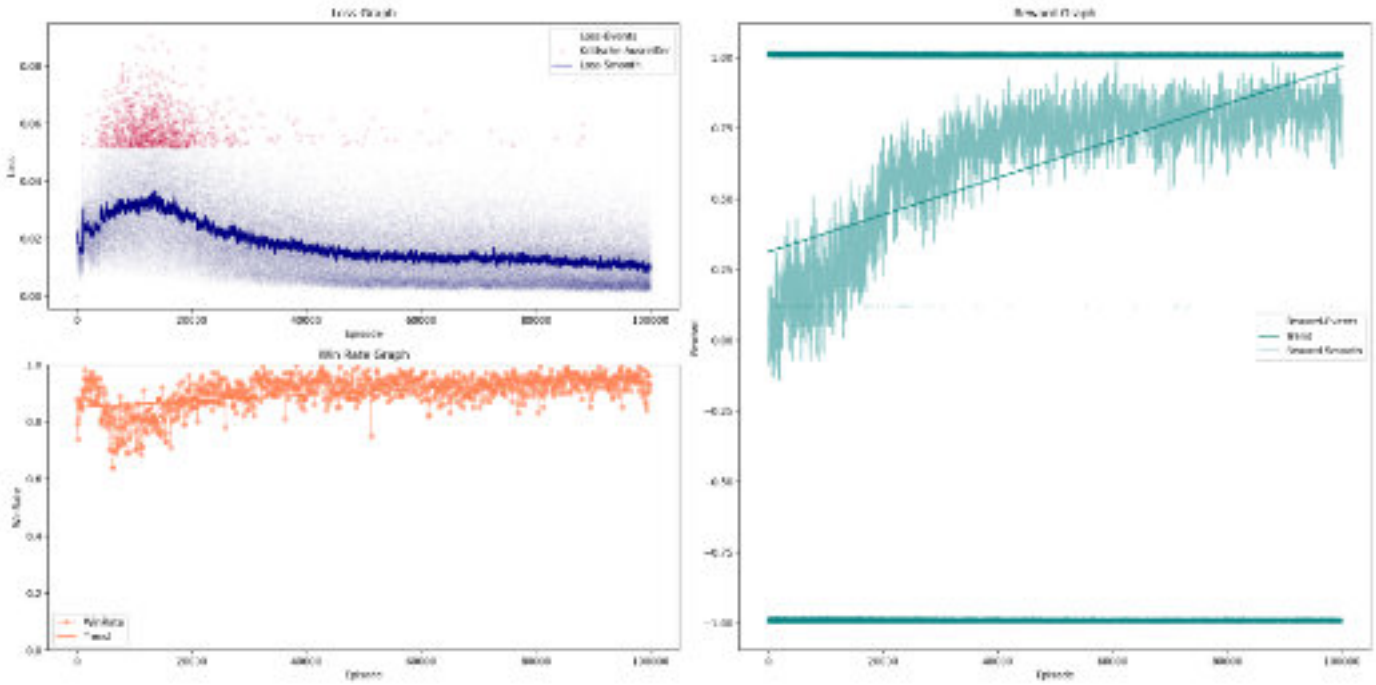
Haperparameter: hidden_size = 256, gamma = 0,95, batch_size = 128, survive_reward = 0,001

Erwartung: Der Agent sollte sich ähnlich zu Trainer 2 verhalten, es werden jedoch mehr Daten im Lernprozess einbezogen, wodurch potentiell weniger relevante Daten miteibezogen werden.

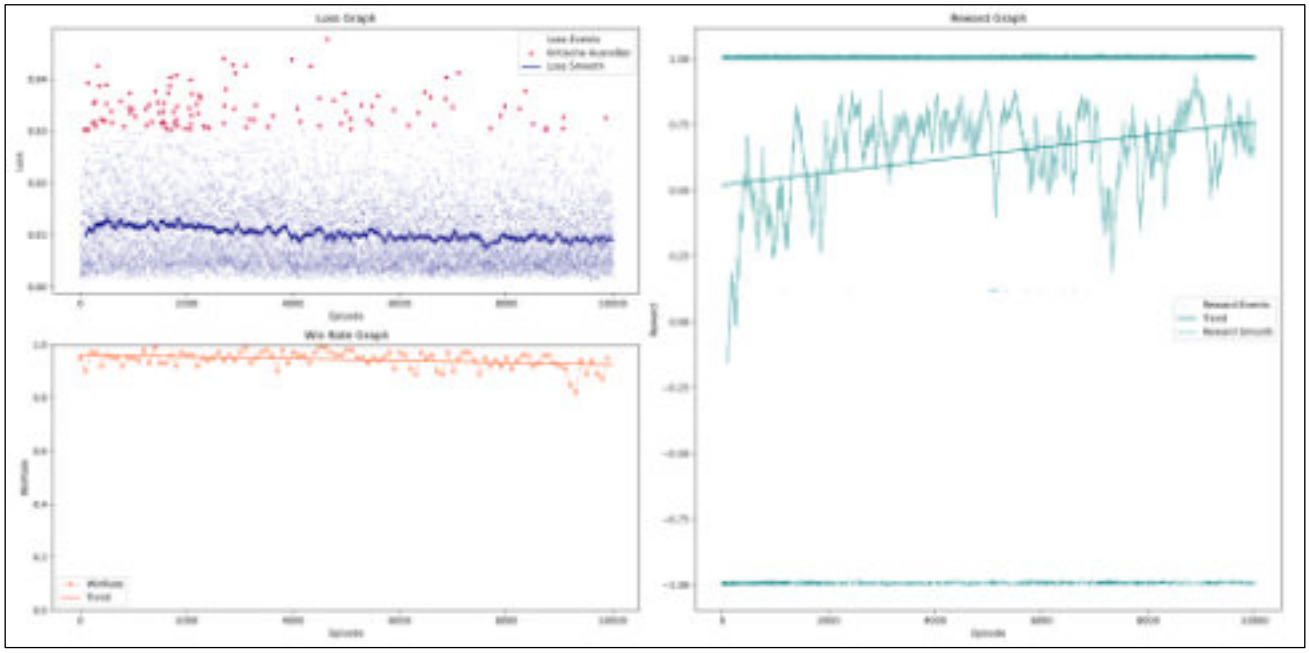
Überblick



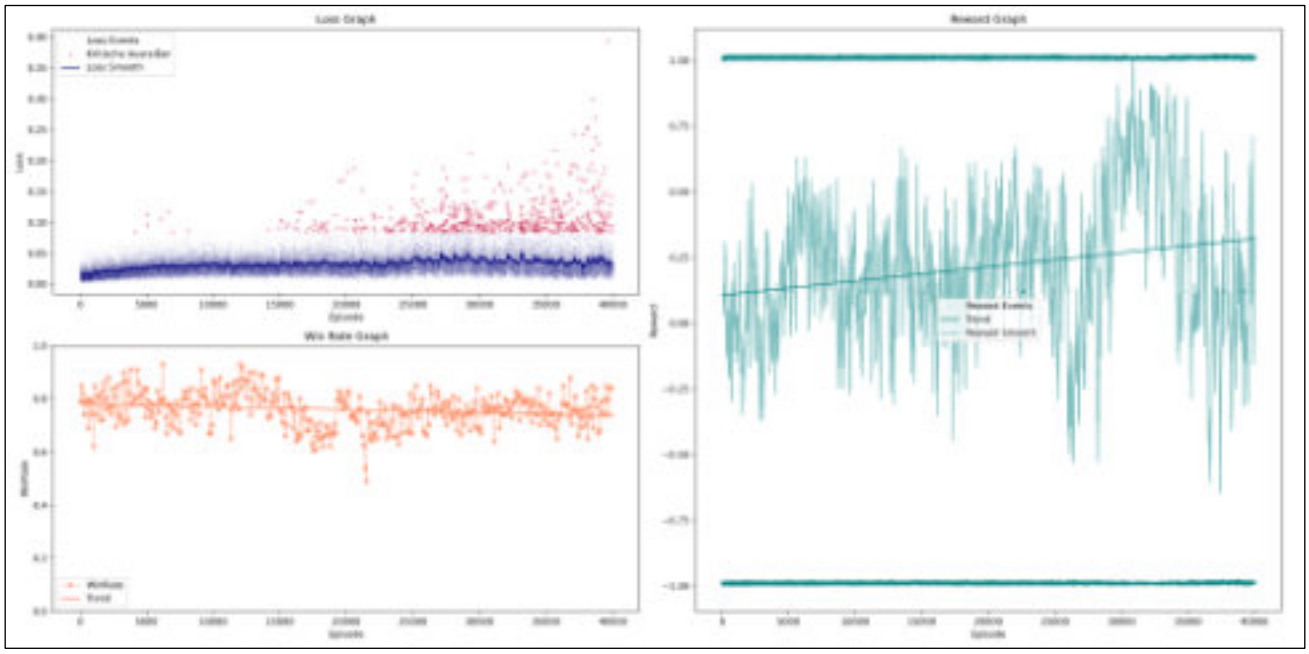
base_model



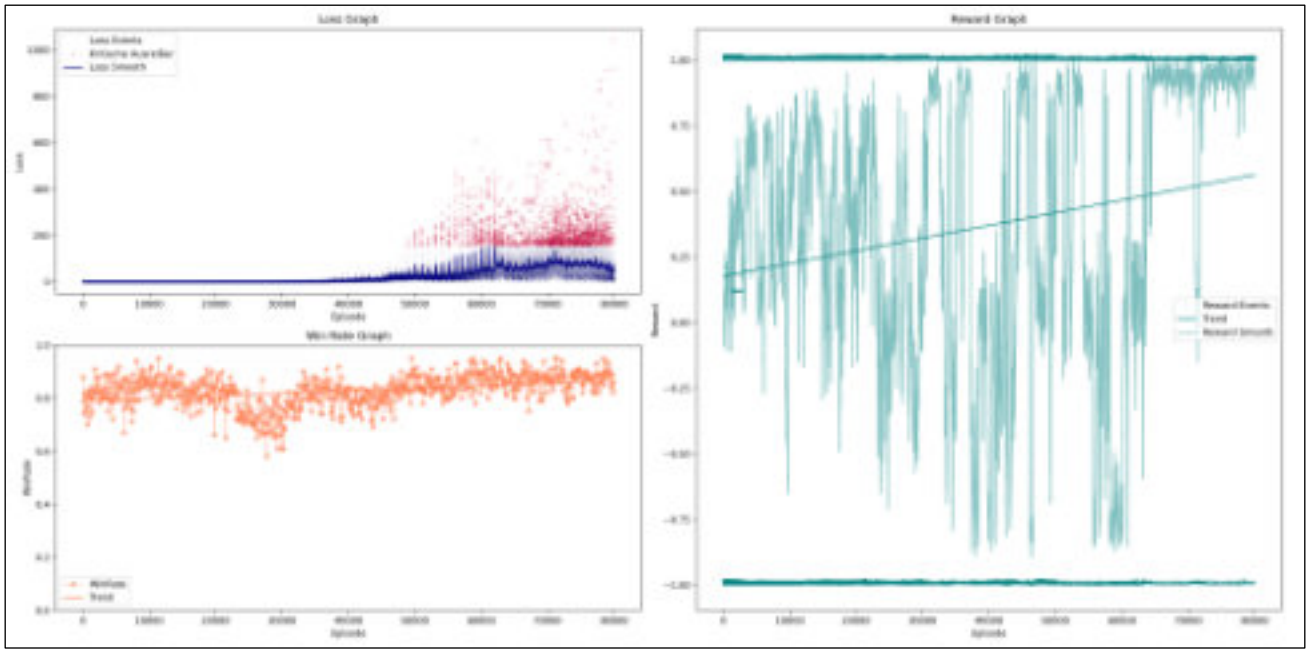
league_play_model_1



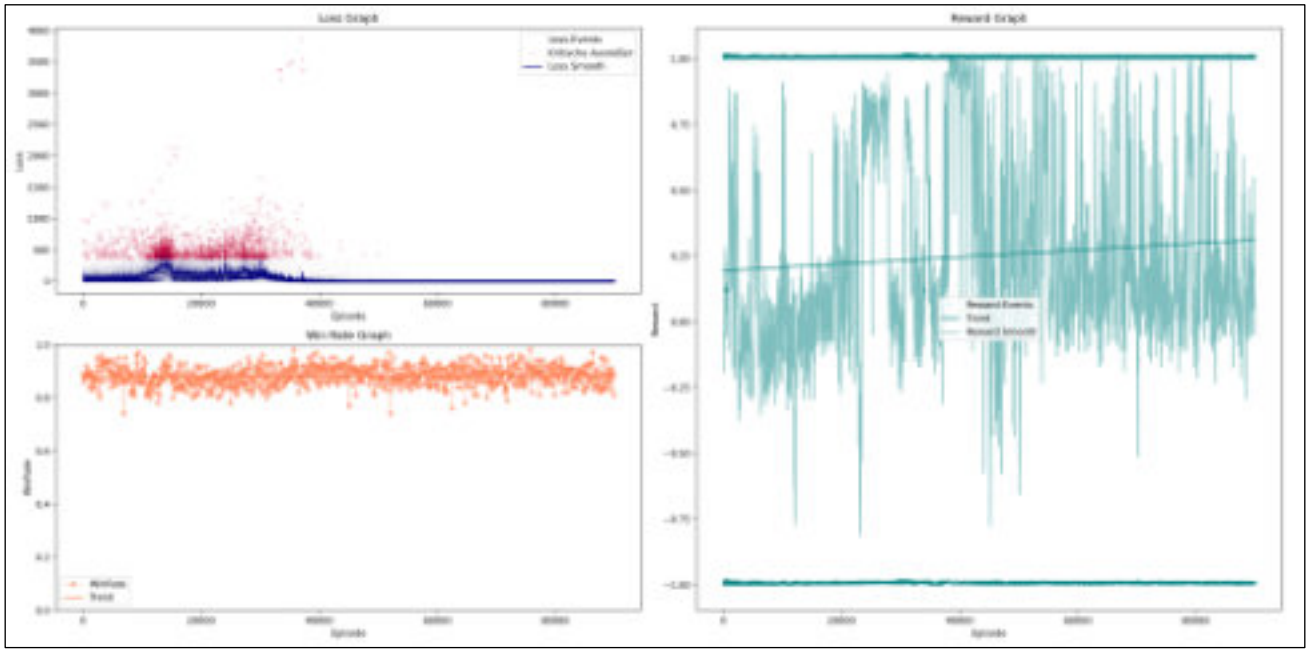
league_play_model_4



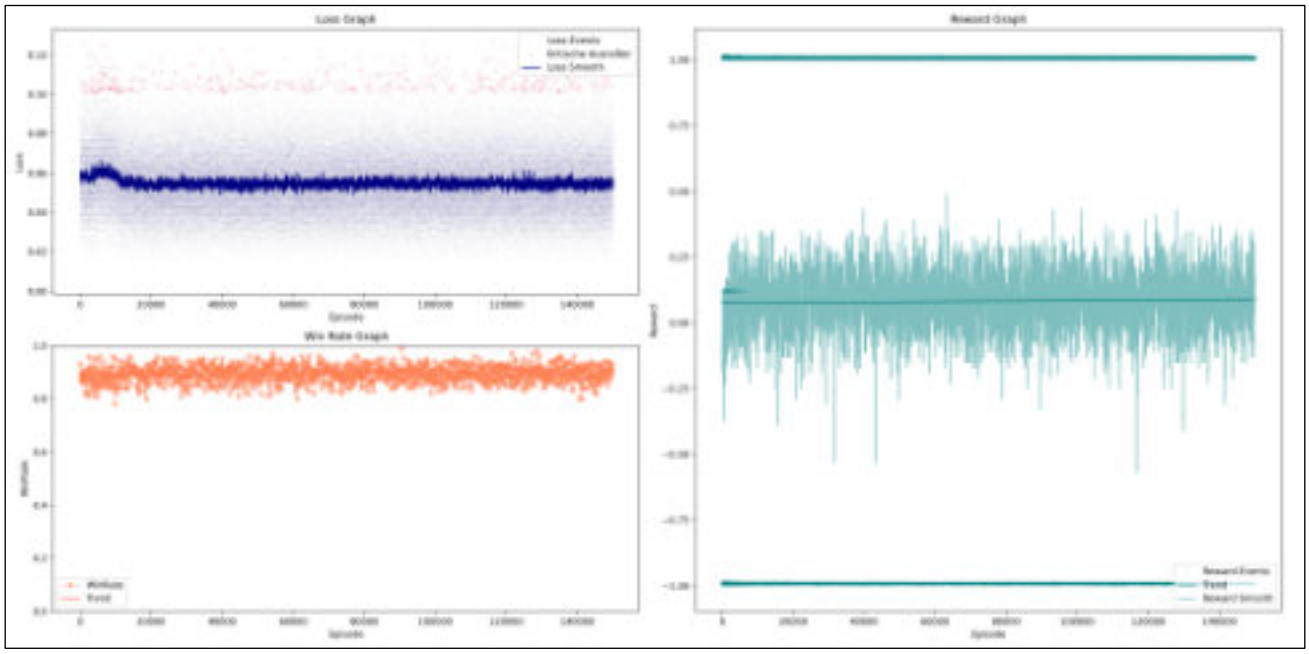
league_play_model_8



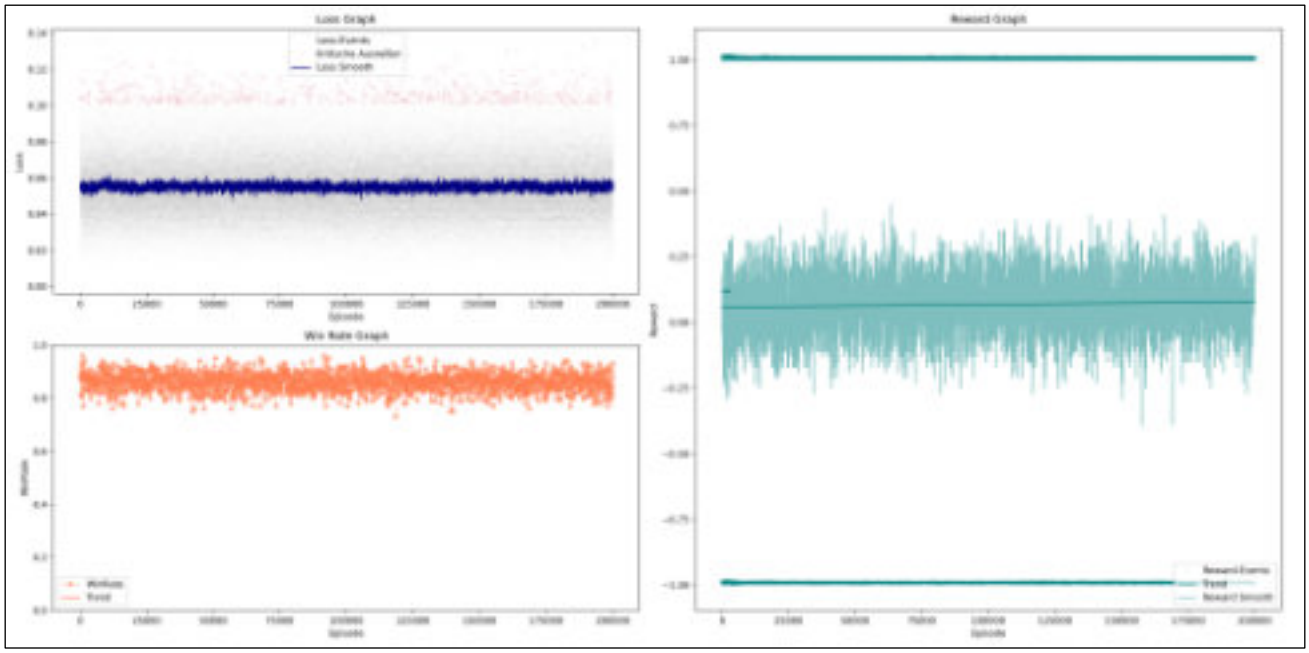
league_play_model_9



league_play_model_15



league_play_model_20

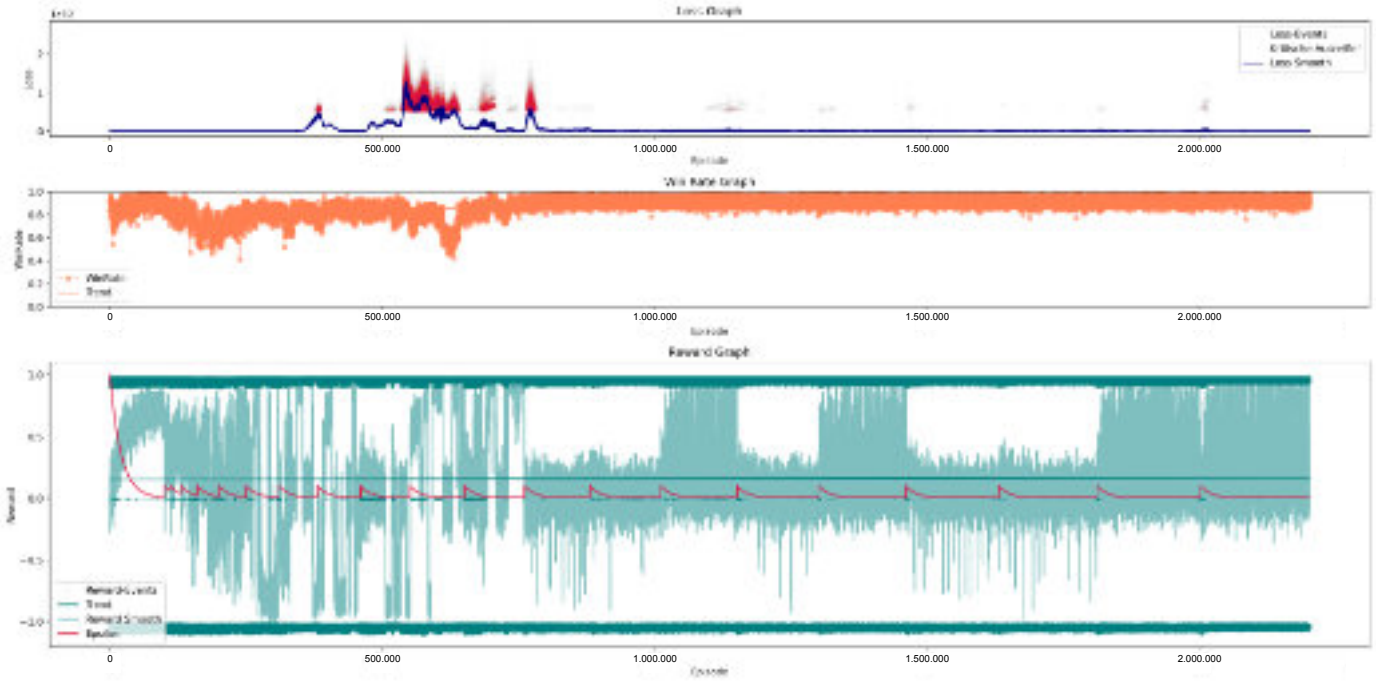


Trainer 5

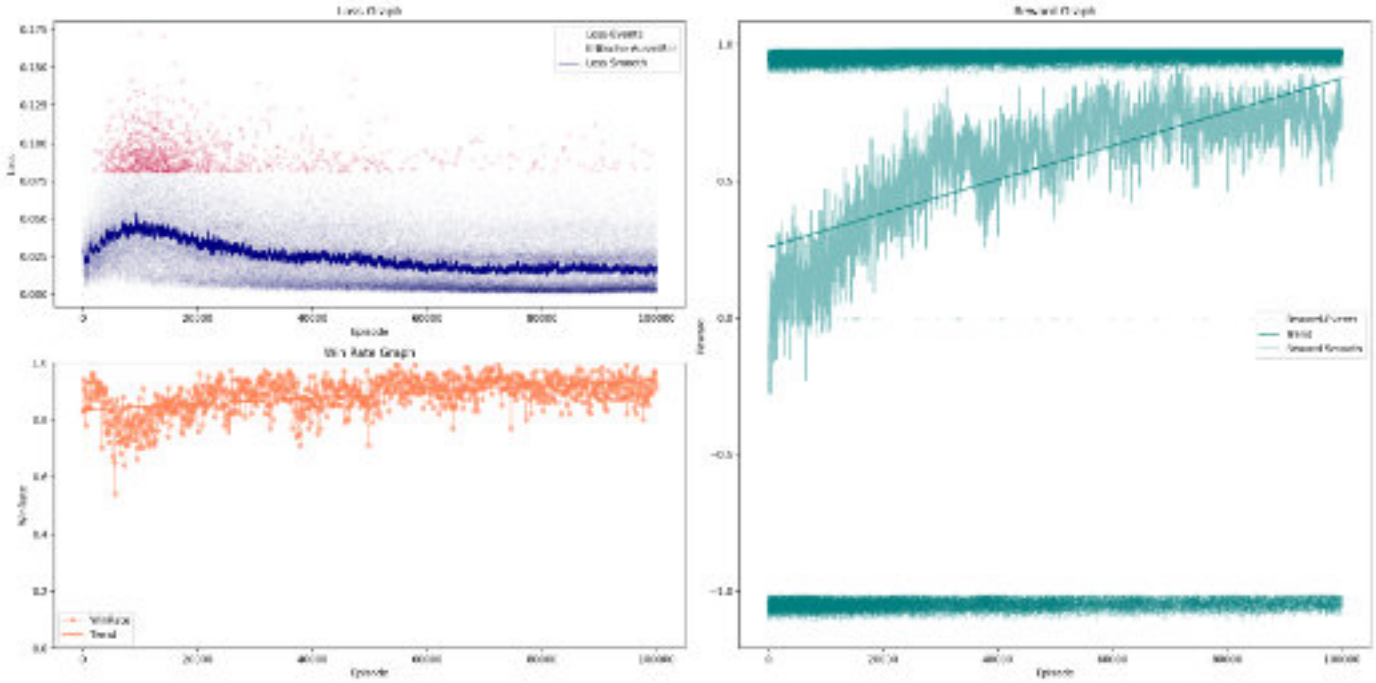
Haperparameter: hidden_size = 256, gamma = 0,99, batch_size = 64, survive_reward = -0,005

Erwartung: Durch den höheren Gamma-Wert in Kombination mit dem survive_reward und der hohen Netzwerkgröße, könnte der Agent vergleichbar mit Trainer 2 oder 4.

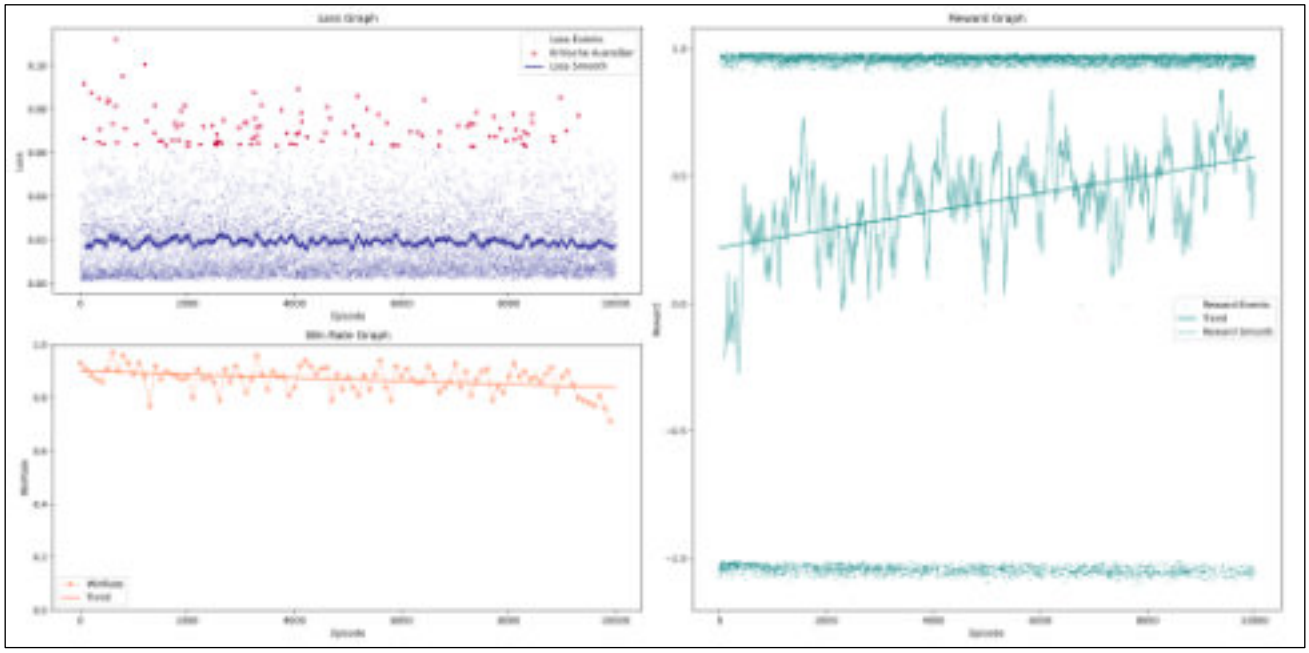
Überblick



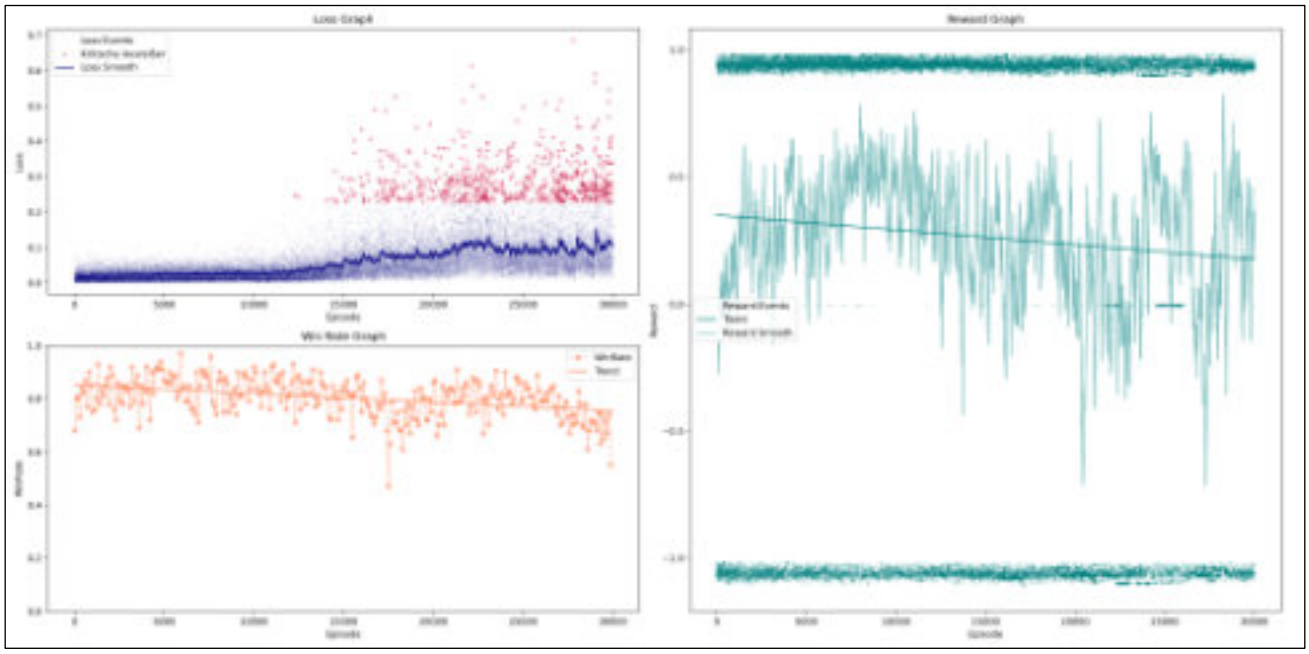
base_model



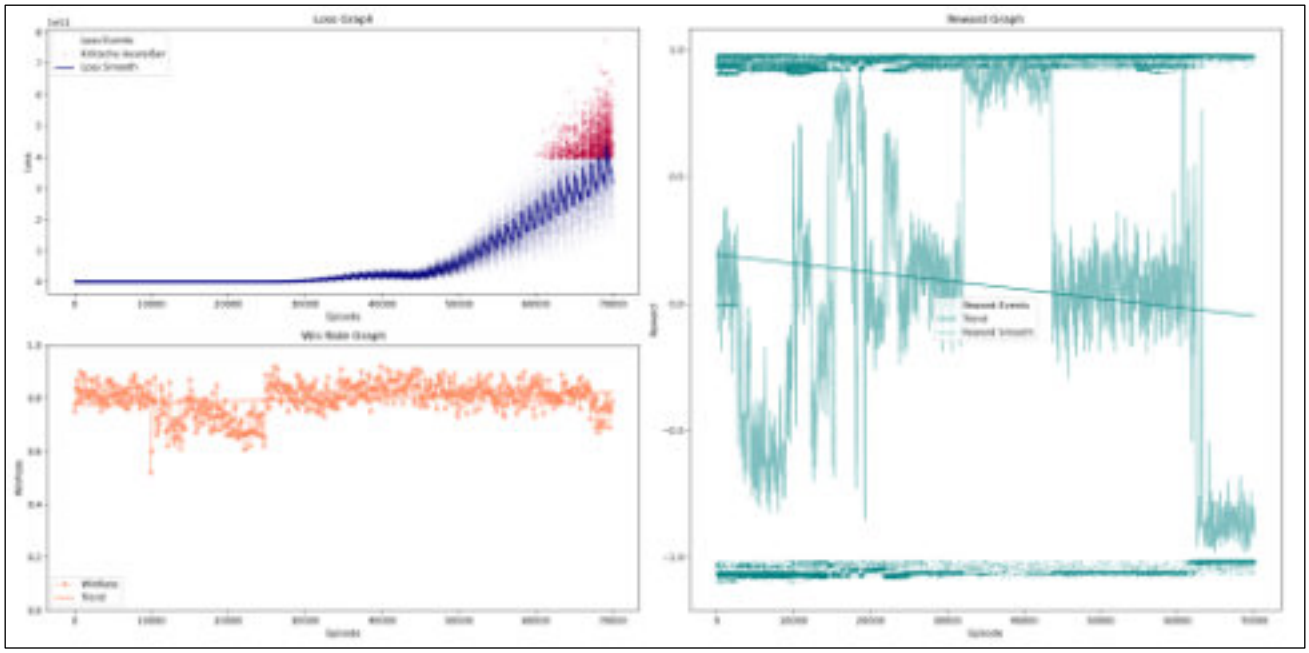
league_play_model_1



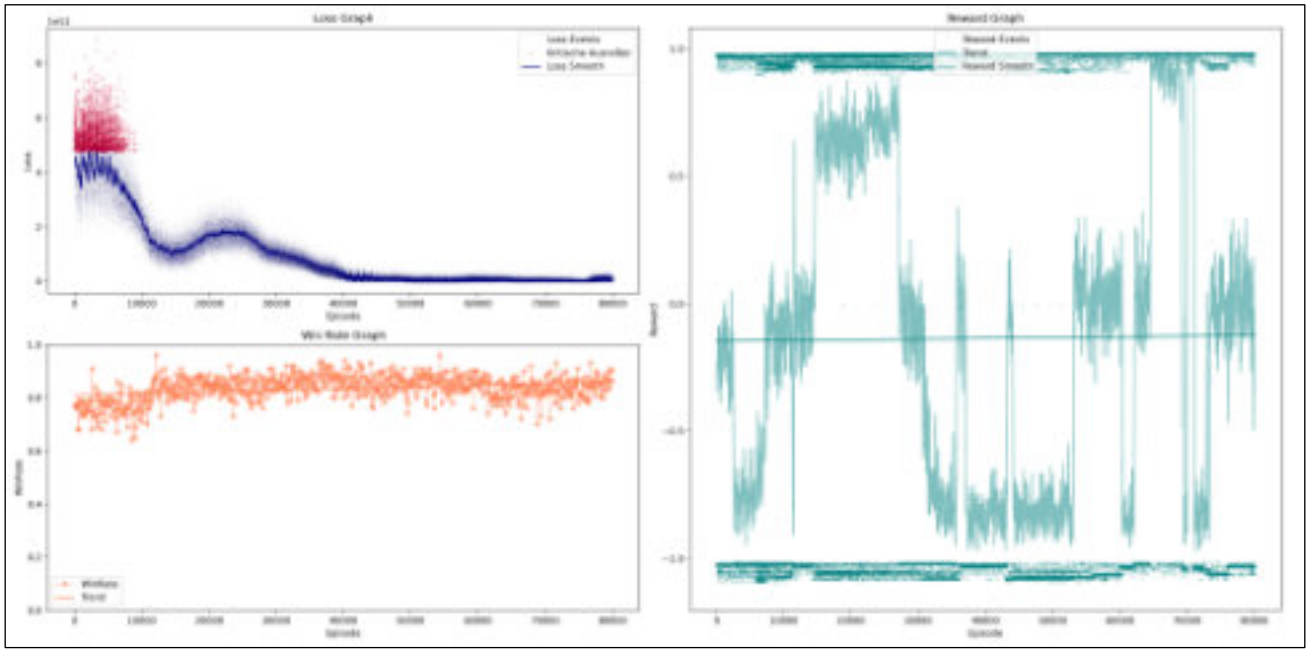
league_play_model_3



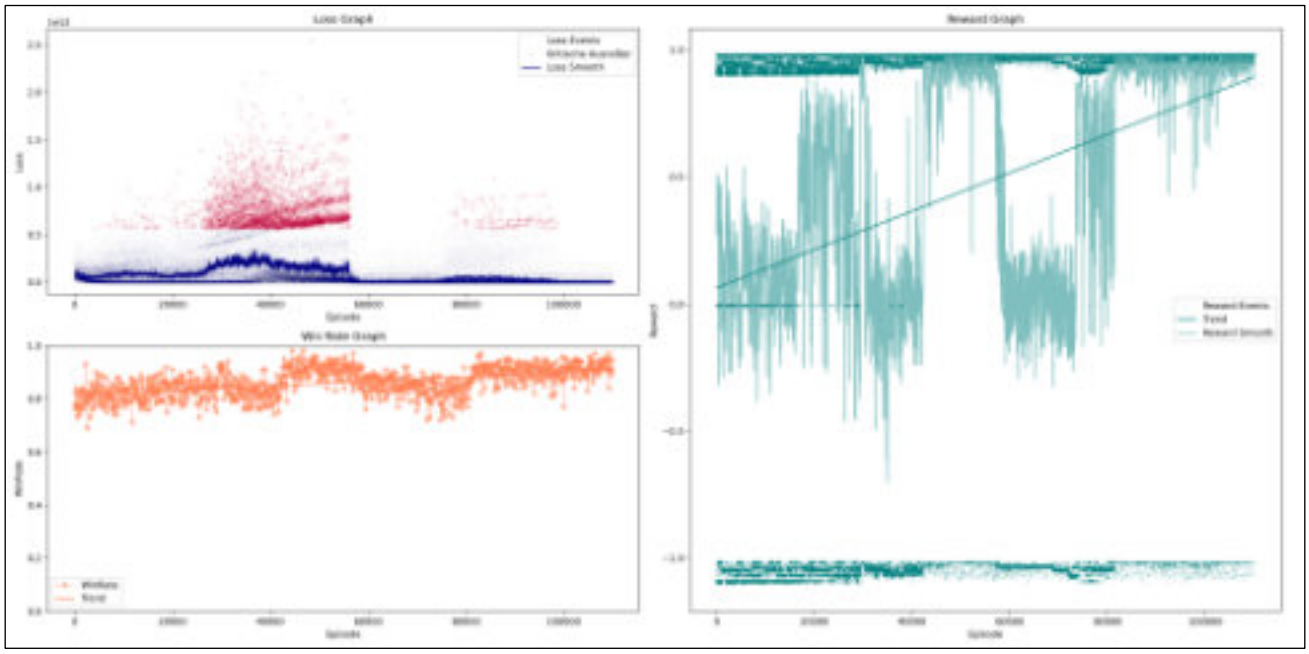
league_play_model_7



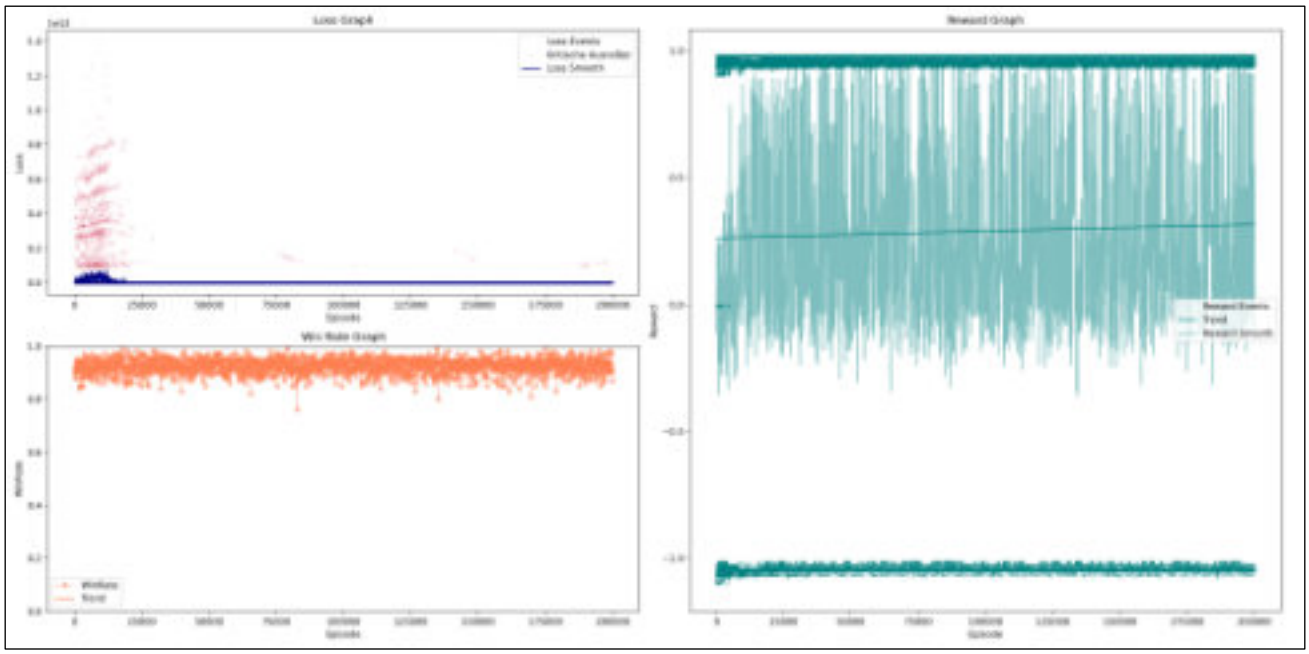
league_play_model_8



league_play_model_12



league_play_model_20

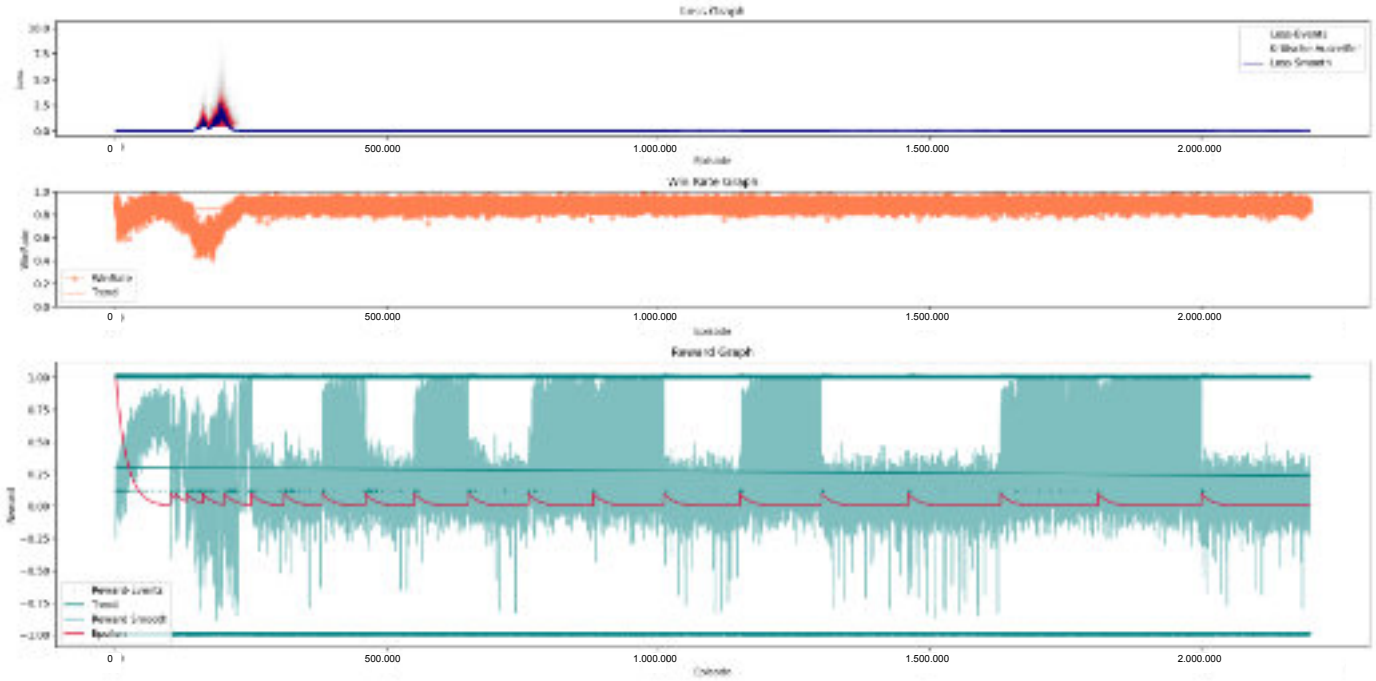


Trainer 6

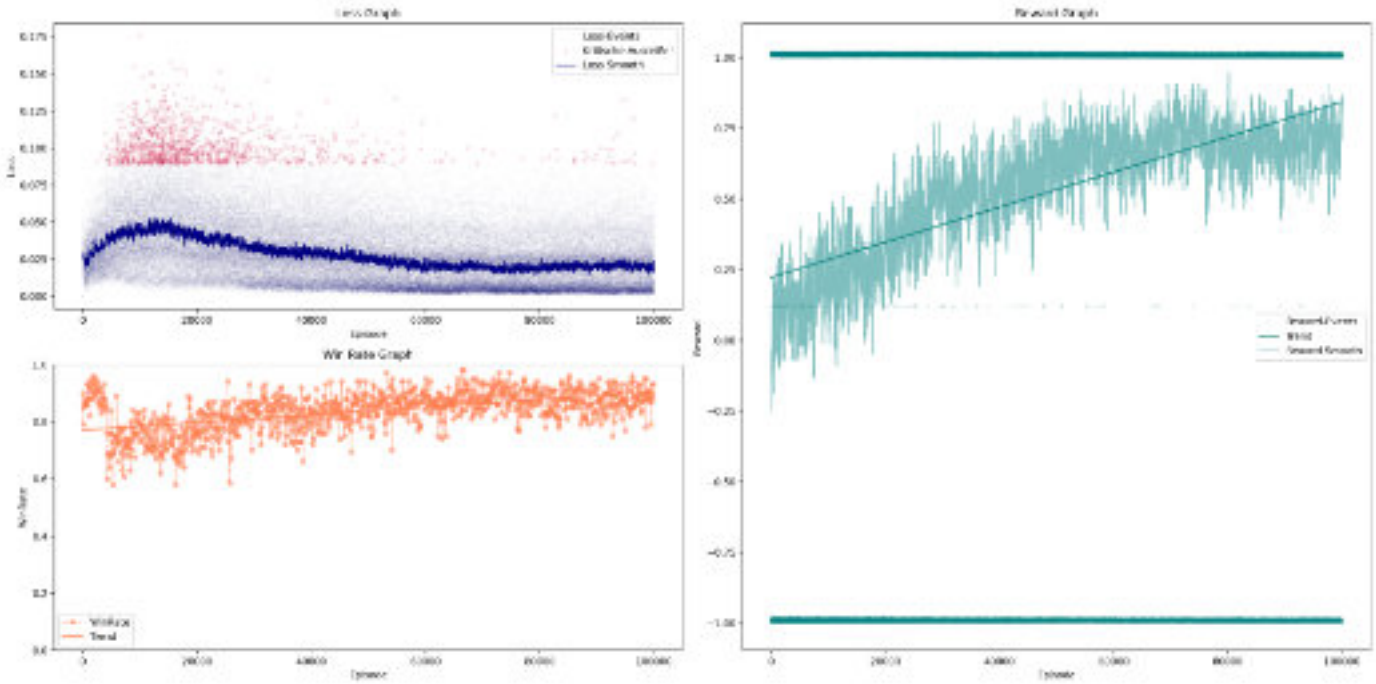
Hyperparameter: hidden_size = 256, gamma = 0,99, batch_size = 64, survive_reward = 0,001

Erwartung: Durch den höheren Gamma-Wert in Kombination mit dem positiven survive_reward und der hohen Netzwerkgröße, sollte der Agent am langfristigsten handeln.

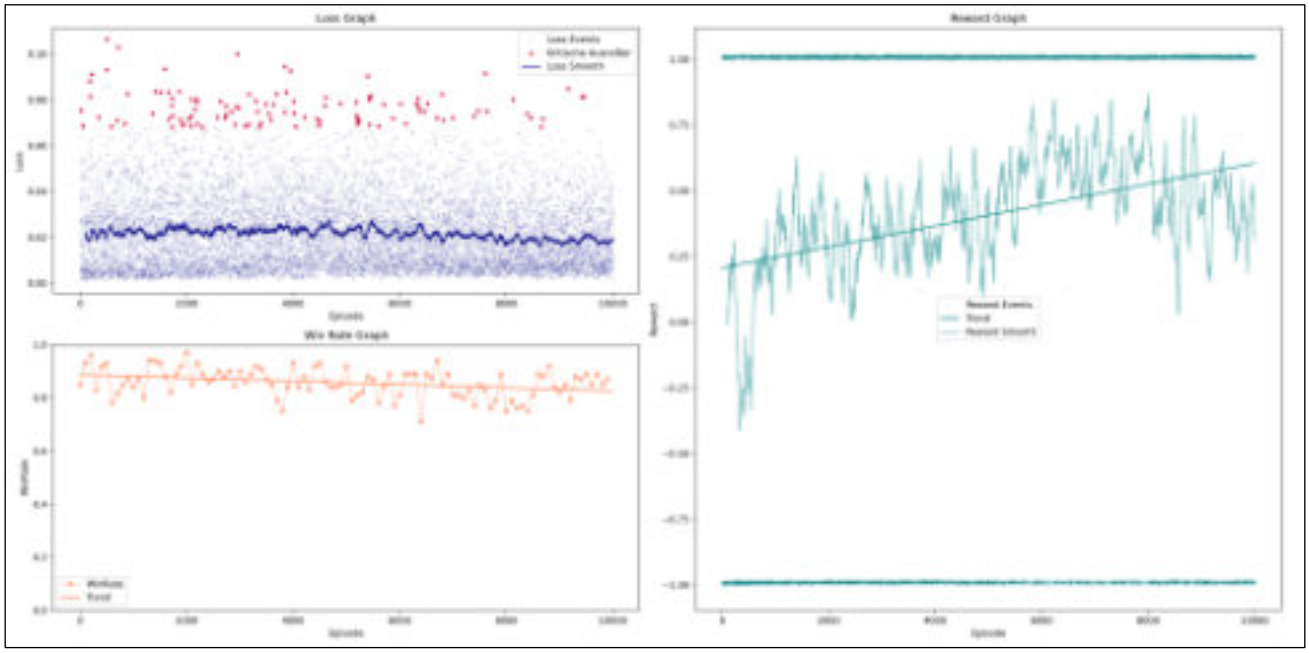
Überblick



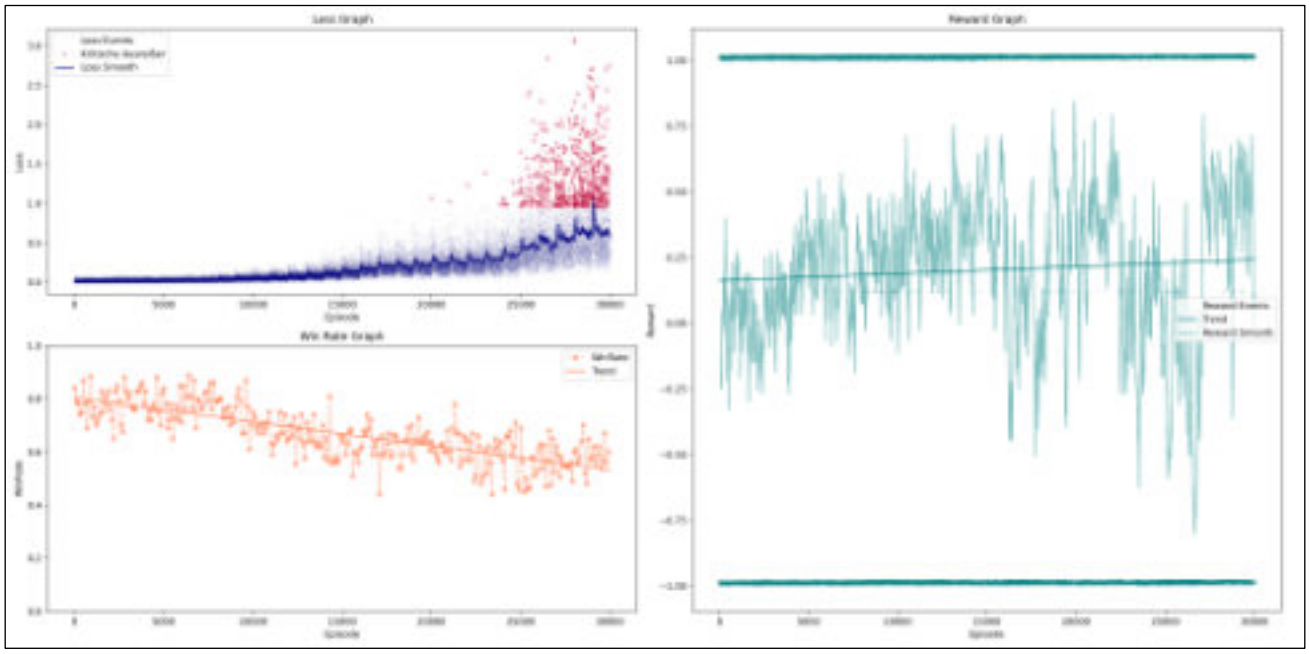
base_model



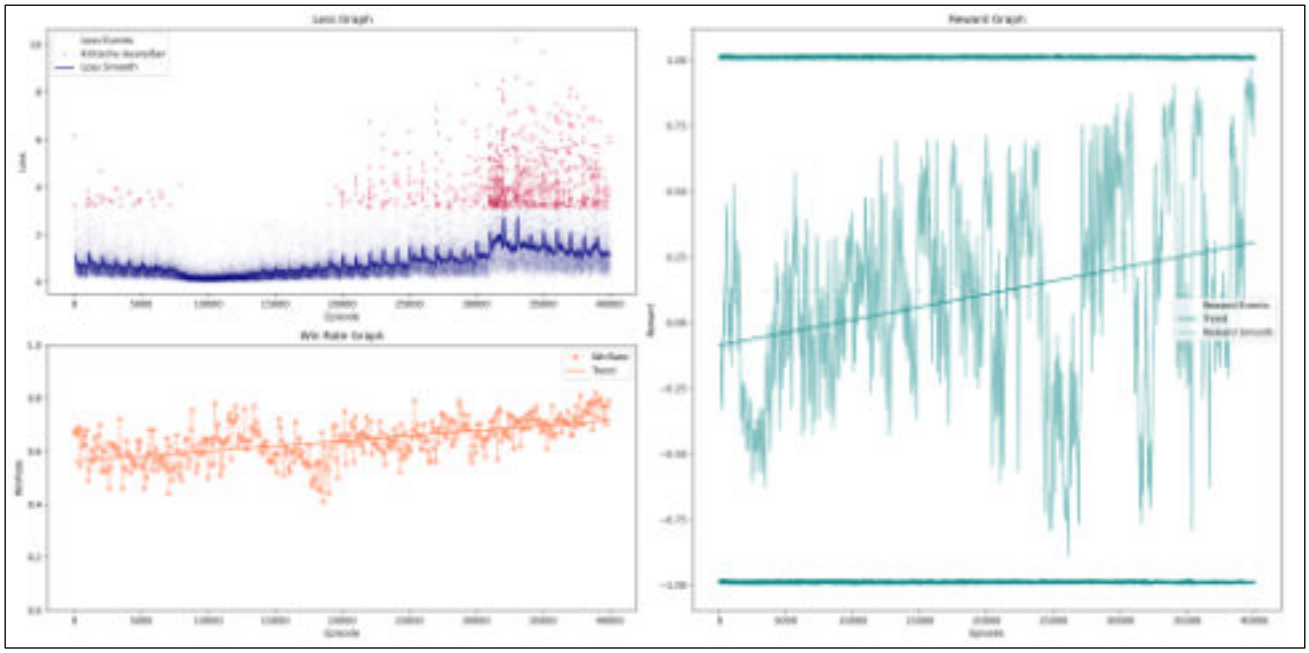
league_play_model_1



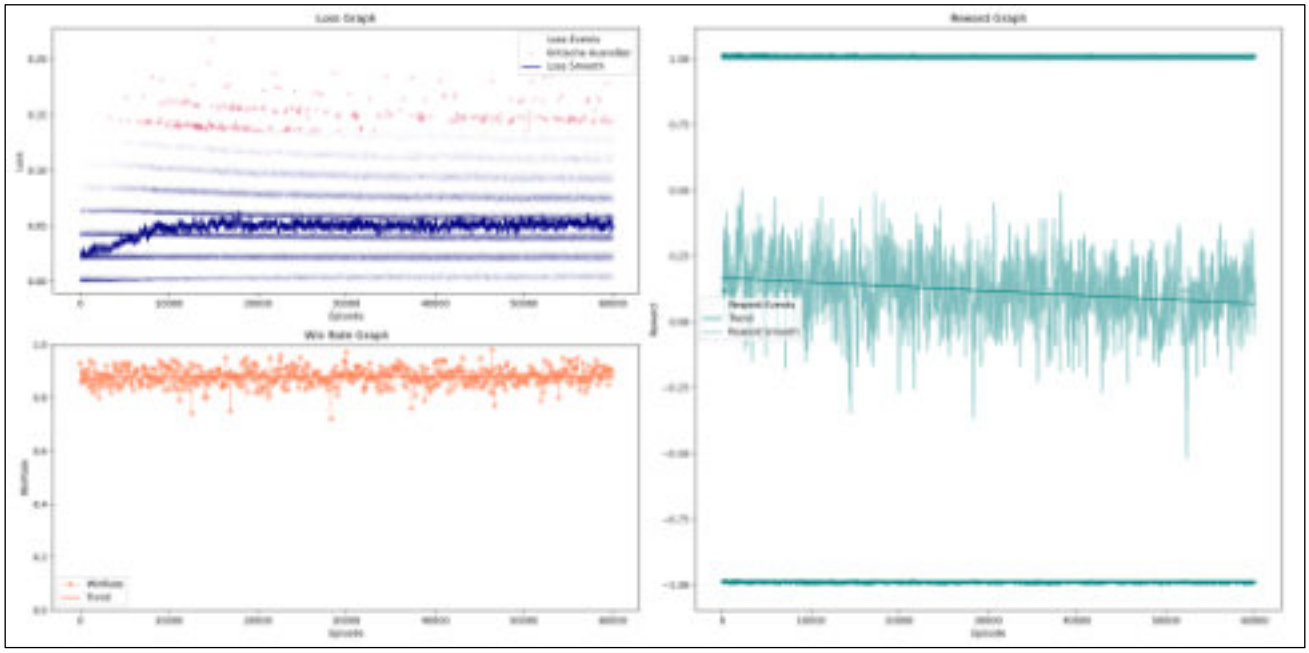
league_play_model_3



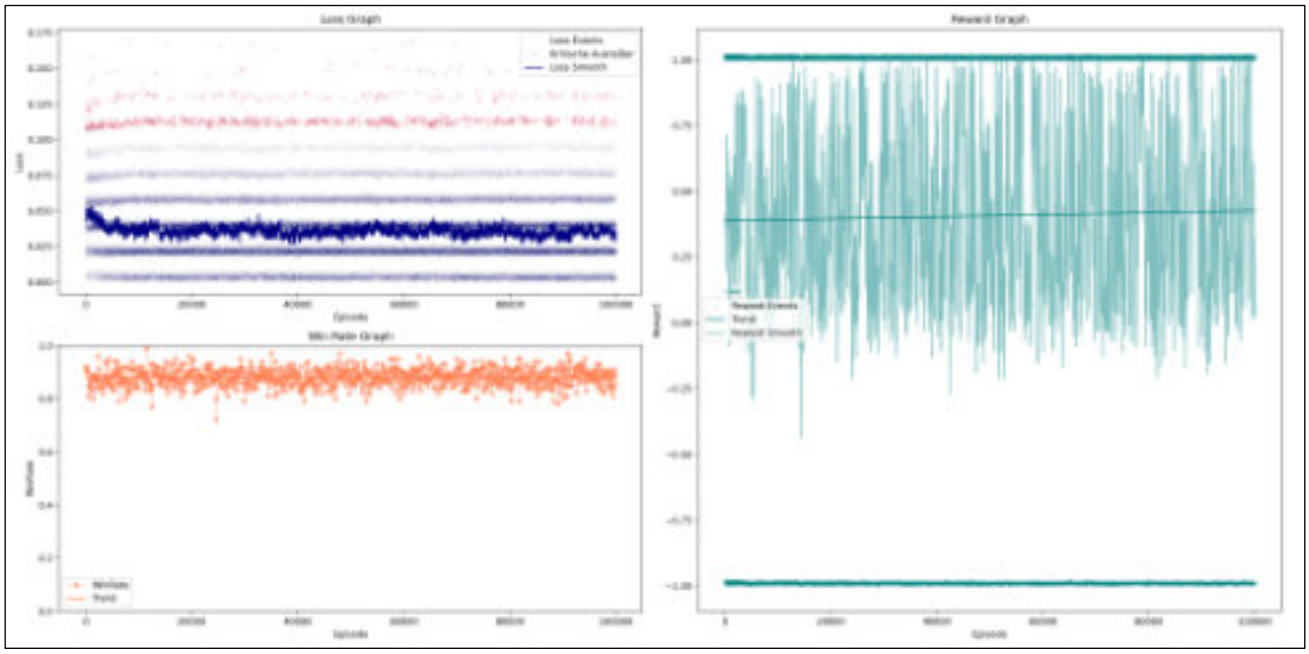
league_play_model_4



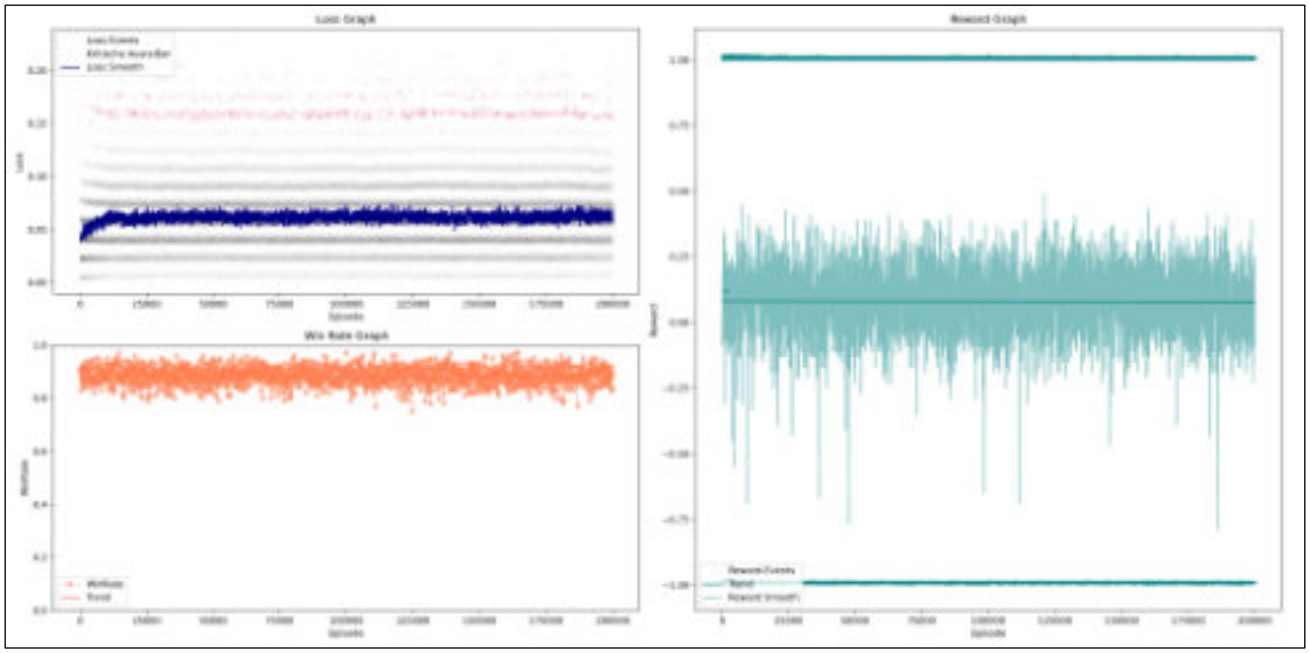
league_play_model_5



league_play_model_10



league_play_model_20

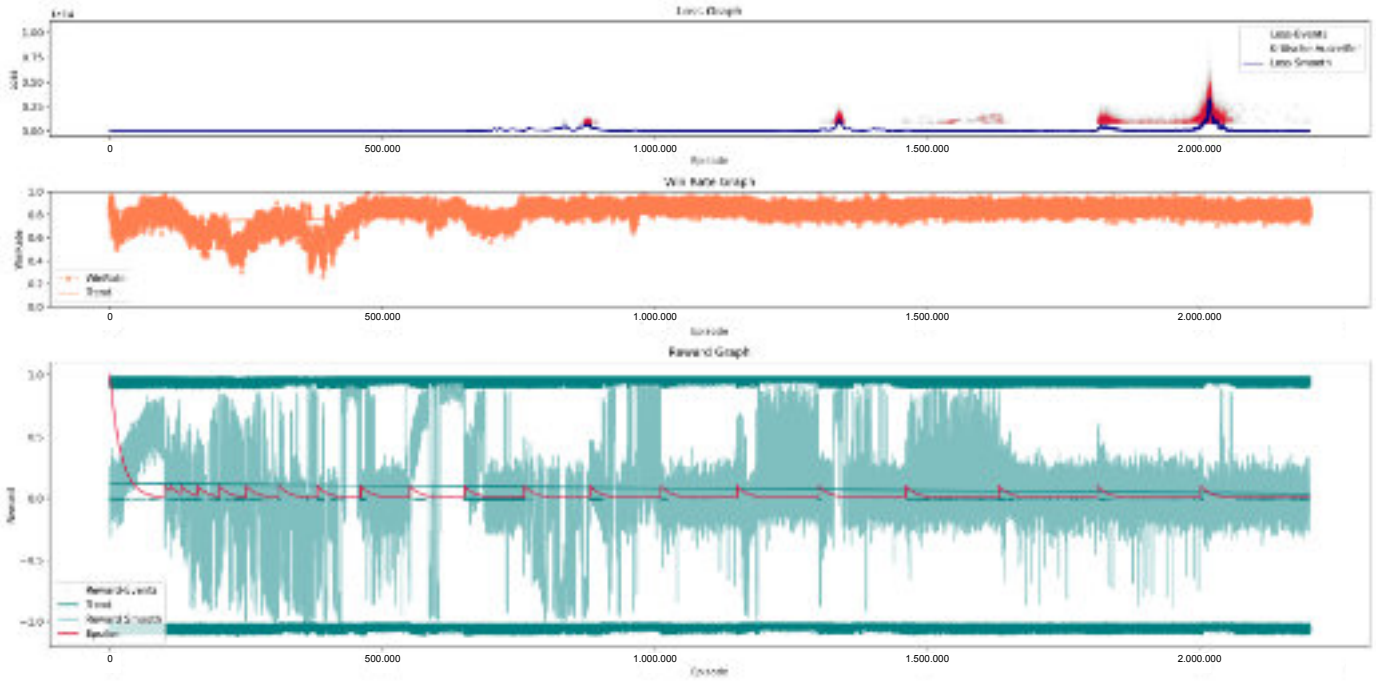


Trainer 7

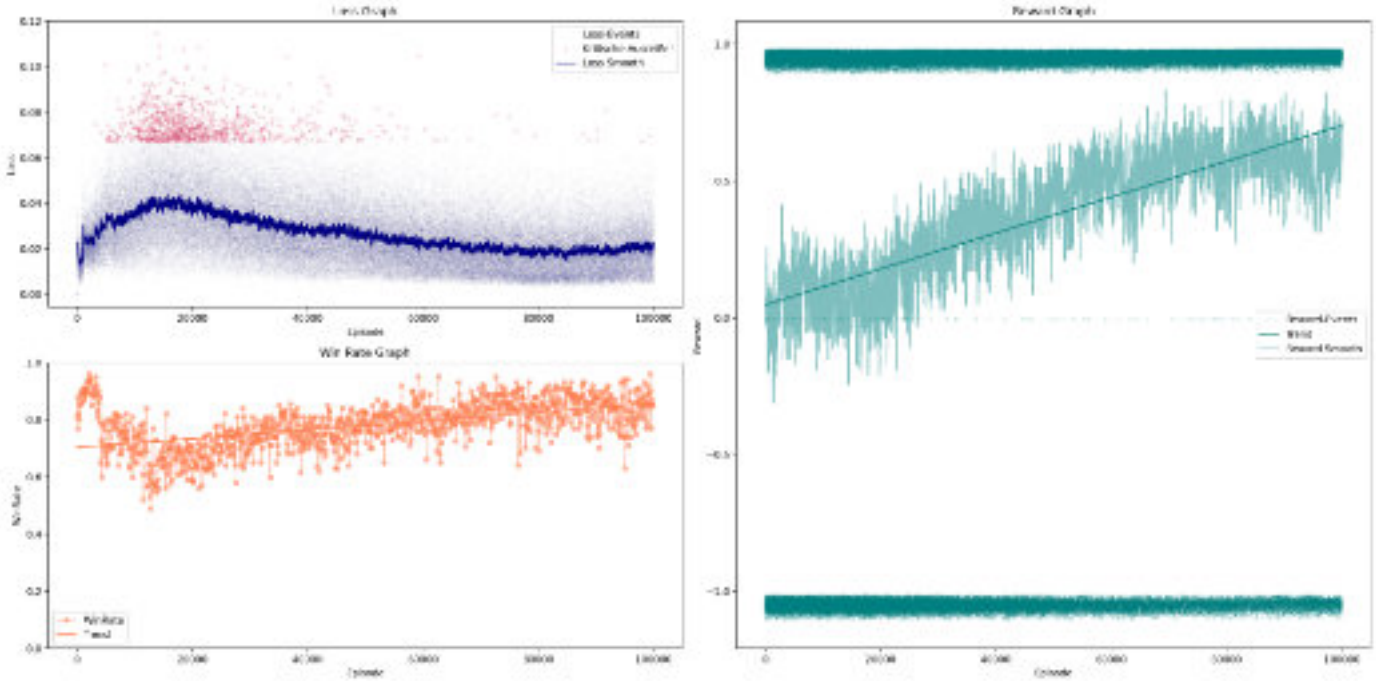
Hyperparameter: hidden_size = 256, gamma = 0,99, batch_size = 128, survive_reward = -0,005

Erwartung: Der Agent ähnelt Trainer 5, nur mit höherer batch_size.

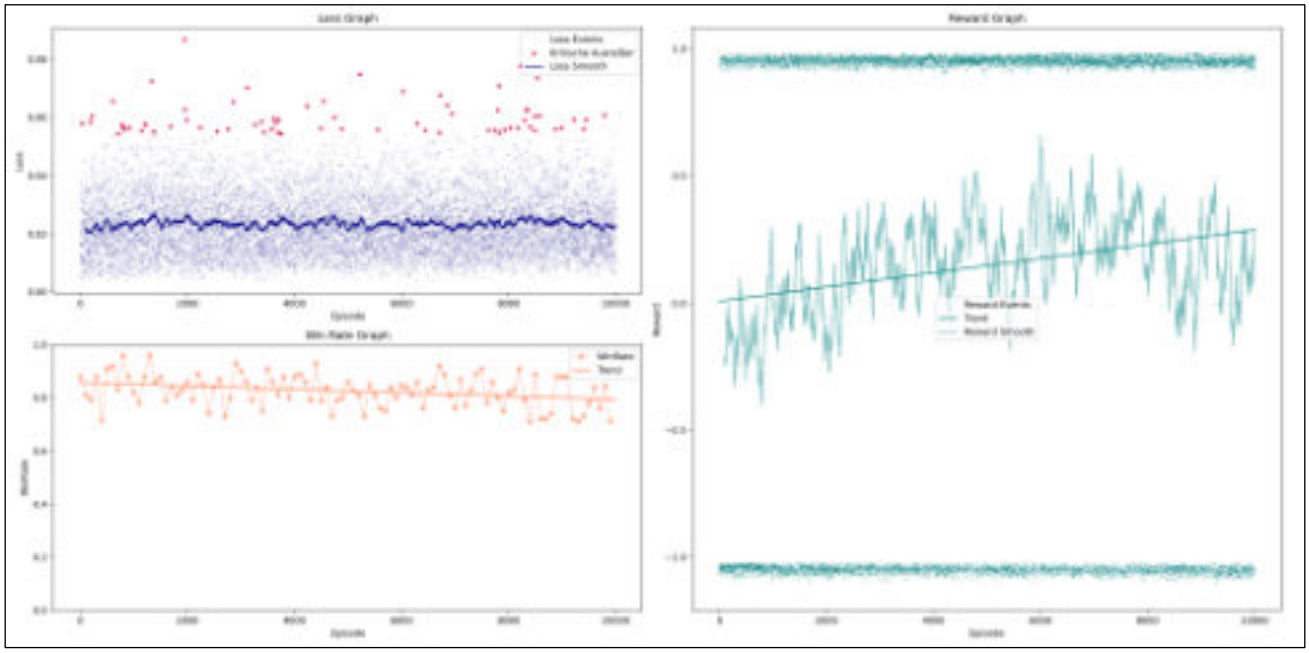
Überblick



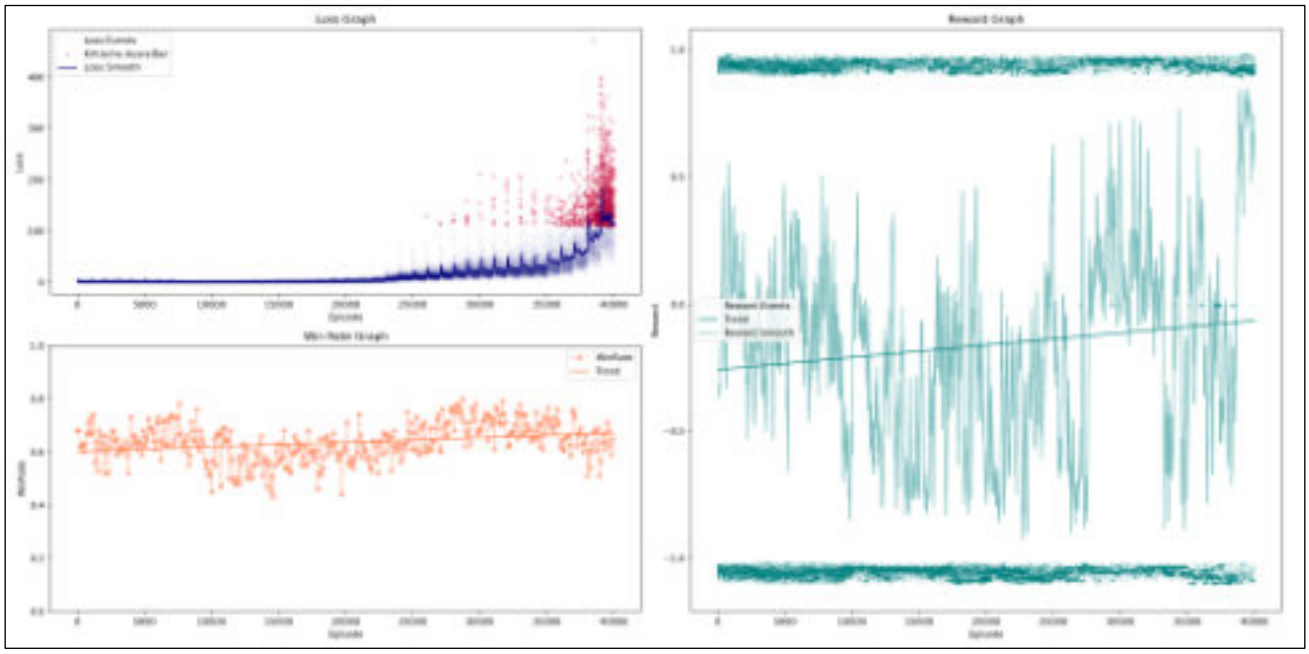
base_model



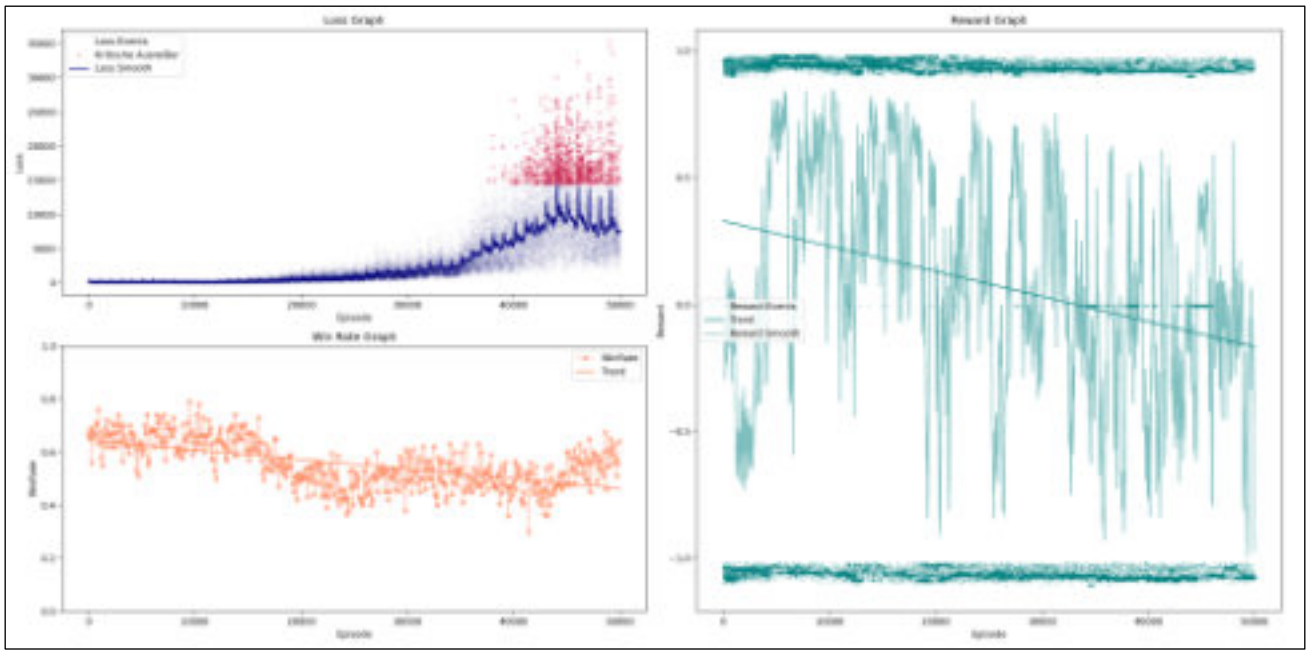
league_play_model_1



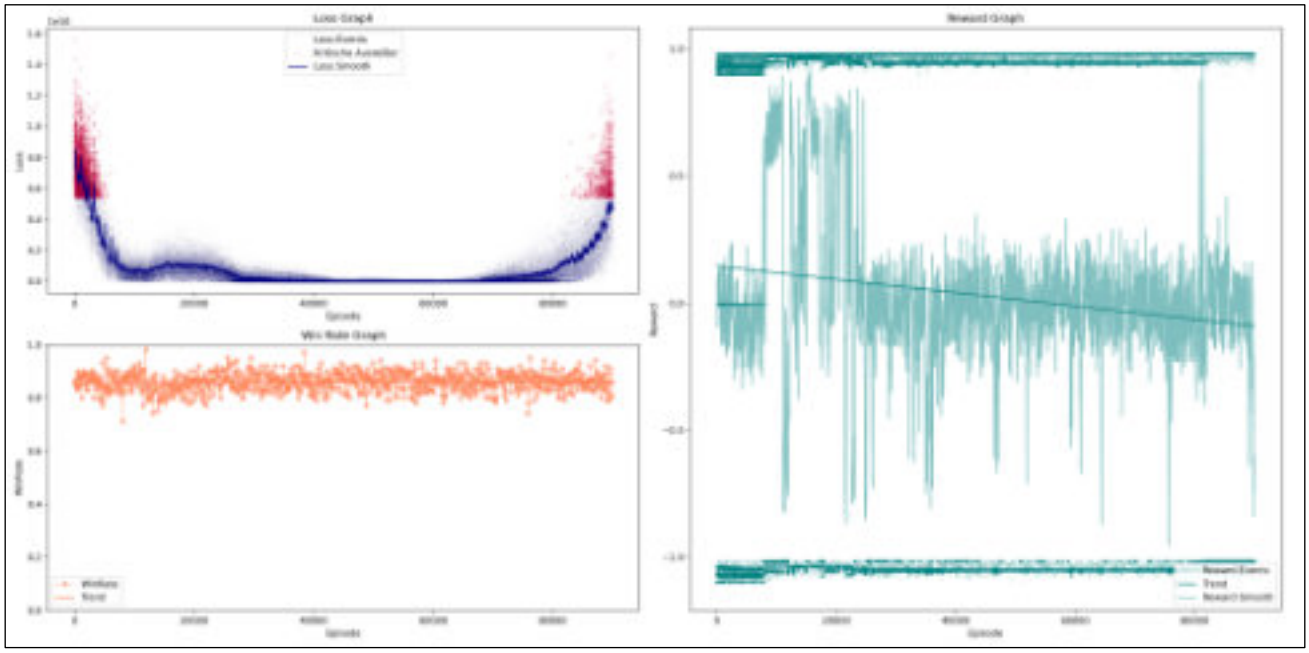
league_play_model_4



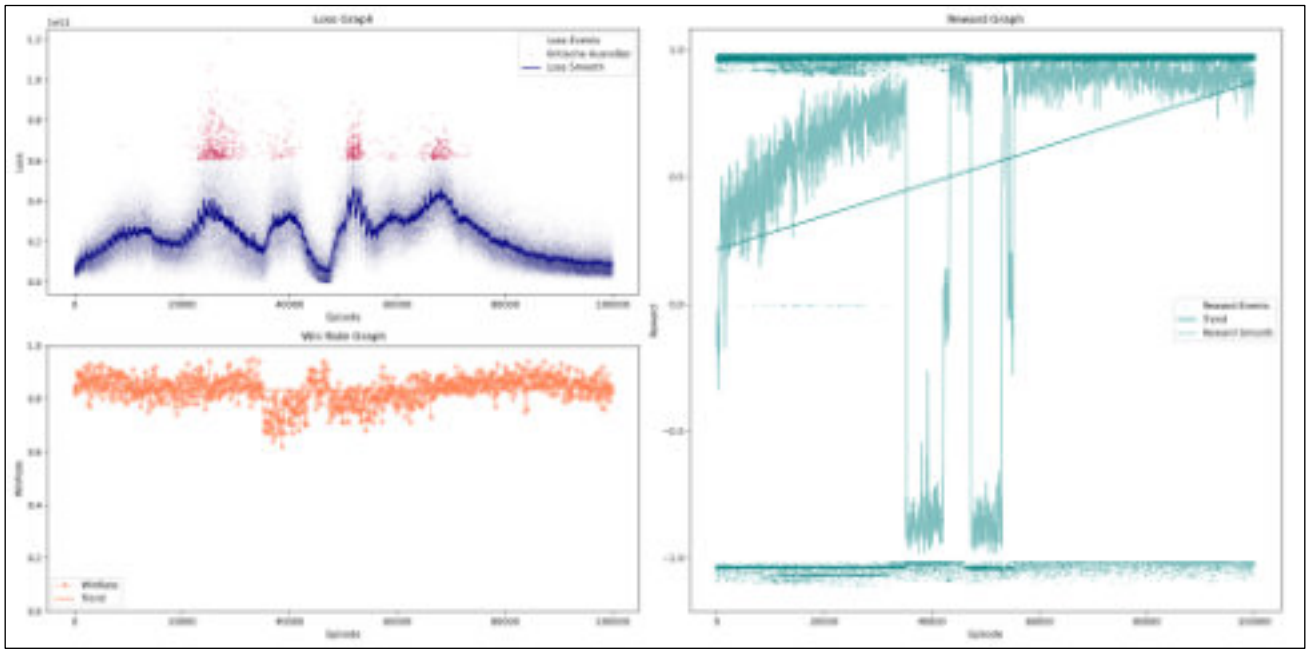
league_play_model_5



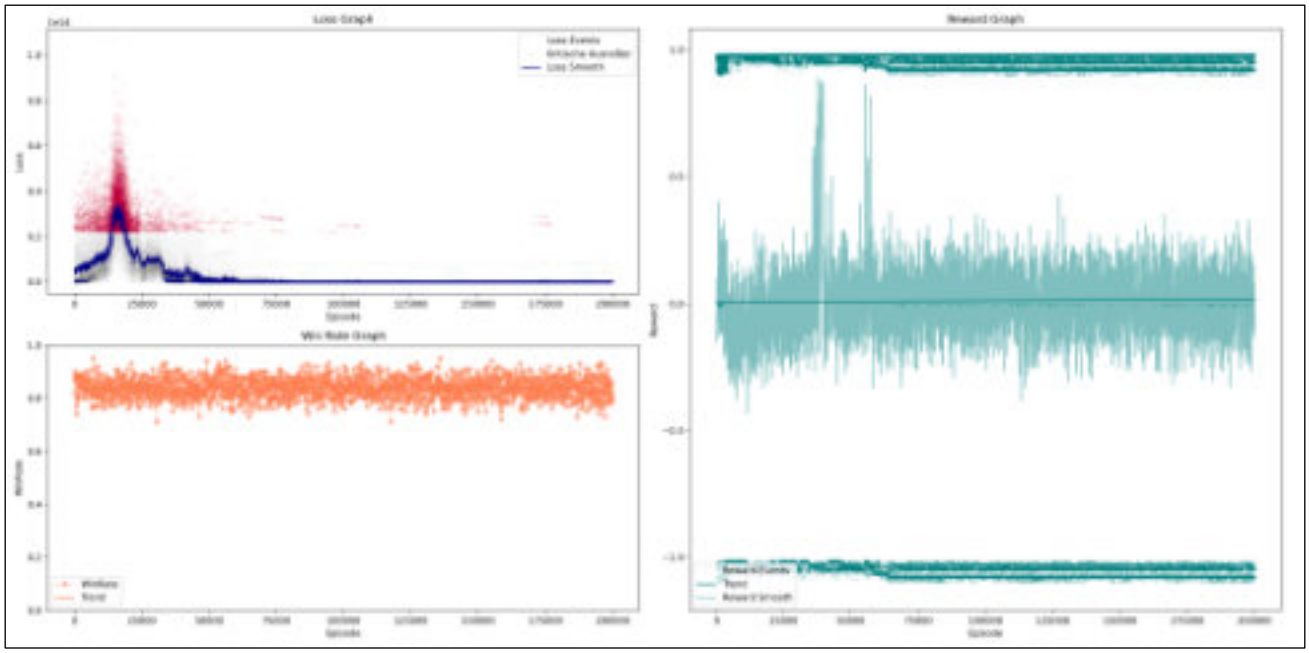
league_play_model_9



league_play_model_10



league_play_model_20

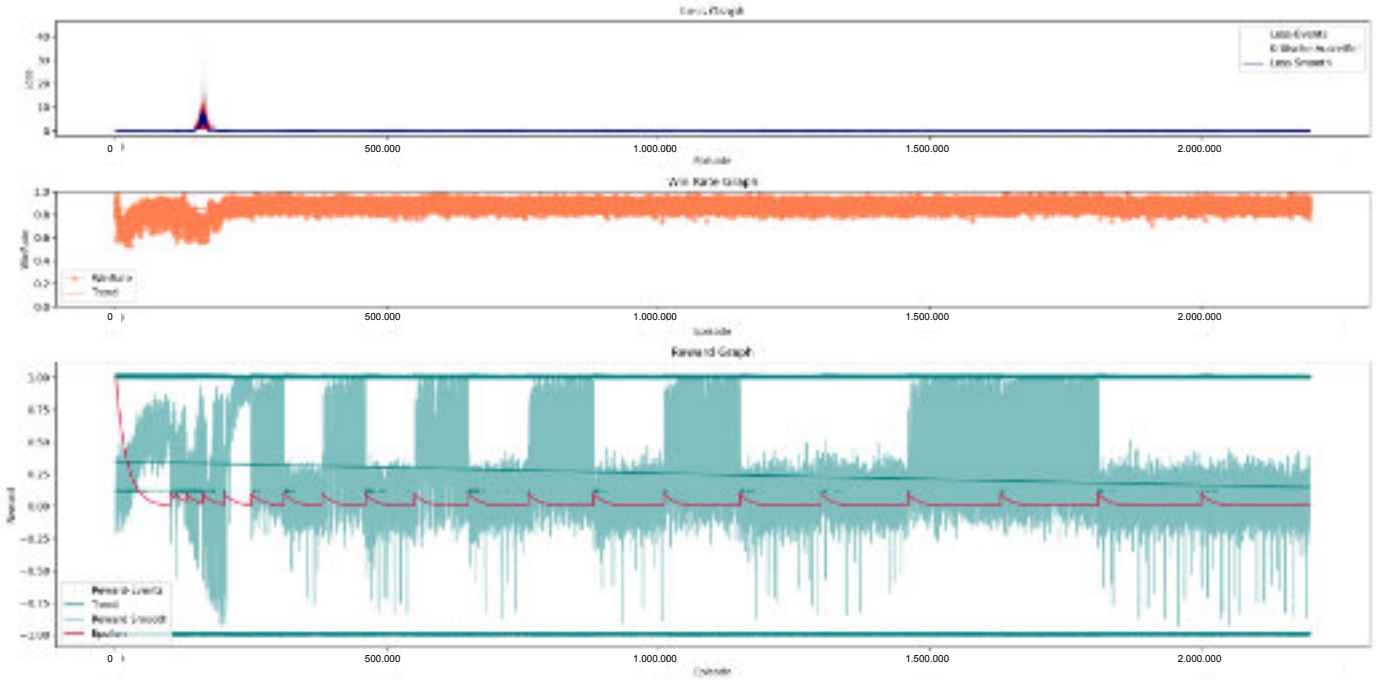


Trainer 8

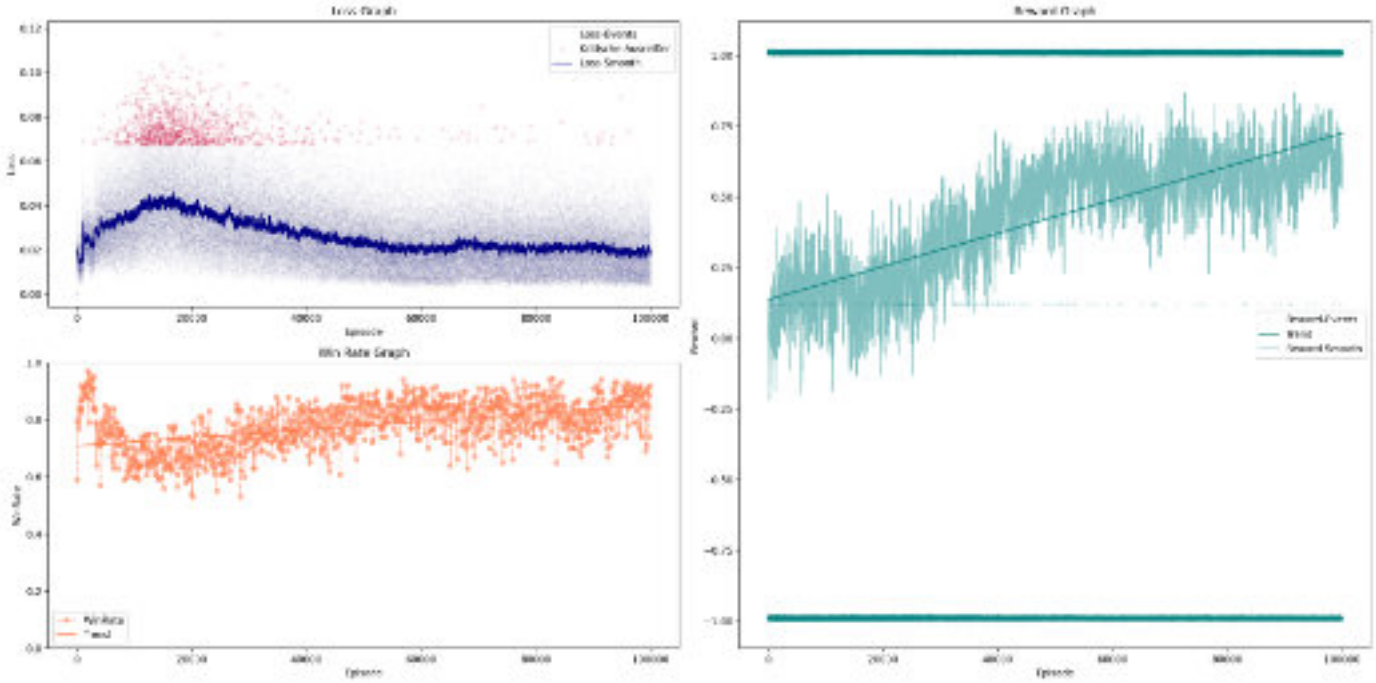
Hyperparameter: hidden_size = 256, gamma = 0,99, batch_size = 128, survive_reward = 0,001

Erwartung: Der Agent ähnelt Trainer 6, nur mit höherer batch_size. Dadurch könnte er noch langristiger handeln.

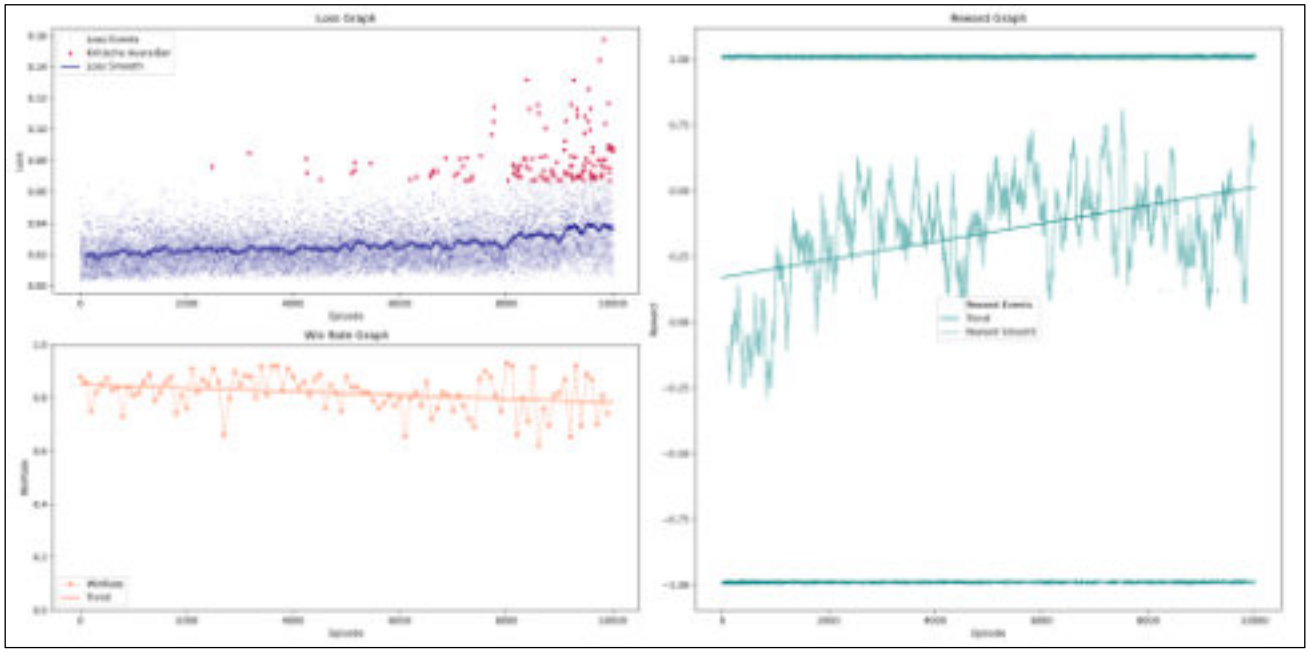
Überblick



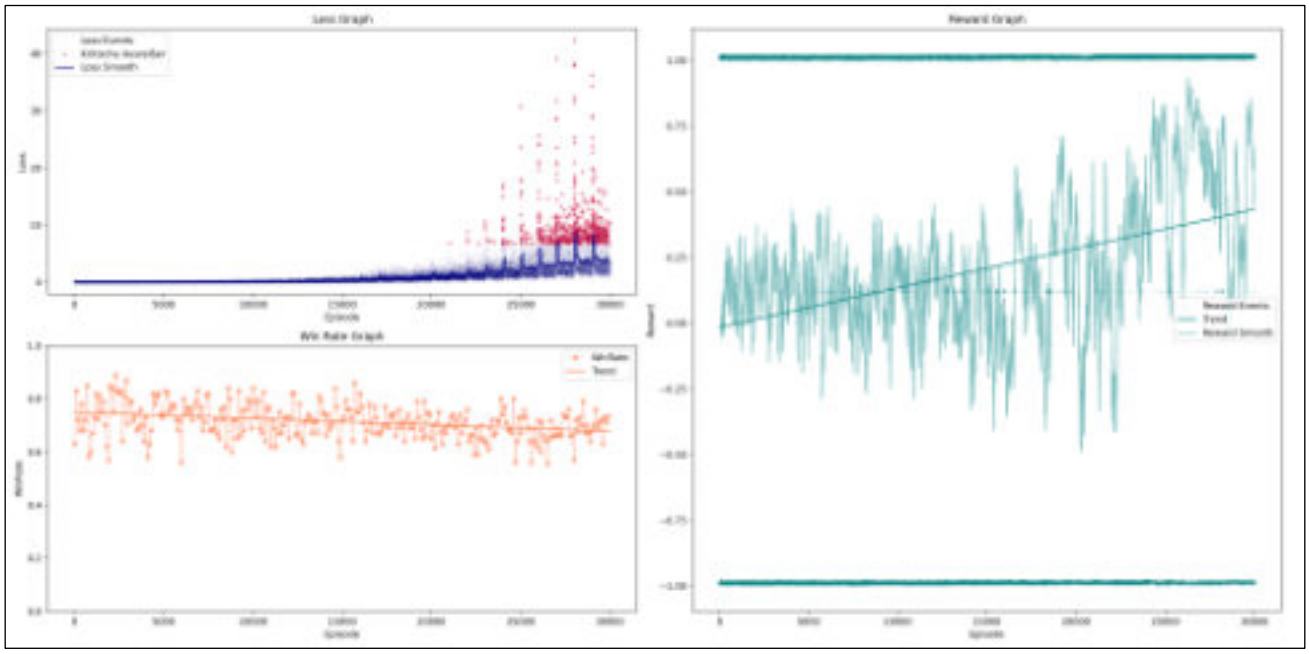
base_model



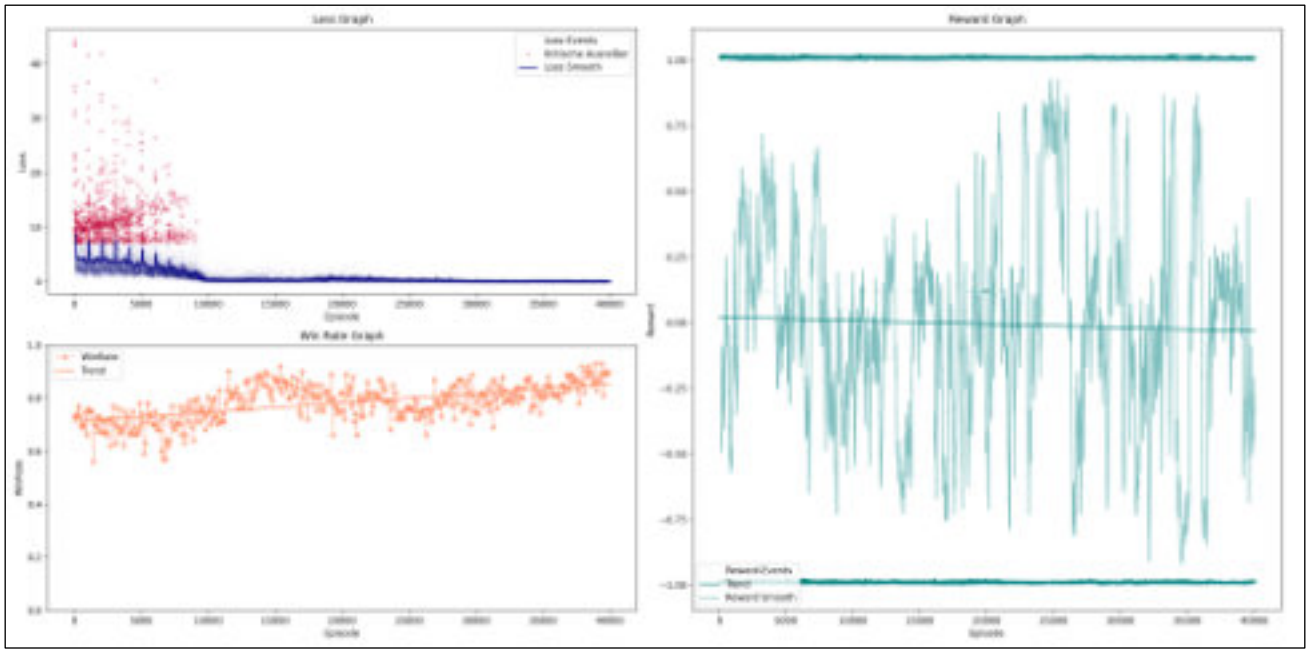
league_play_model_1



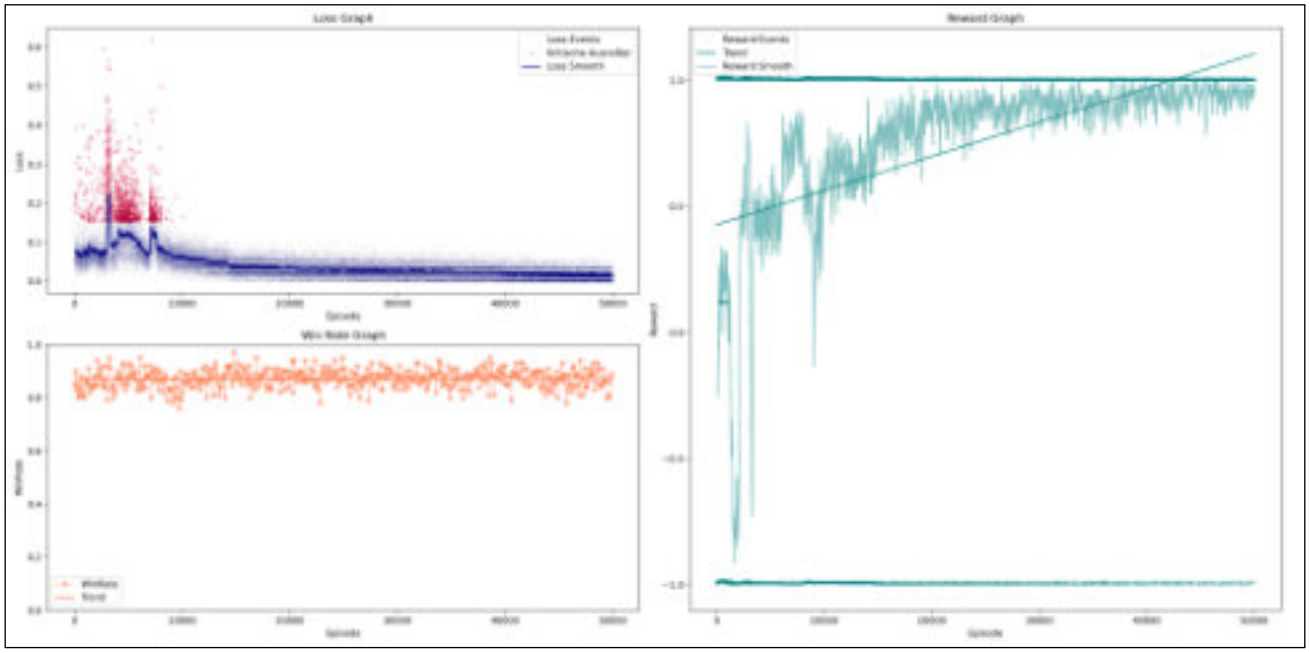
league_play_model_3



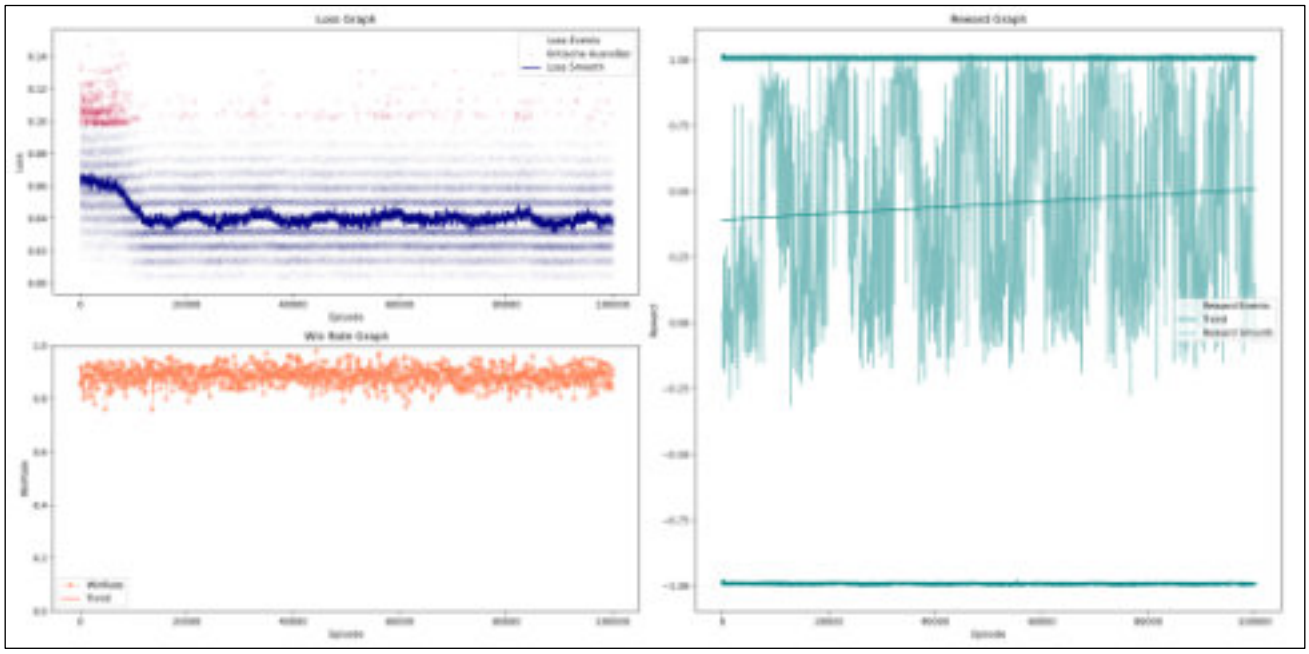
league_play_model_4



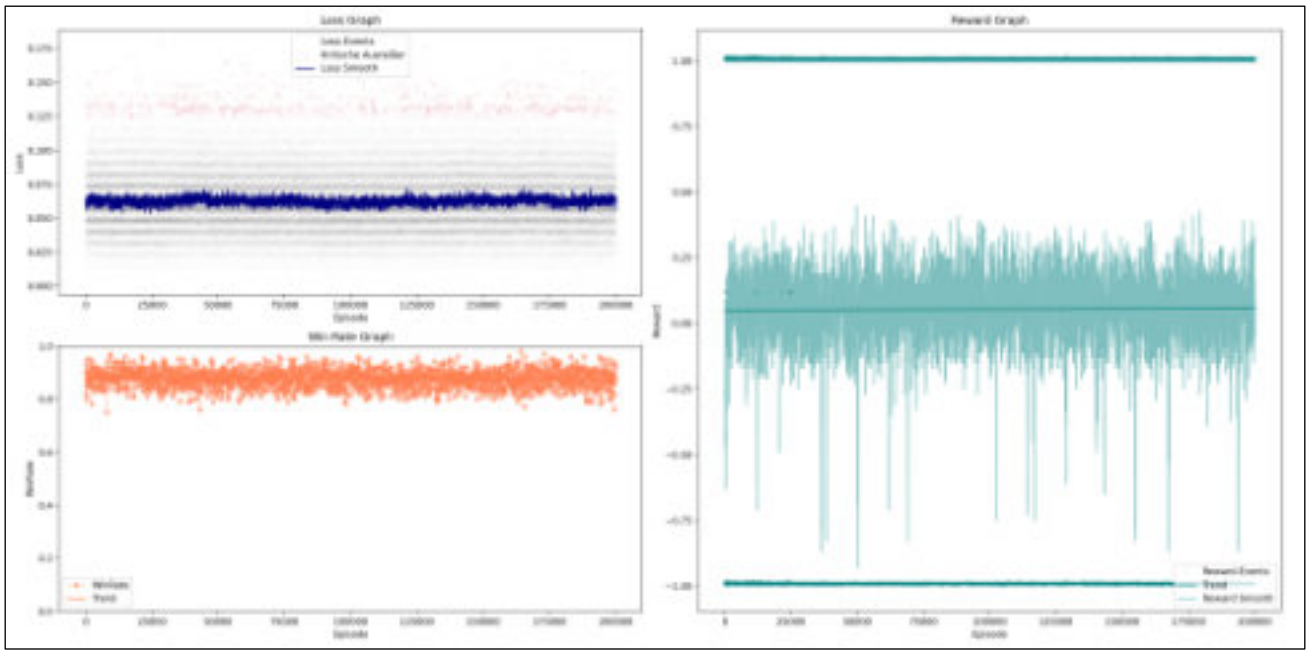
league_play_model_5



league_play_model_10



league_play_model_20

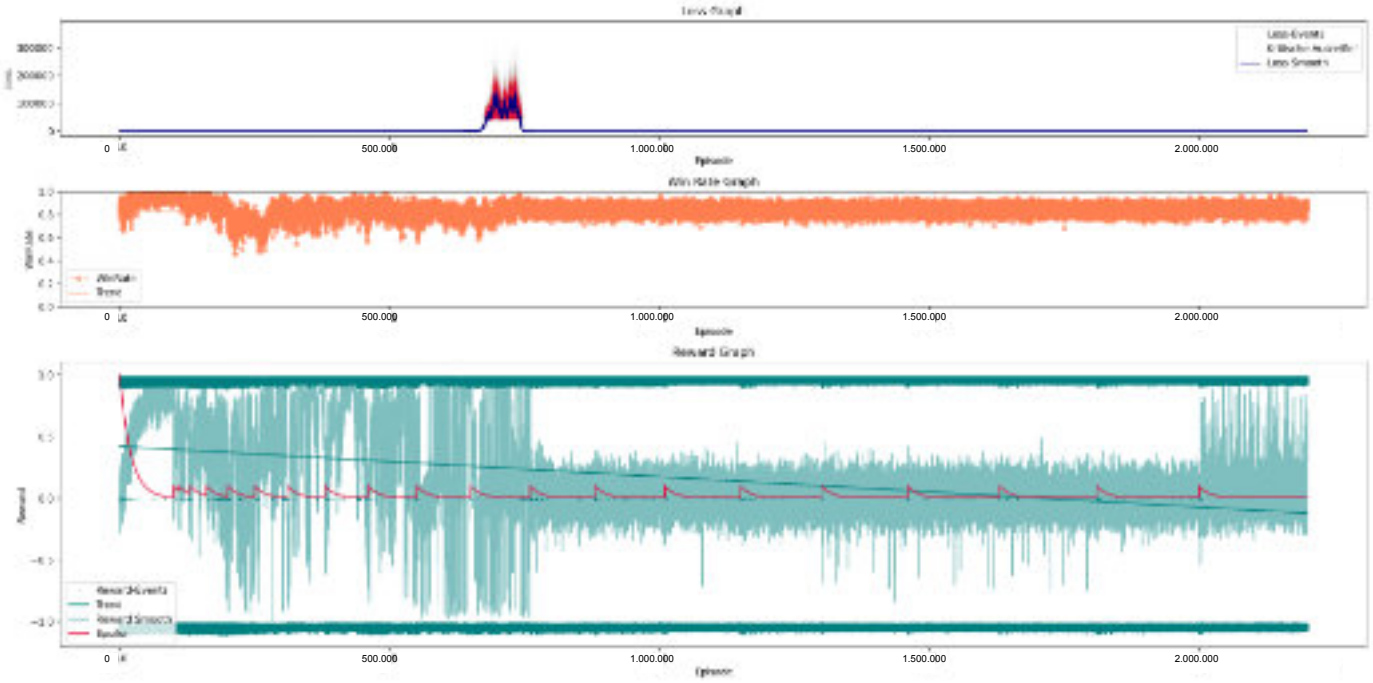


Trainer 9

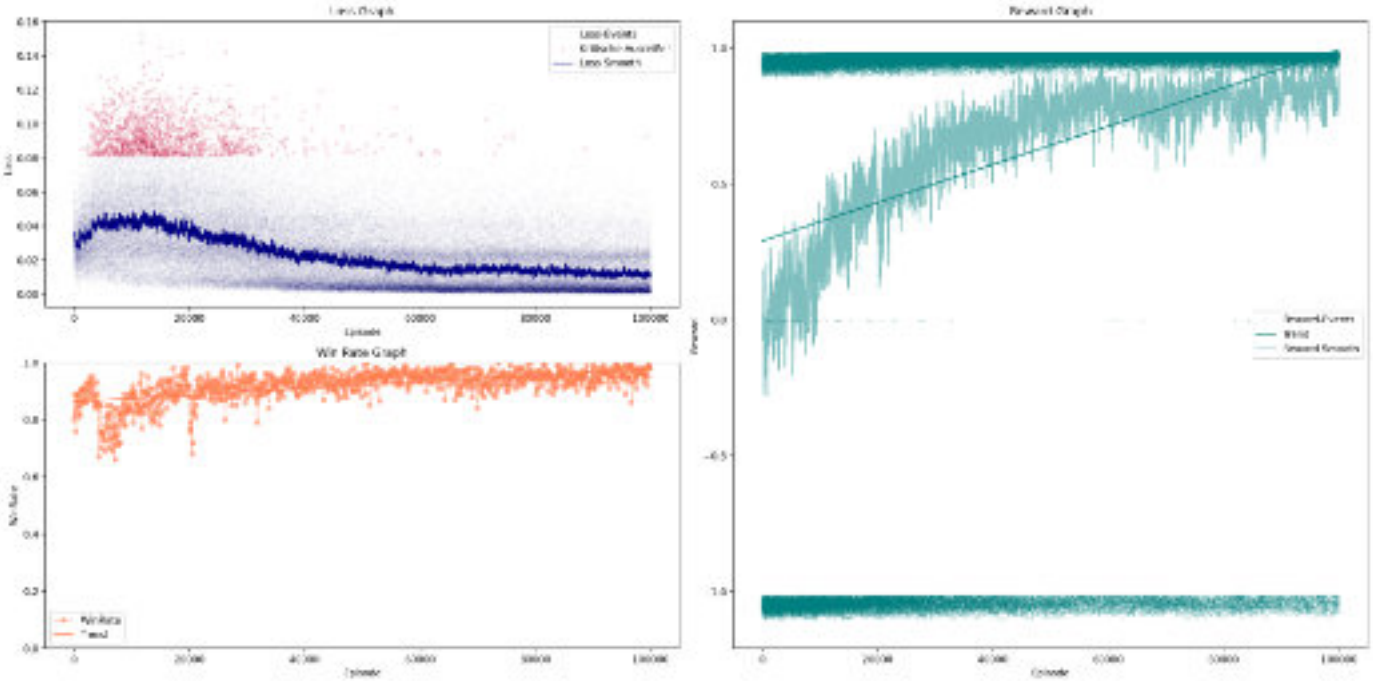
Haperparameter: hidden_size = 128, gamma = 0,95, batch_size = 64, survive_reward = -0,005

Erwartung: Ähnelt Trainer 1 mit verringerter Netzwerkgröße: wahrscheinlich kurzfristigeres Handeln.

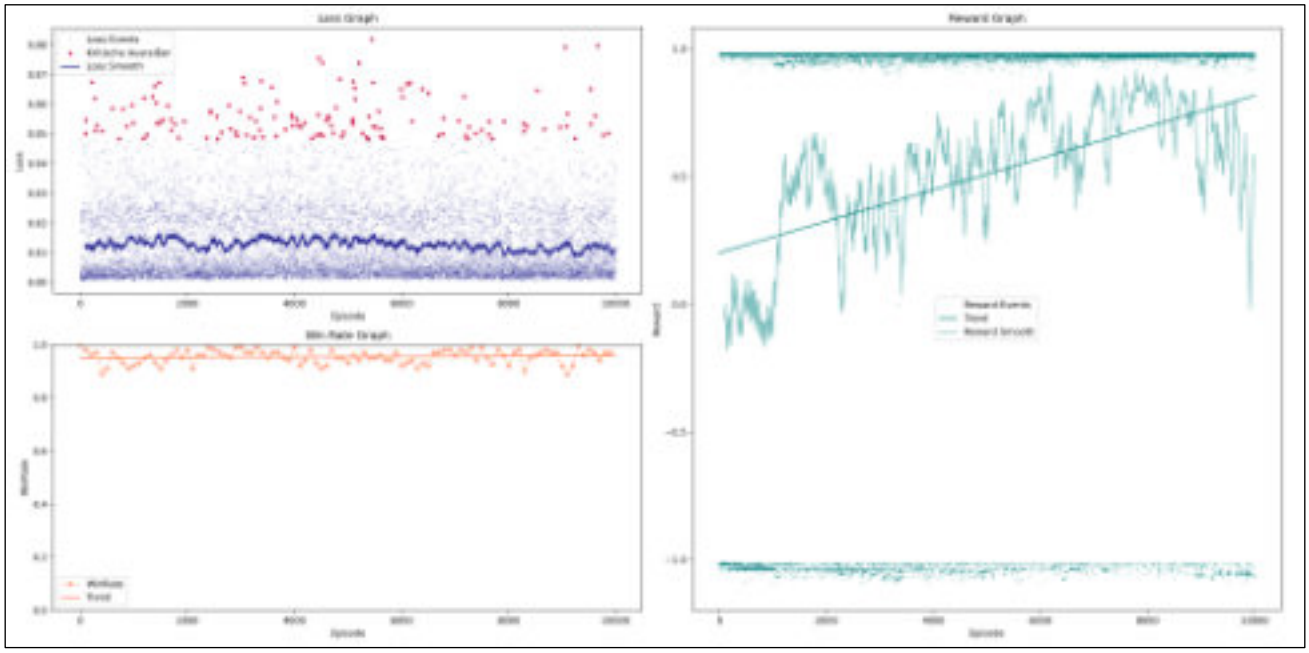
Überblick



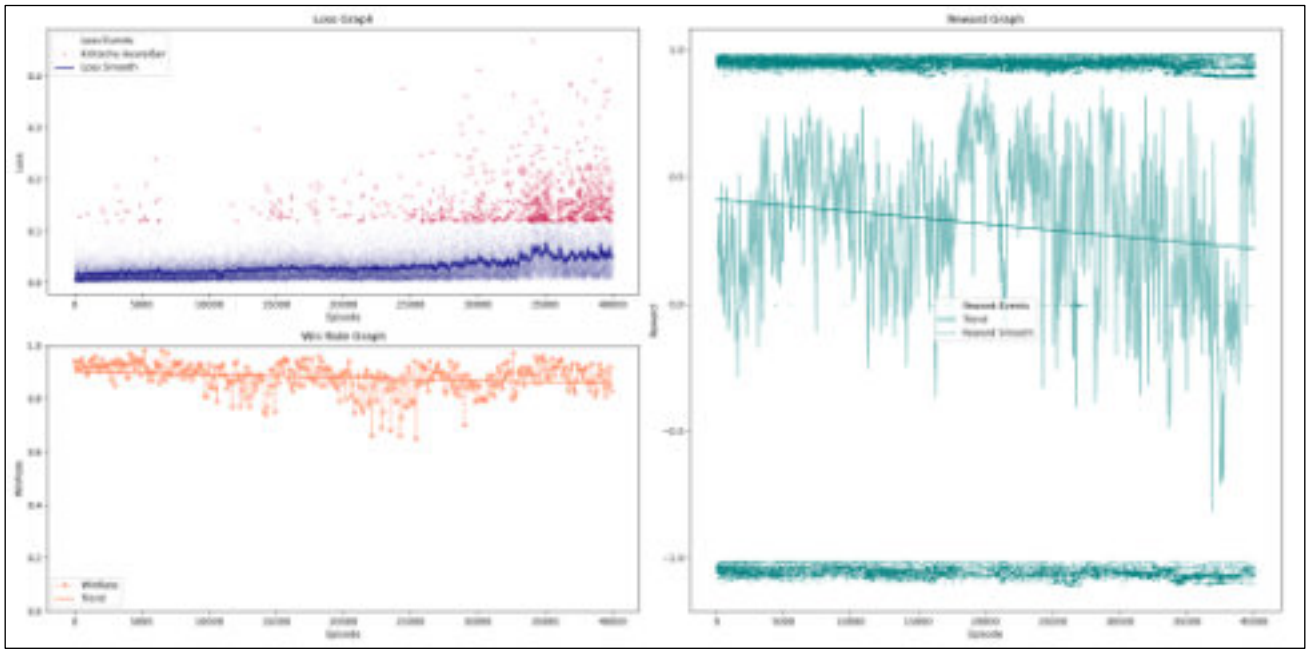
base_model



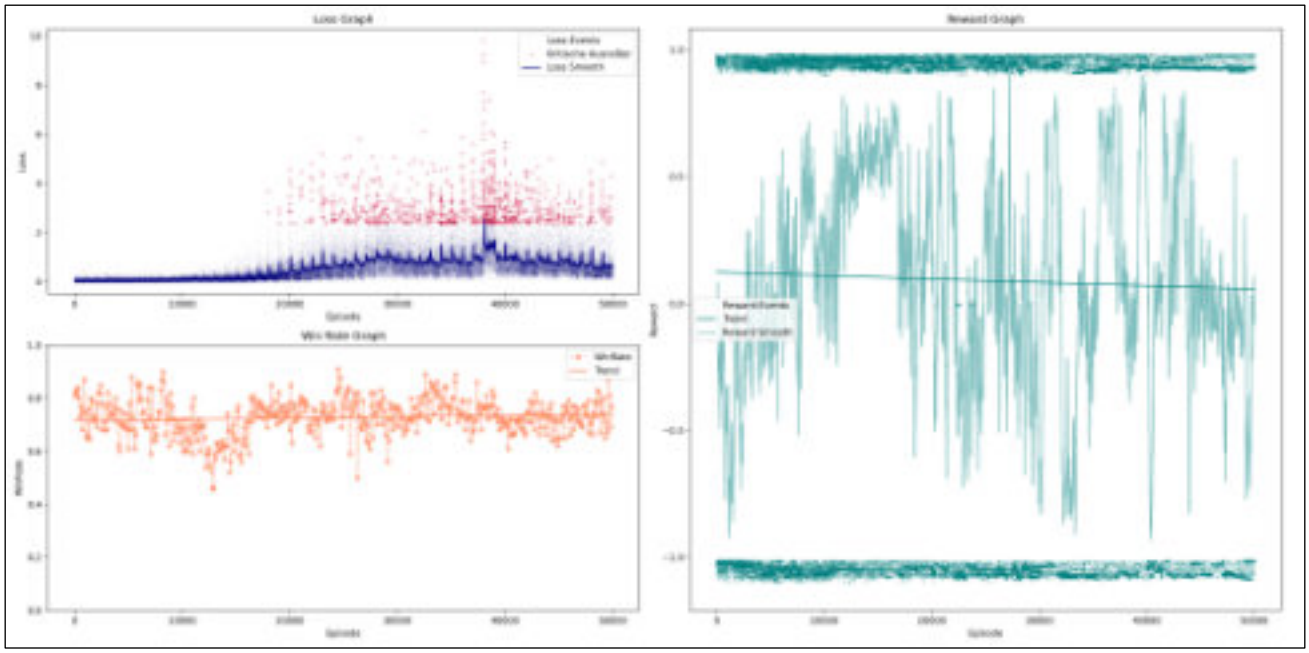
league_play_model_1



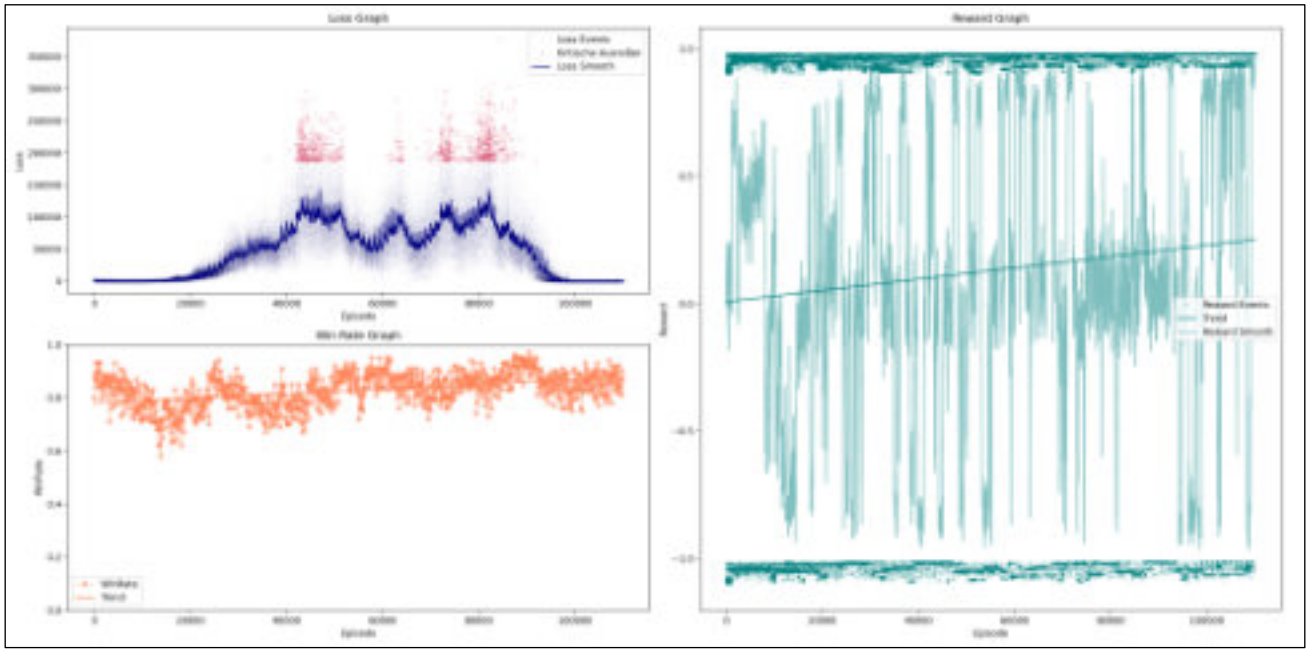
league_play_model_4



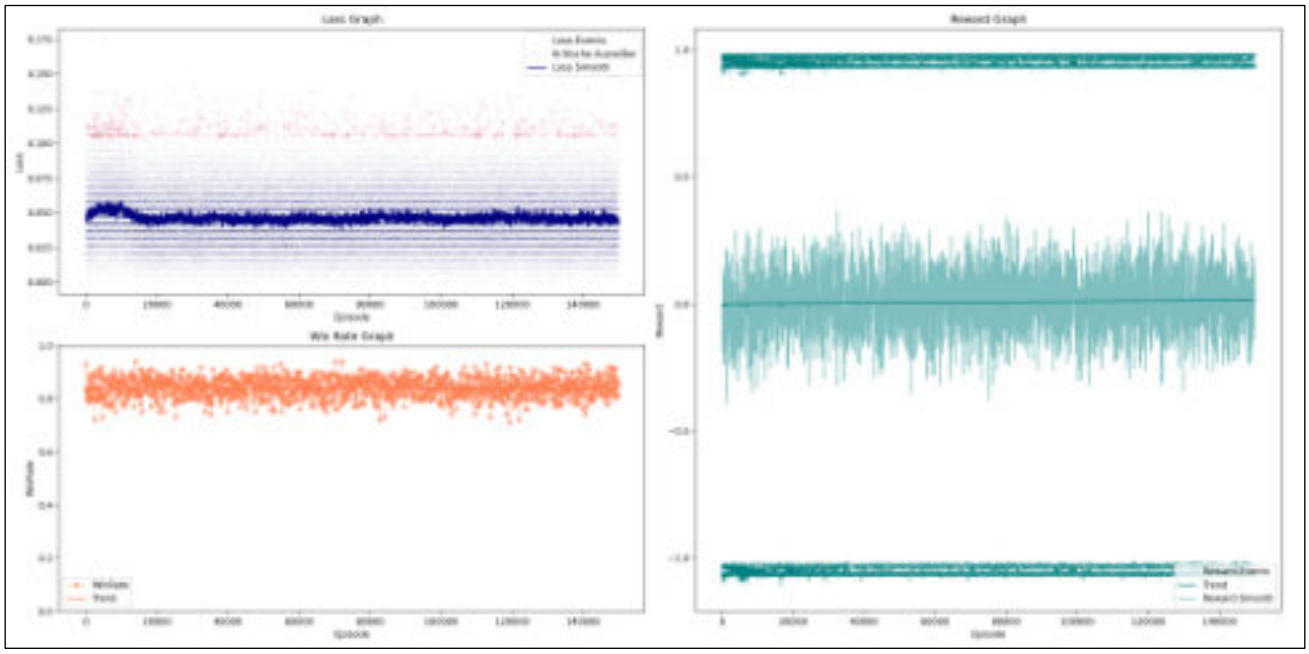
league_play_model_5



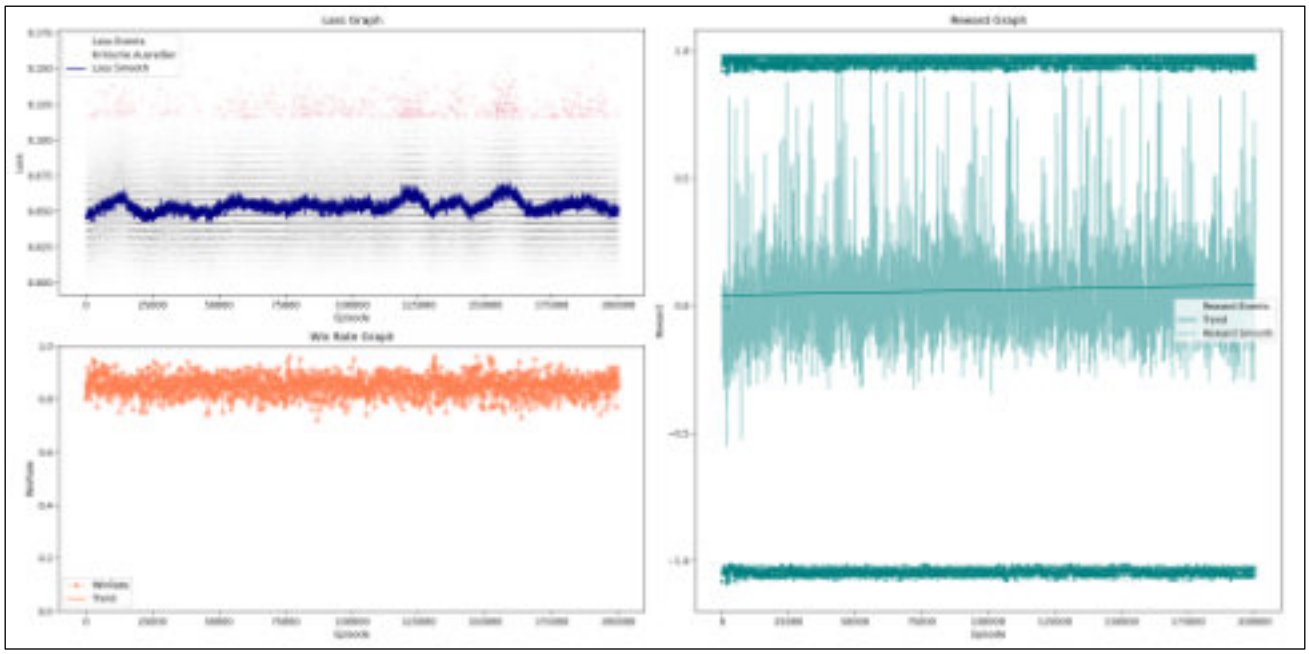
league_play_model_11



league_play_model_15



league_play_model_20

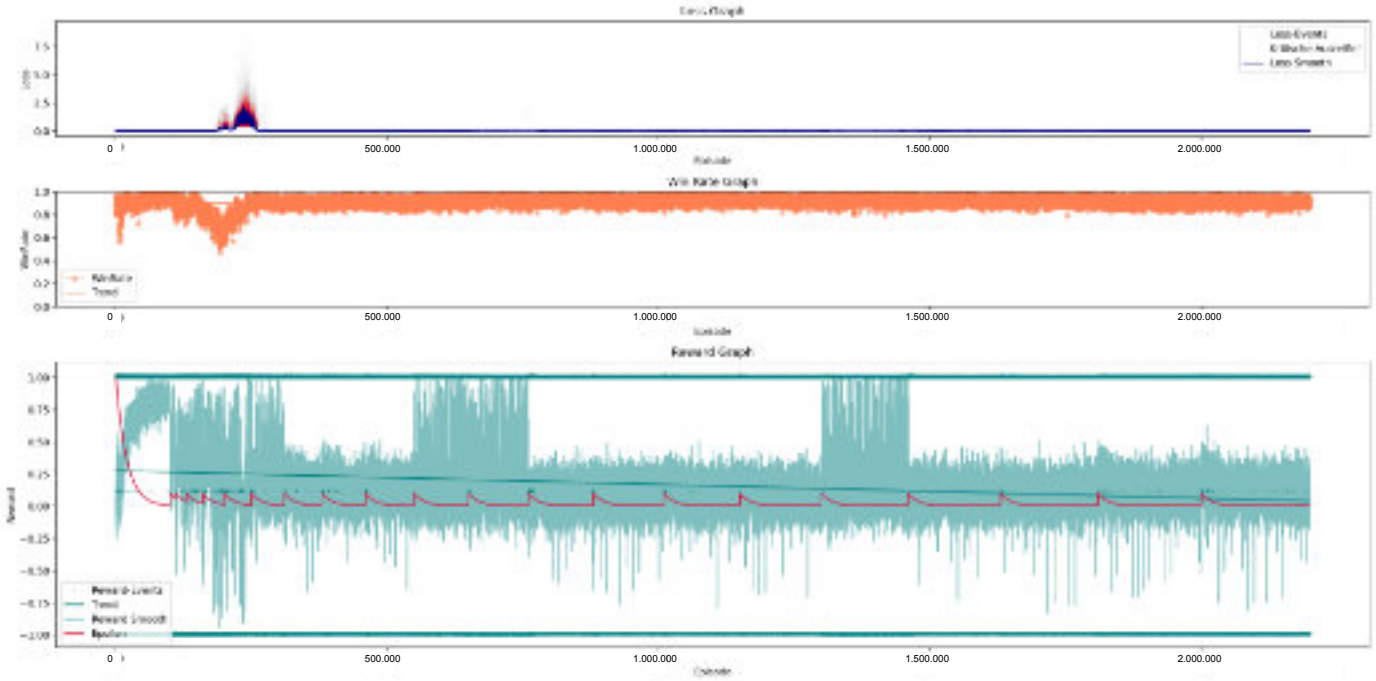


Trainer 10

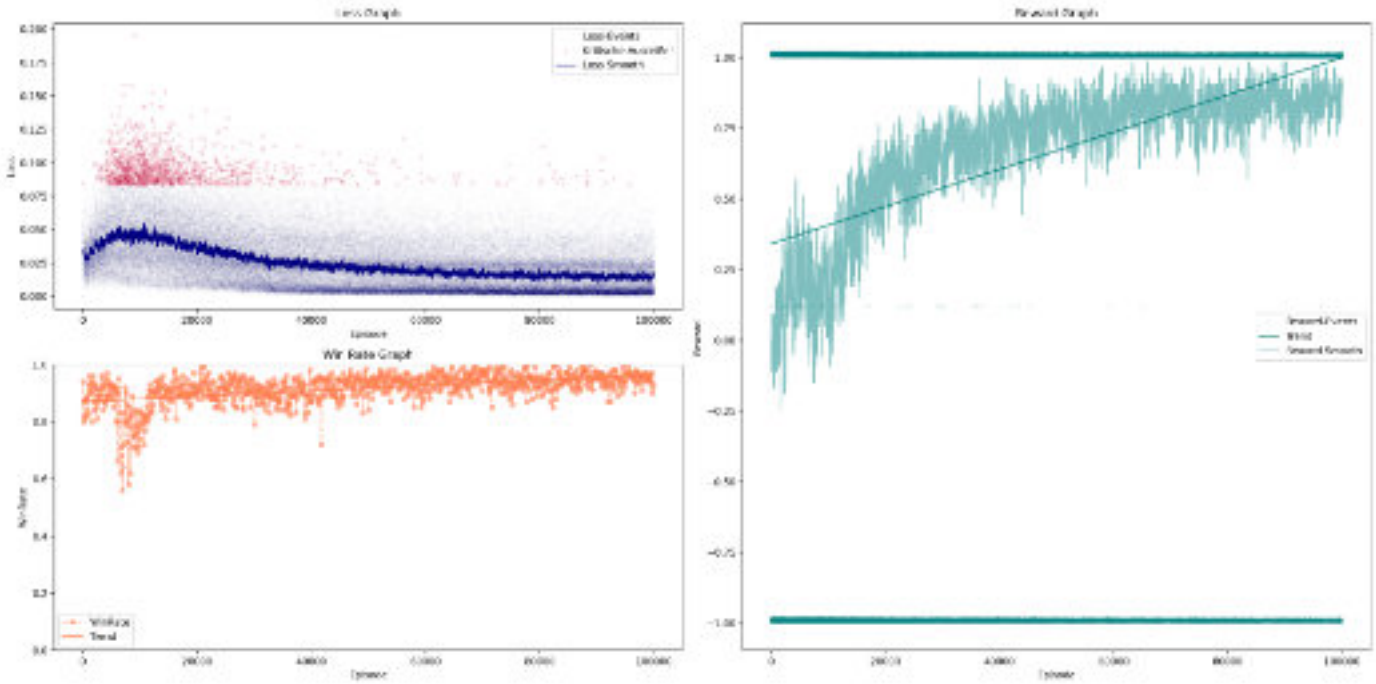
Haperparameter: hidden_size = 128, gamma = 0,95, batch_size = 64, survive_reward = 0,001

Erwartung: Ähnelt Trainer 2.

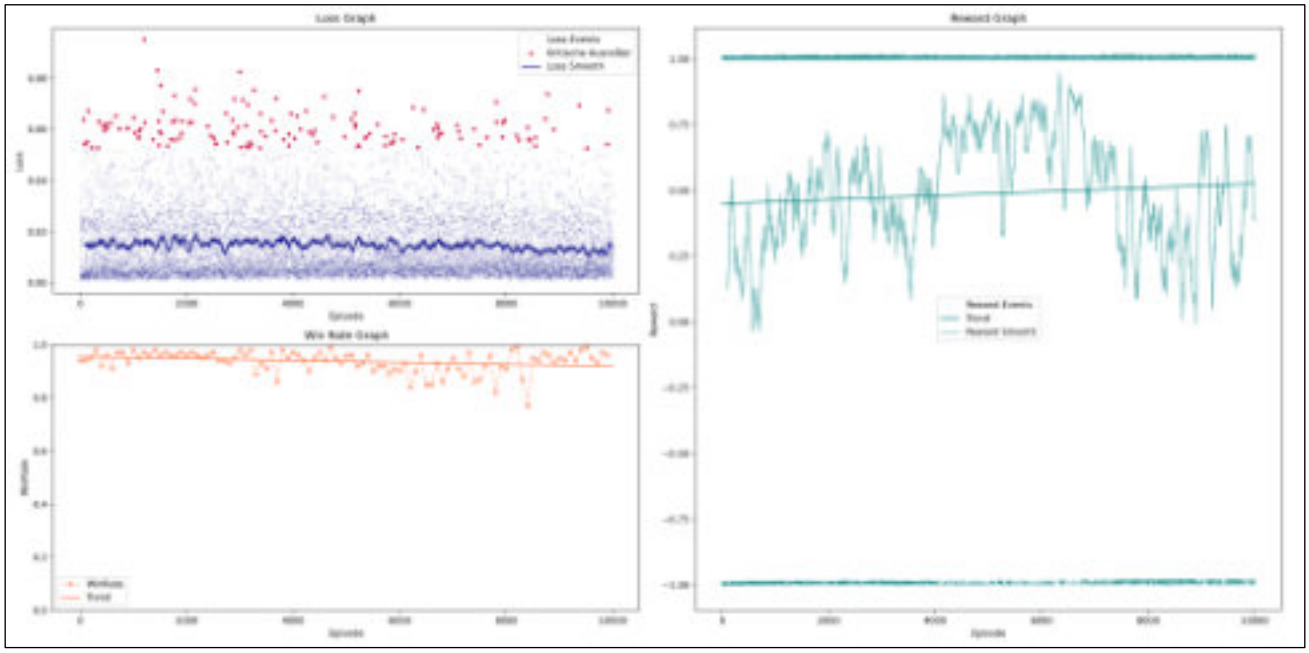
Überblick



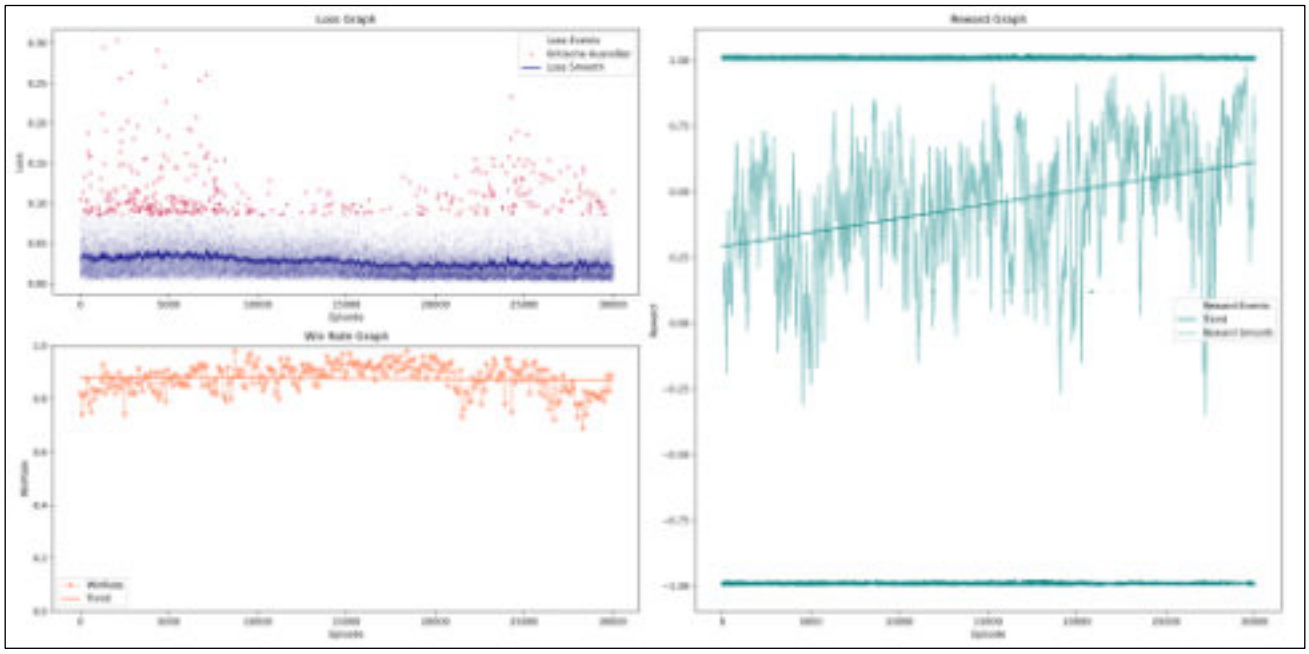
base_model



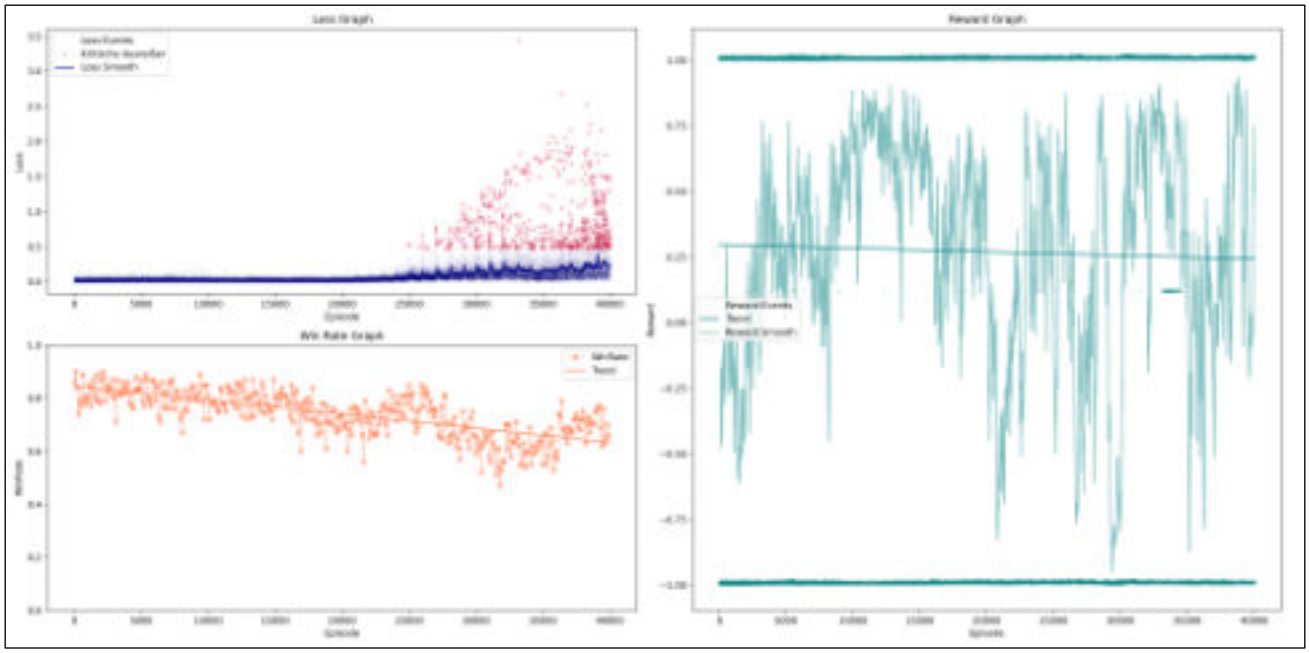
league_play_model_1



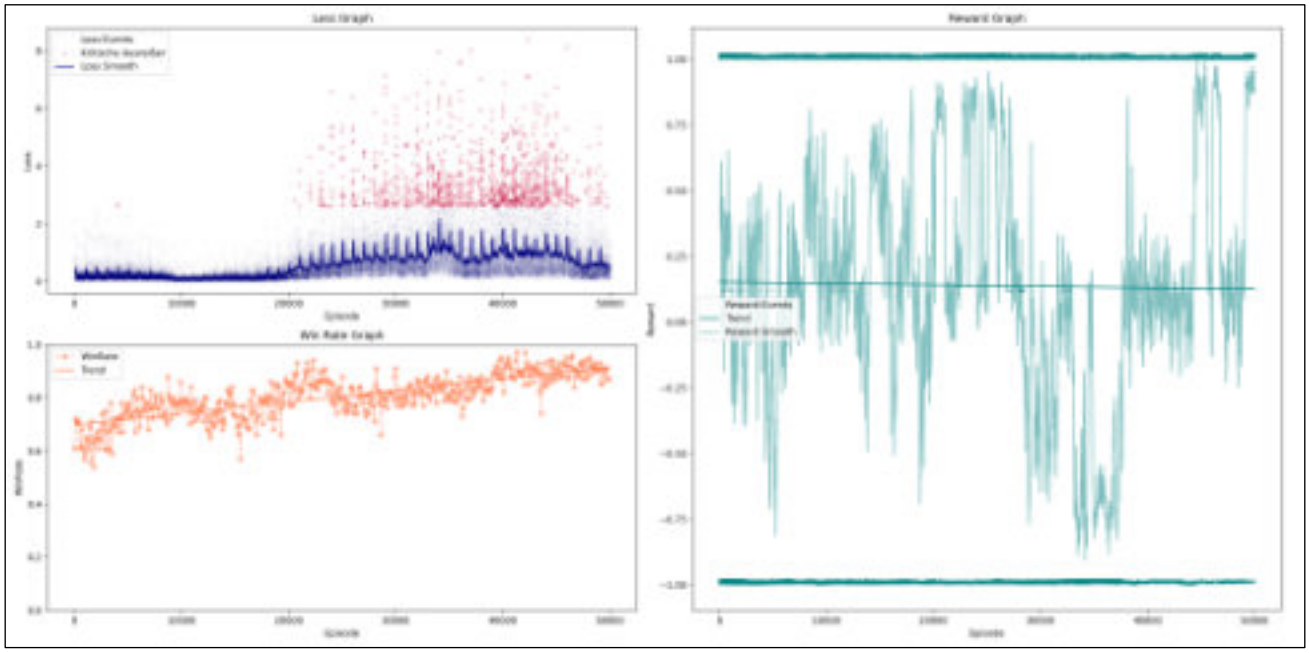
league_play_model_3



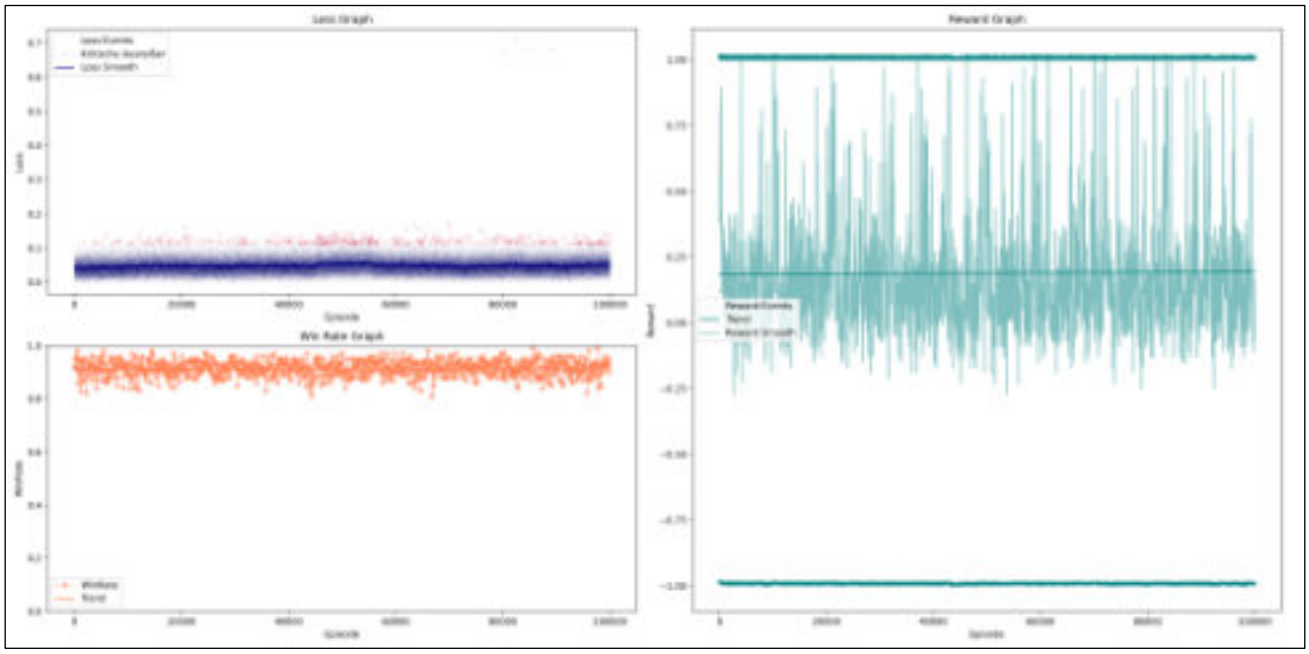
league_play_model_4



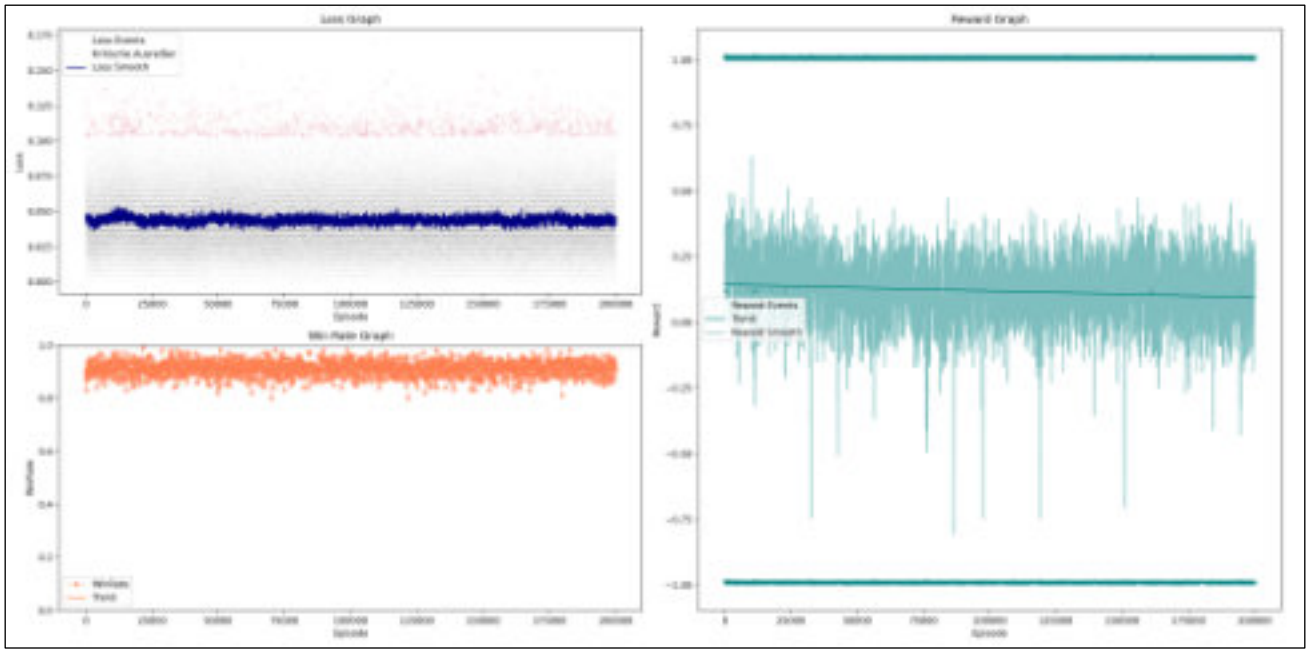
league_play_model_5



league_play_model_10



league_play_model_20

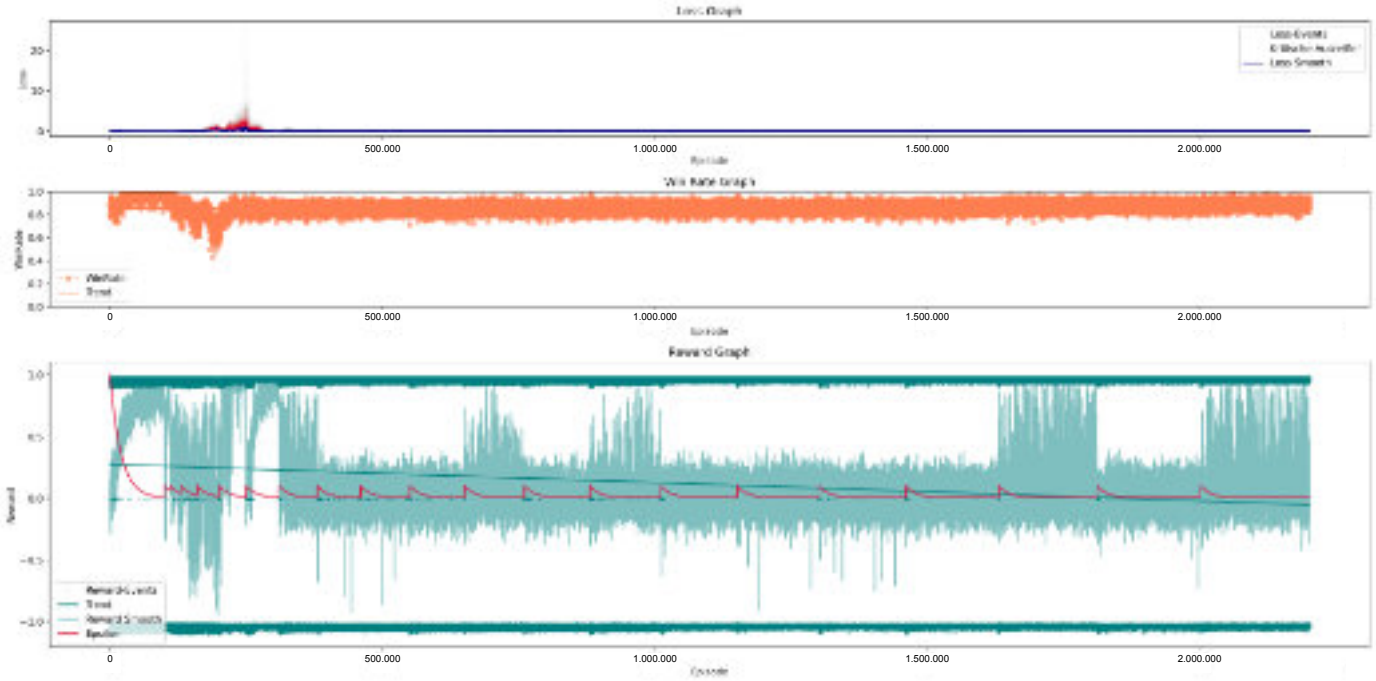


Trainer 11

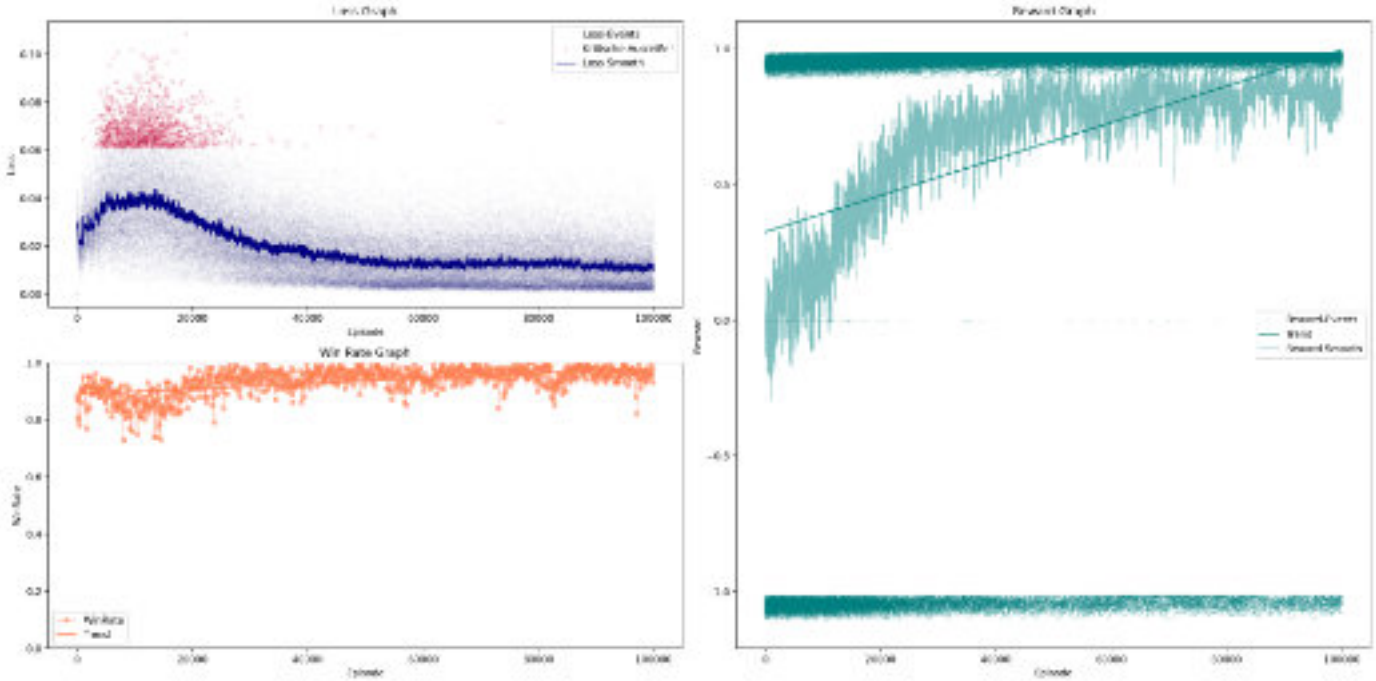
Haperparameter: hidden_size = 128, gamma = 0,95, batch_size = 128, survive_reward = -0,005

Erwartung: Der Agent sollte sich auch ähnlich zu Trainer 9 verhalten, es werden jedoch mehr Daten im Lernprozess einbezogen, wodurch potentiell weniger relevante Daten miteibezogen werden.

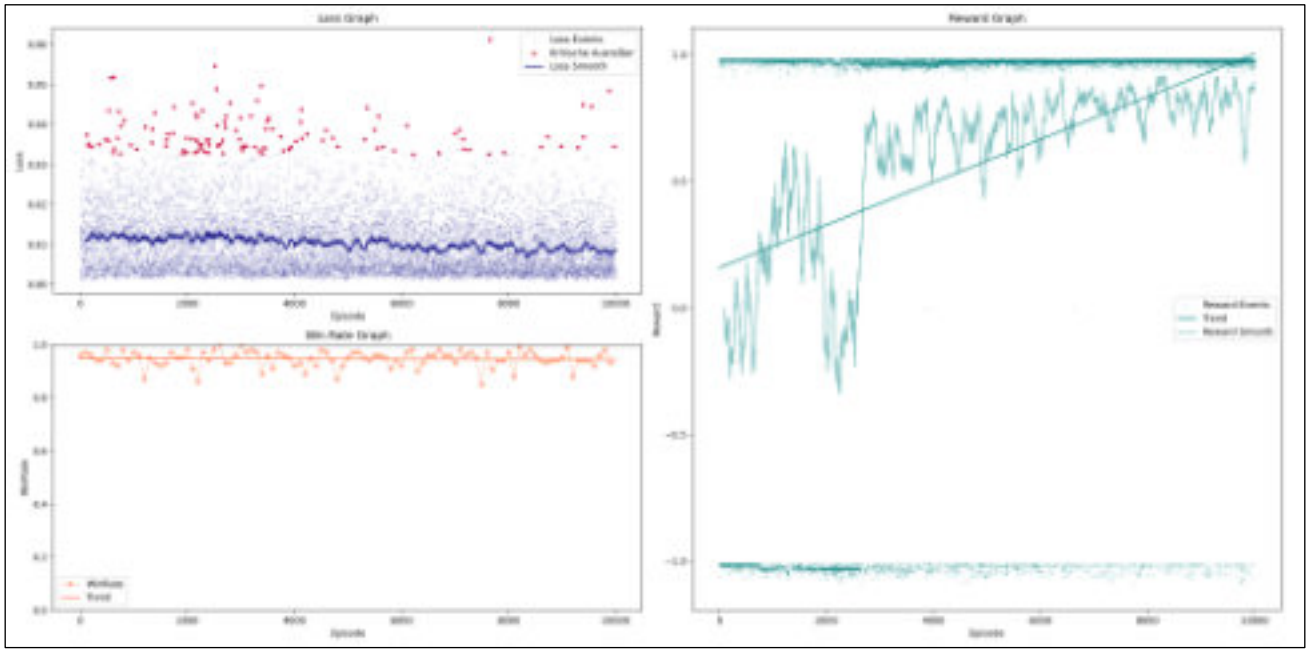
Überblick



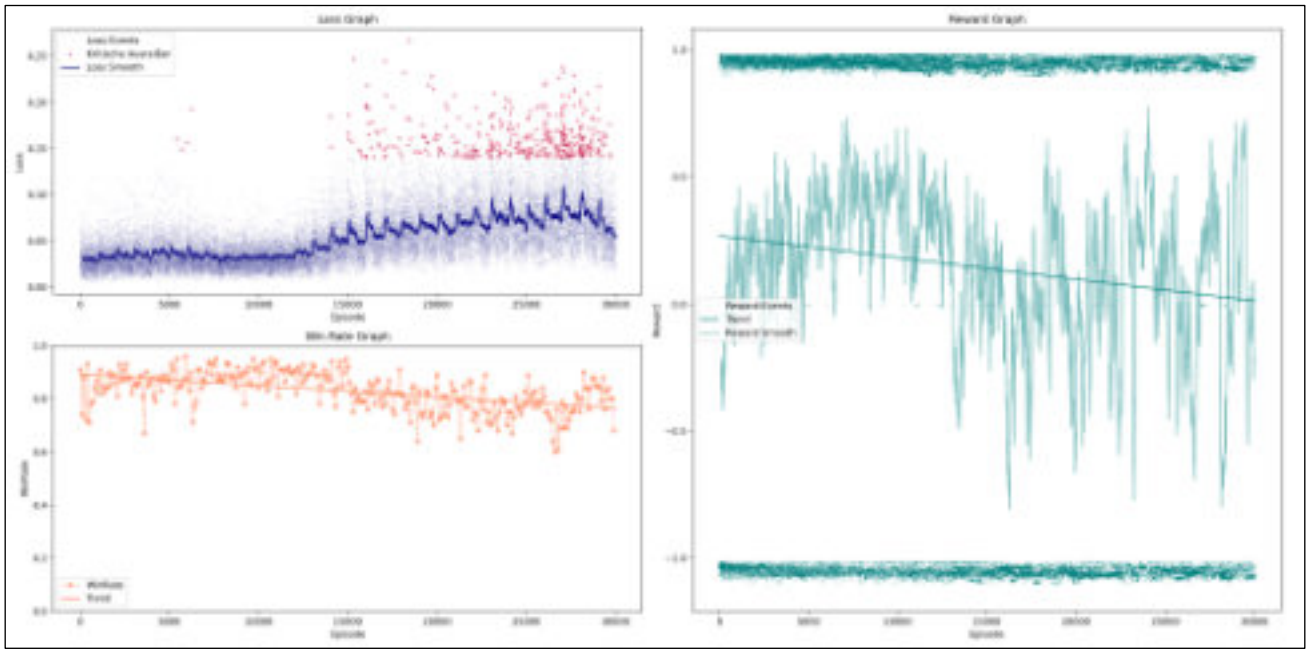
base_model



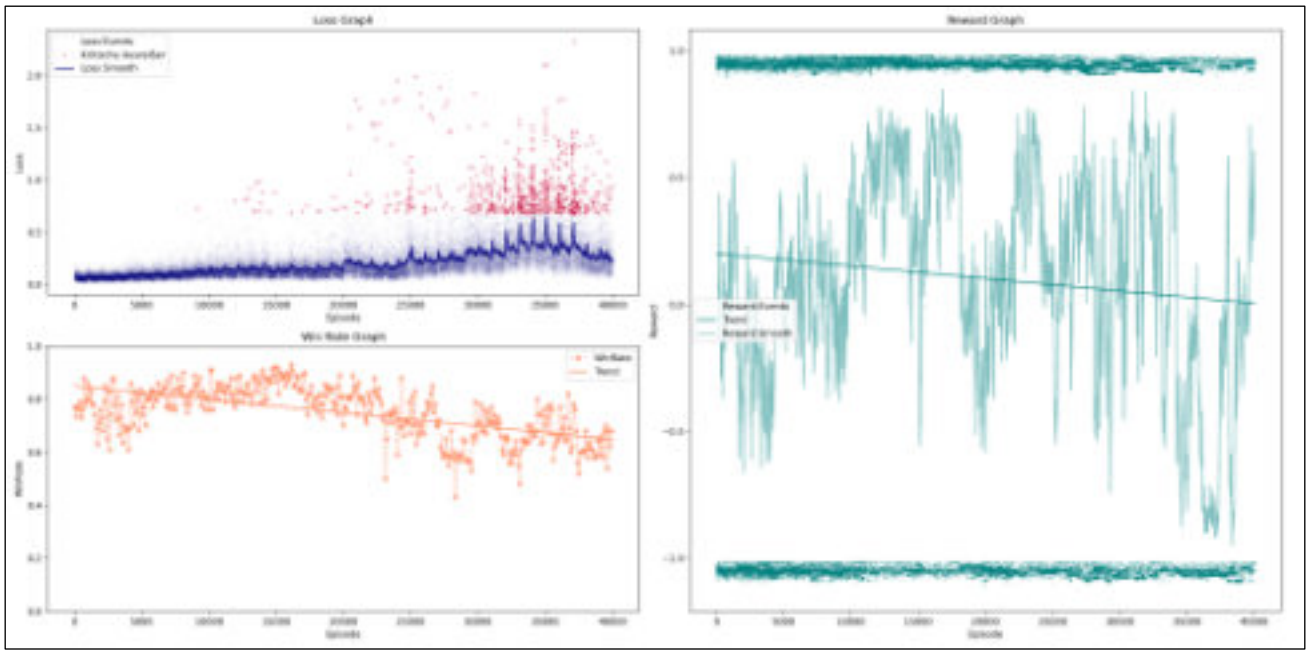
league_play_model_1



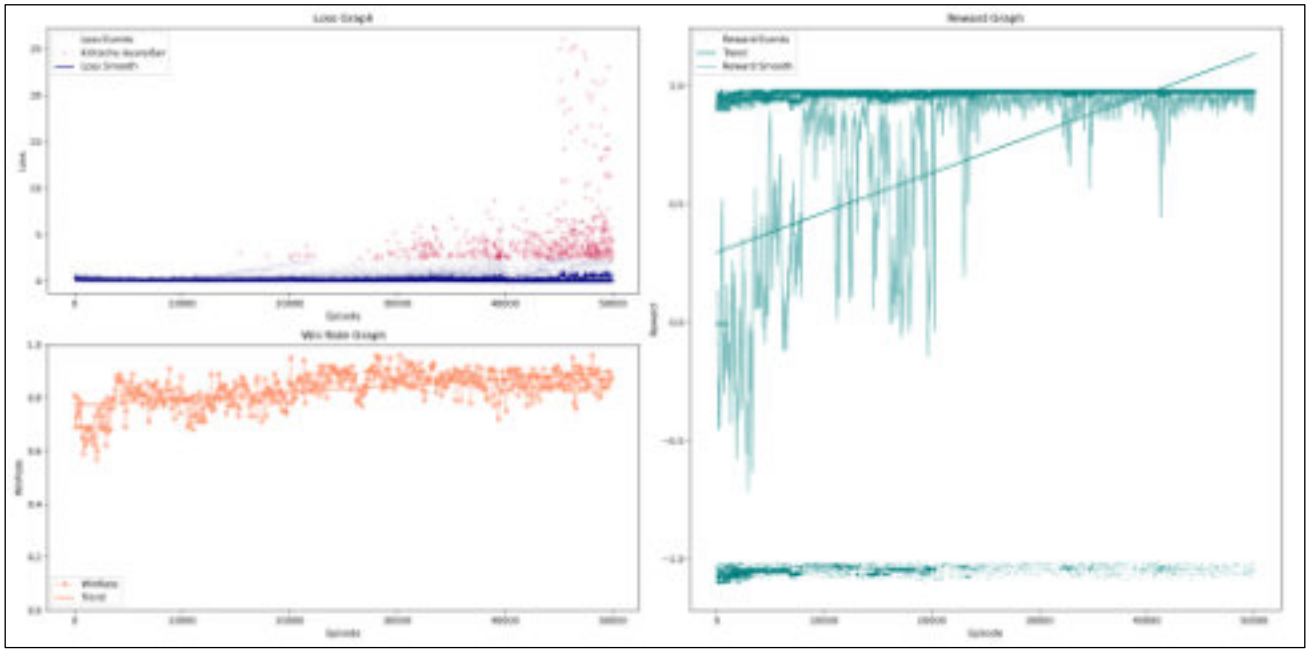
league_play_model_3



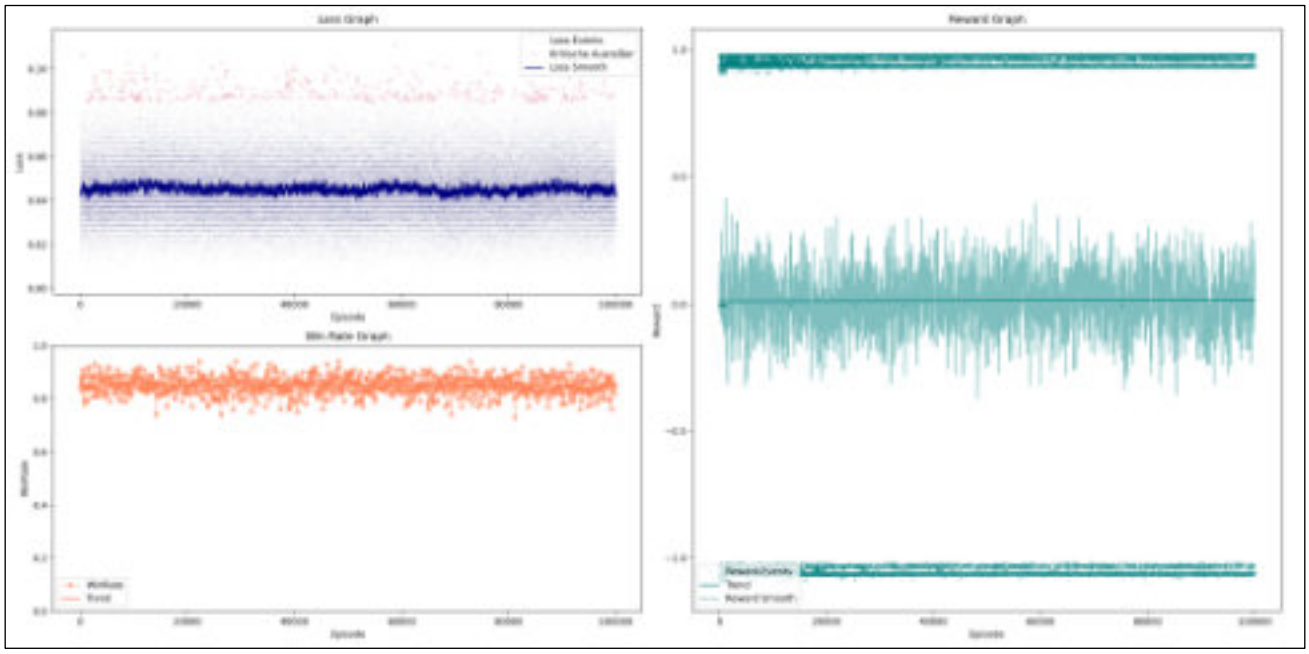
league_play_model_4



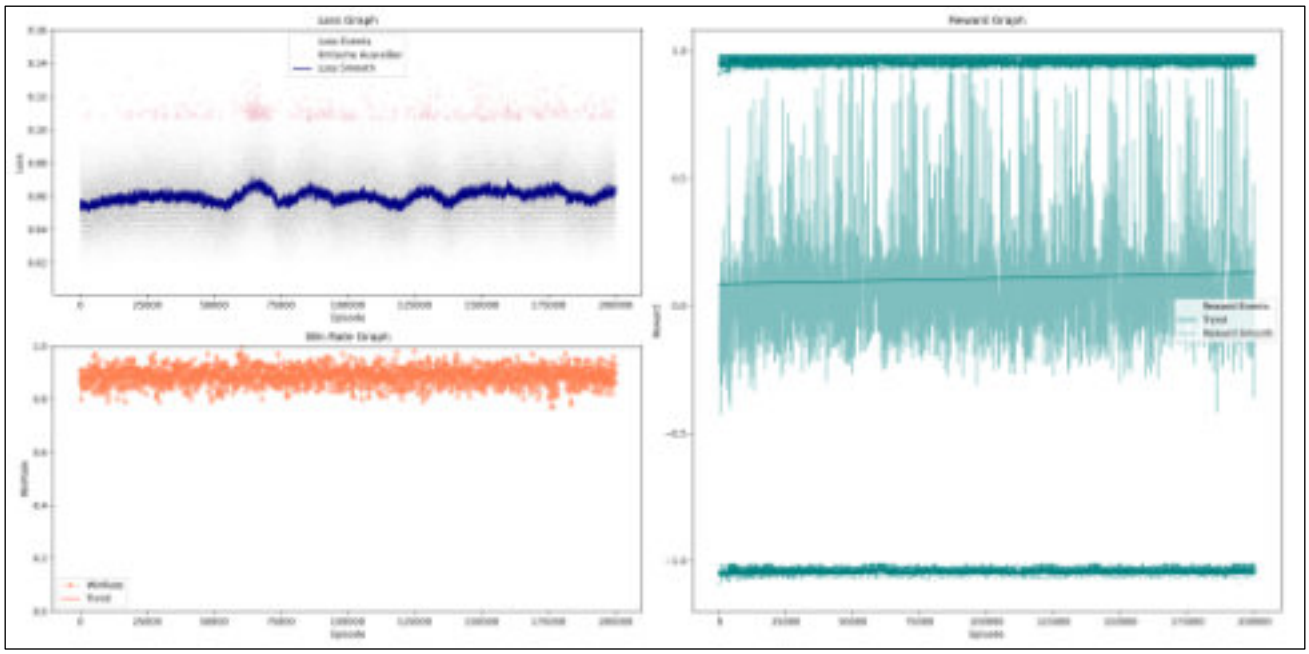
league_play_model_5



league_play_model_10



league_play_model_20

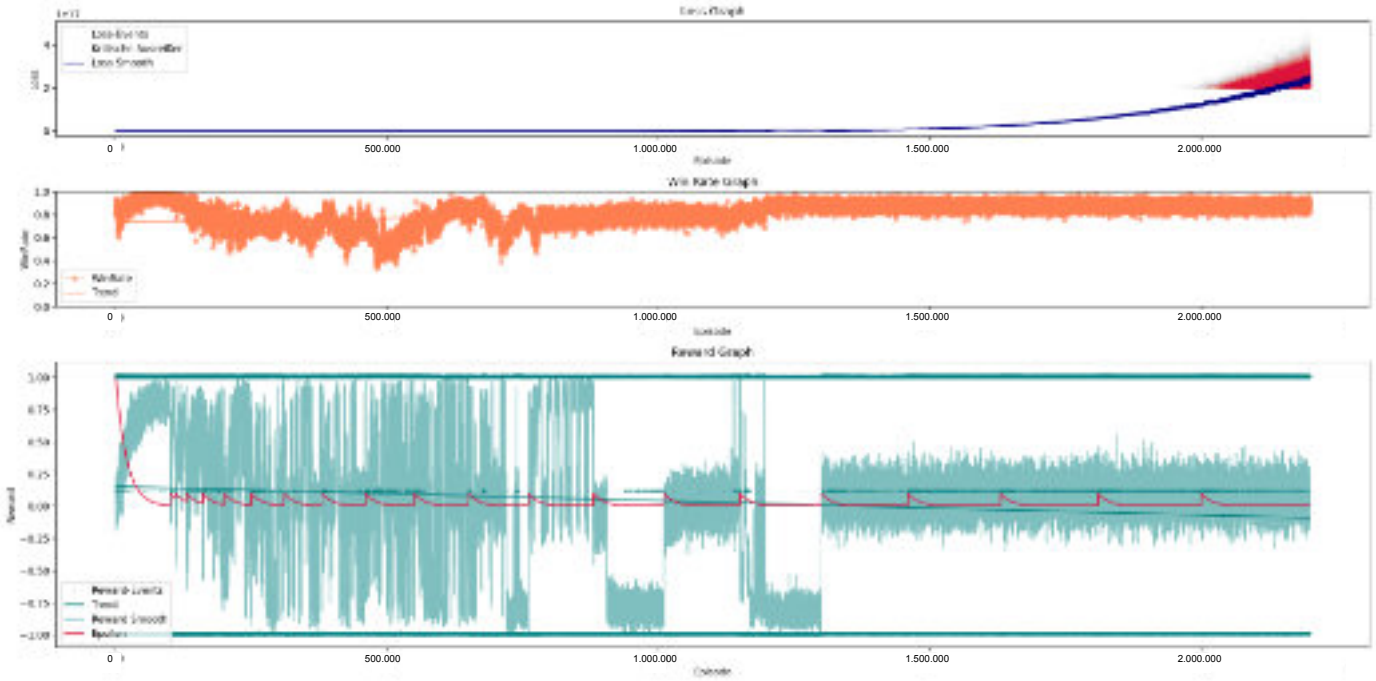


Trainer 12

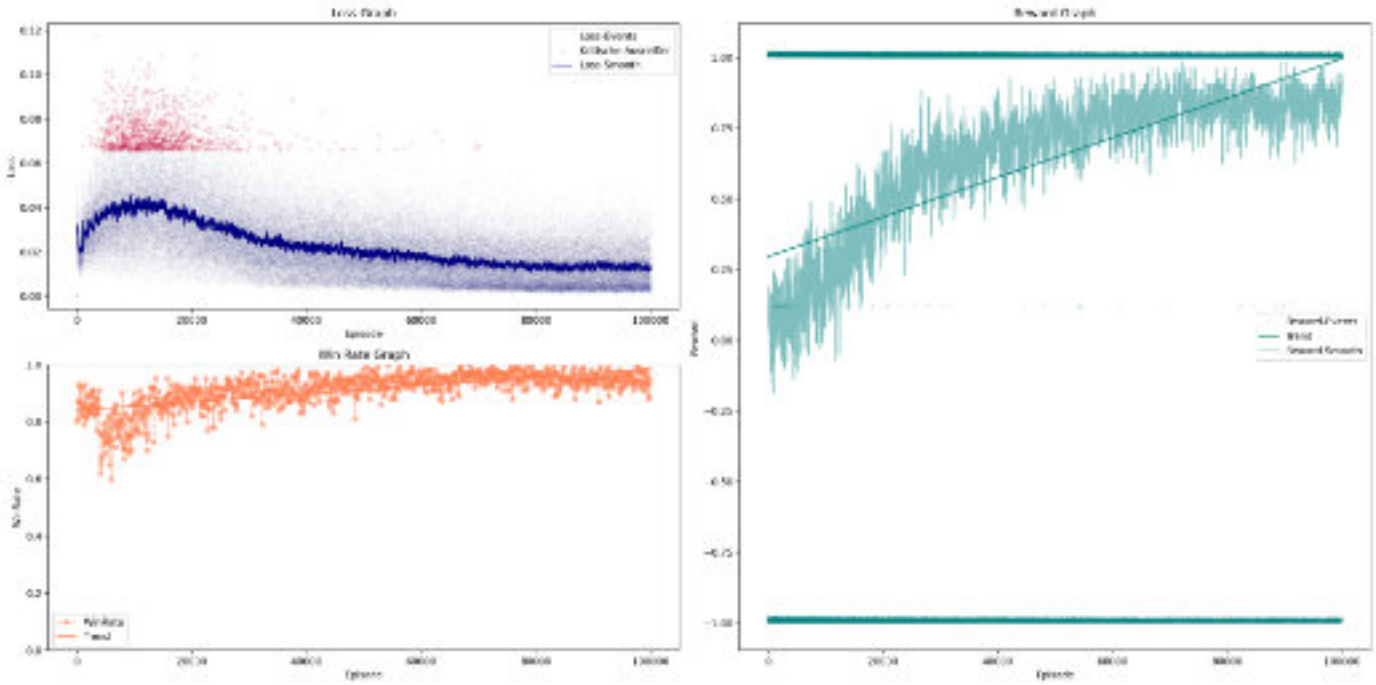
Haperparameter: hidden_size = 128, gamma = 0,95, batch_size = 128, survive_reward = 0,001

Erwartung: Der Agent sollte sich ähnlich zu Trainer 4 verhalten, jedoch mit kleinerer Netzwerkgröße.

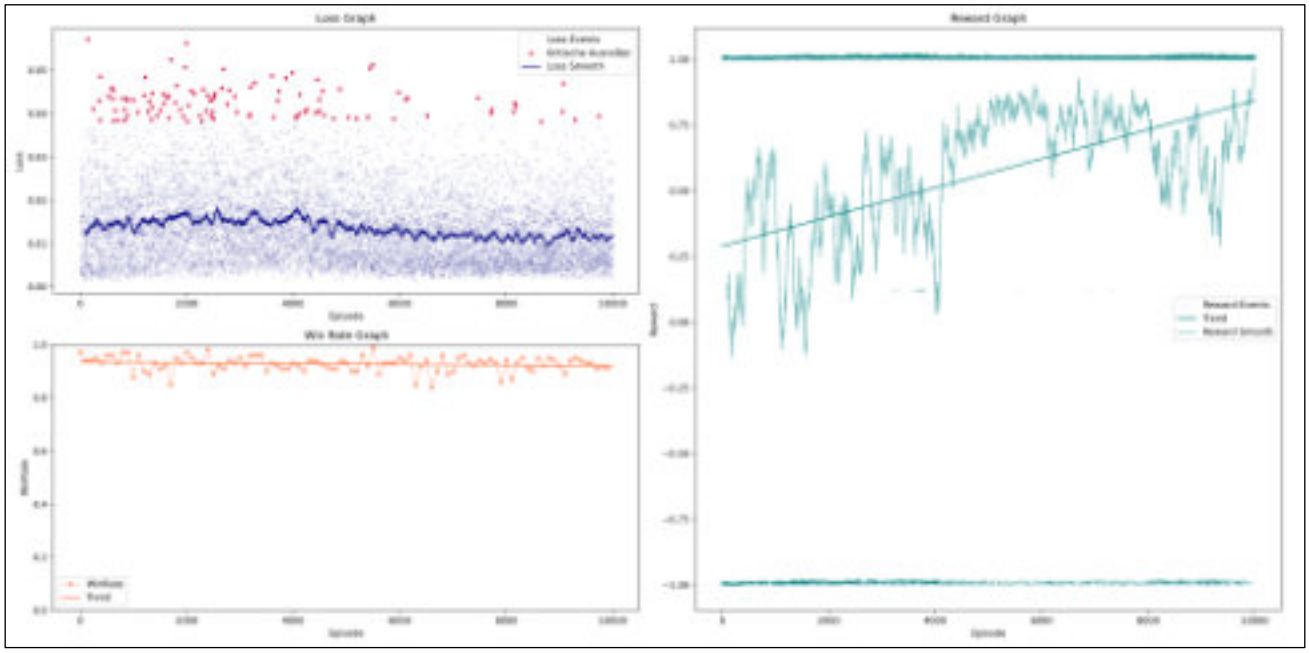
Überblick



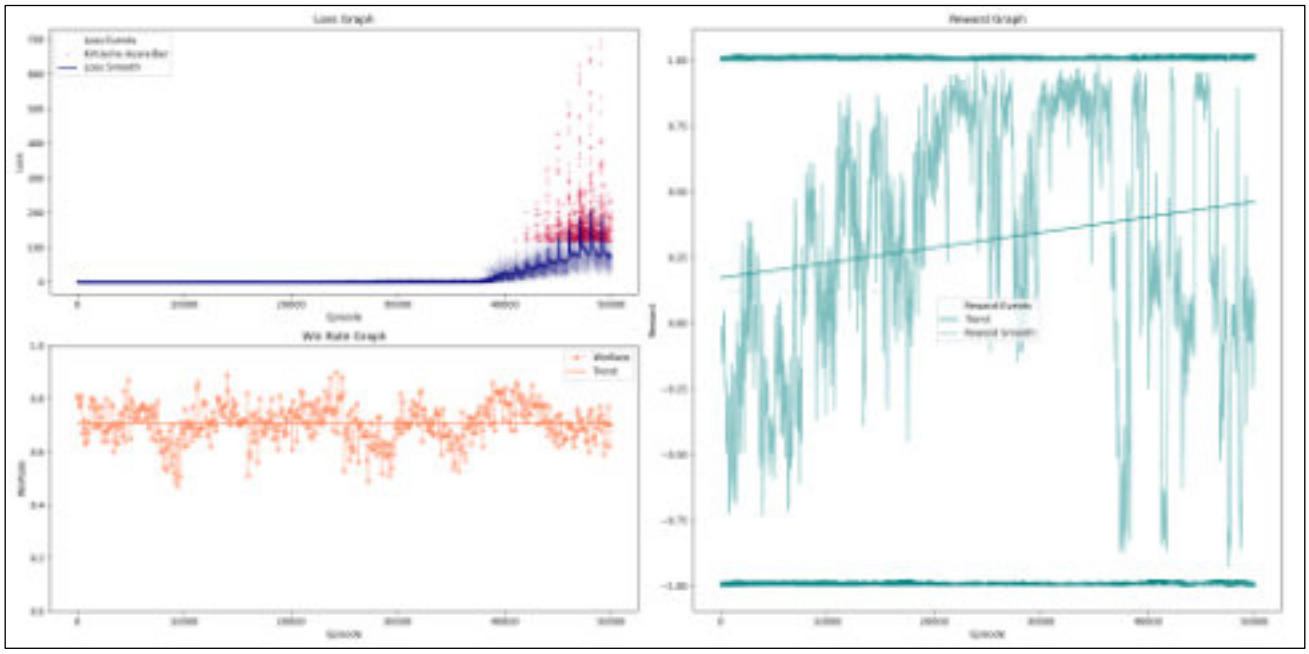
base_model



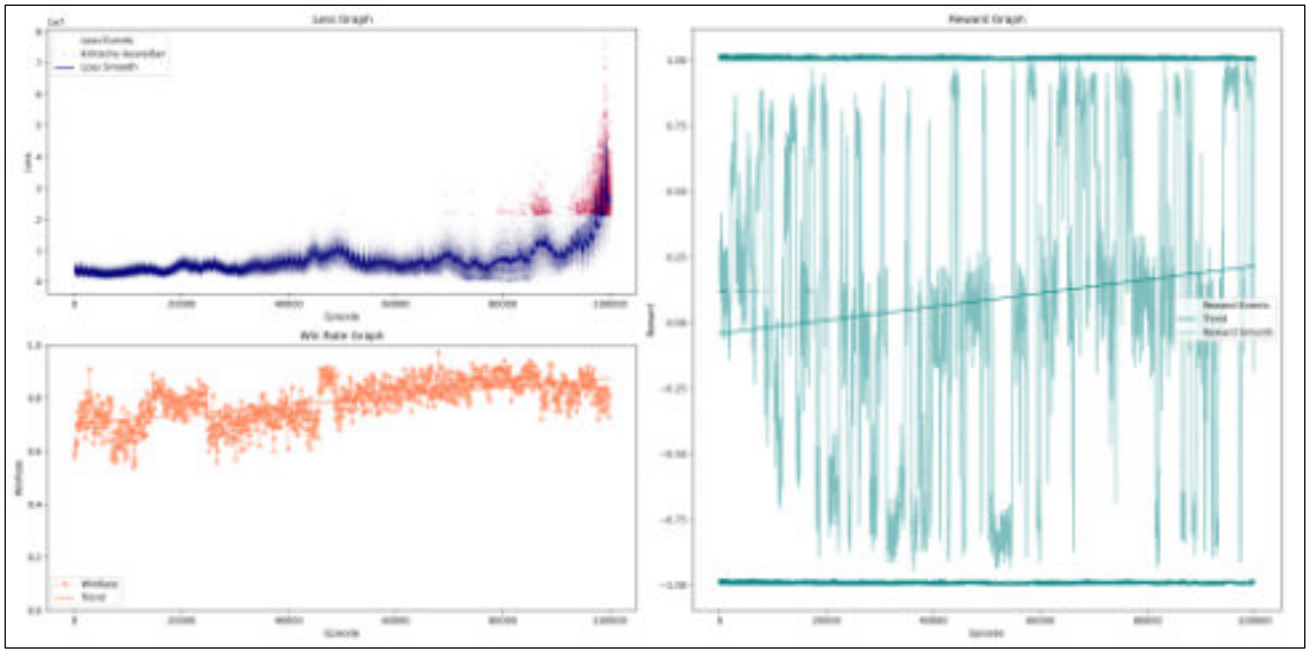
league_play_model_1



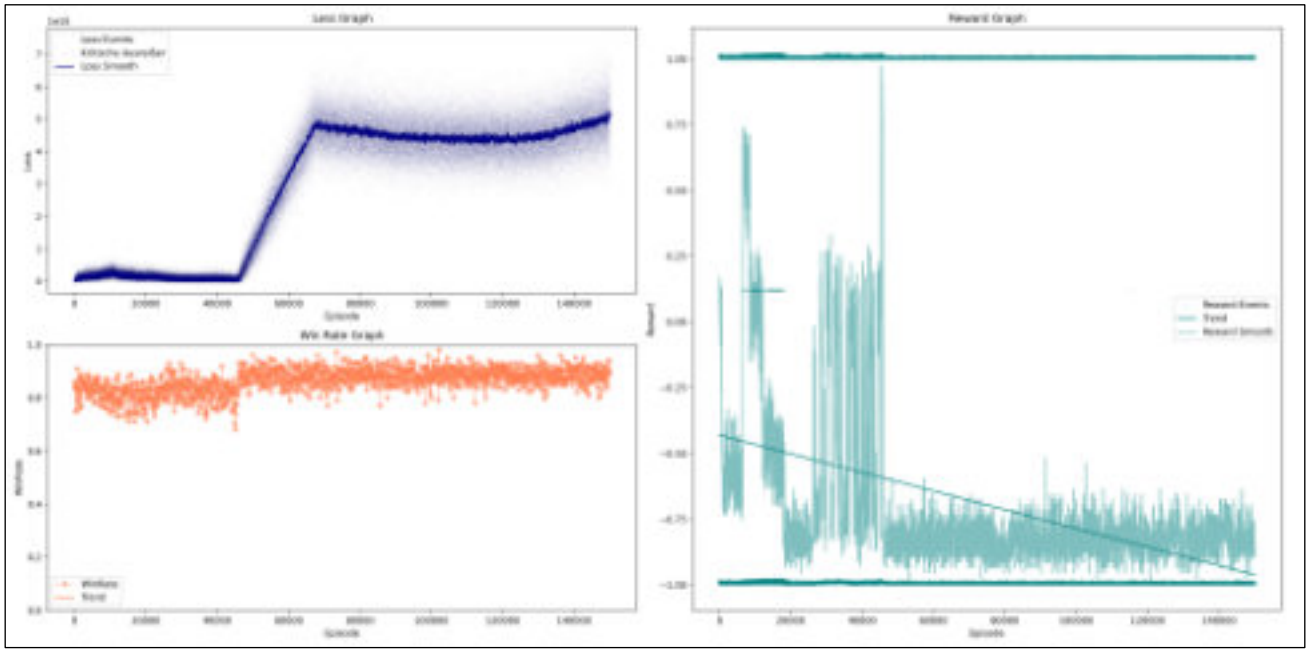
league_play_model_5



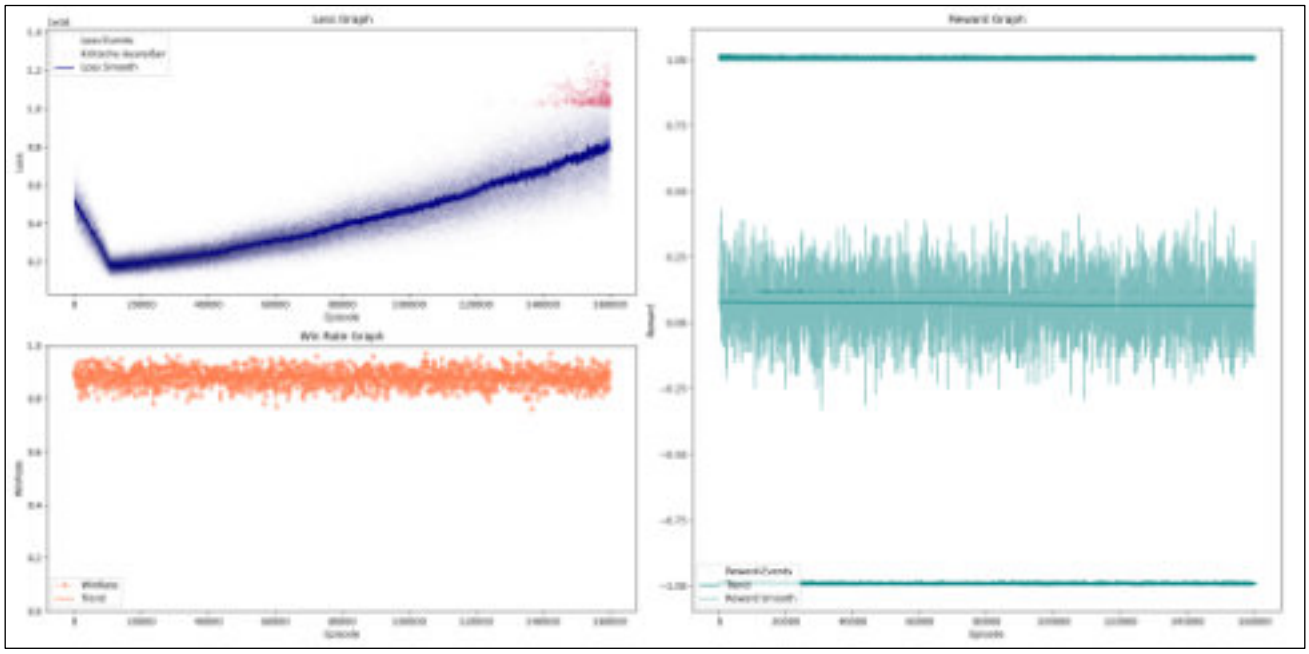
league_play_model_10



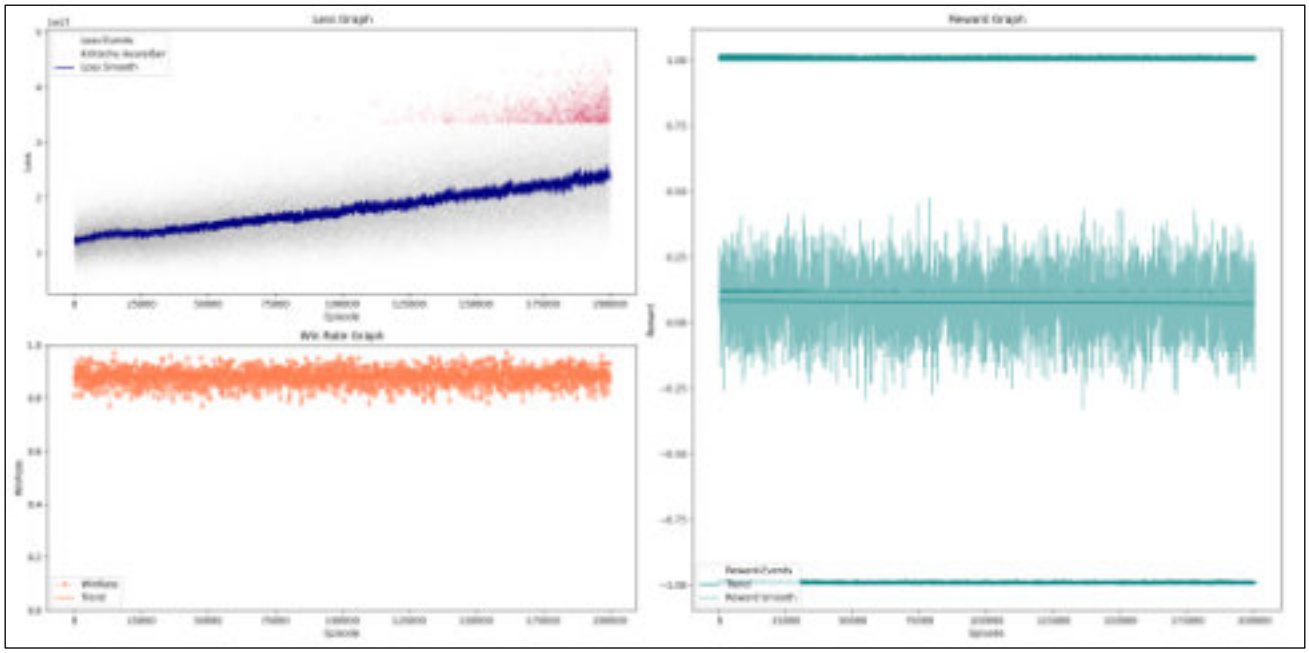
league_play_model_15



league_play_model_16



league_play_model_20

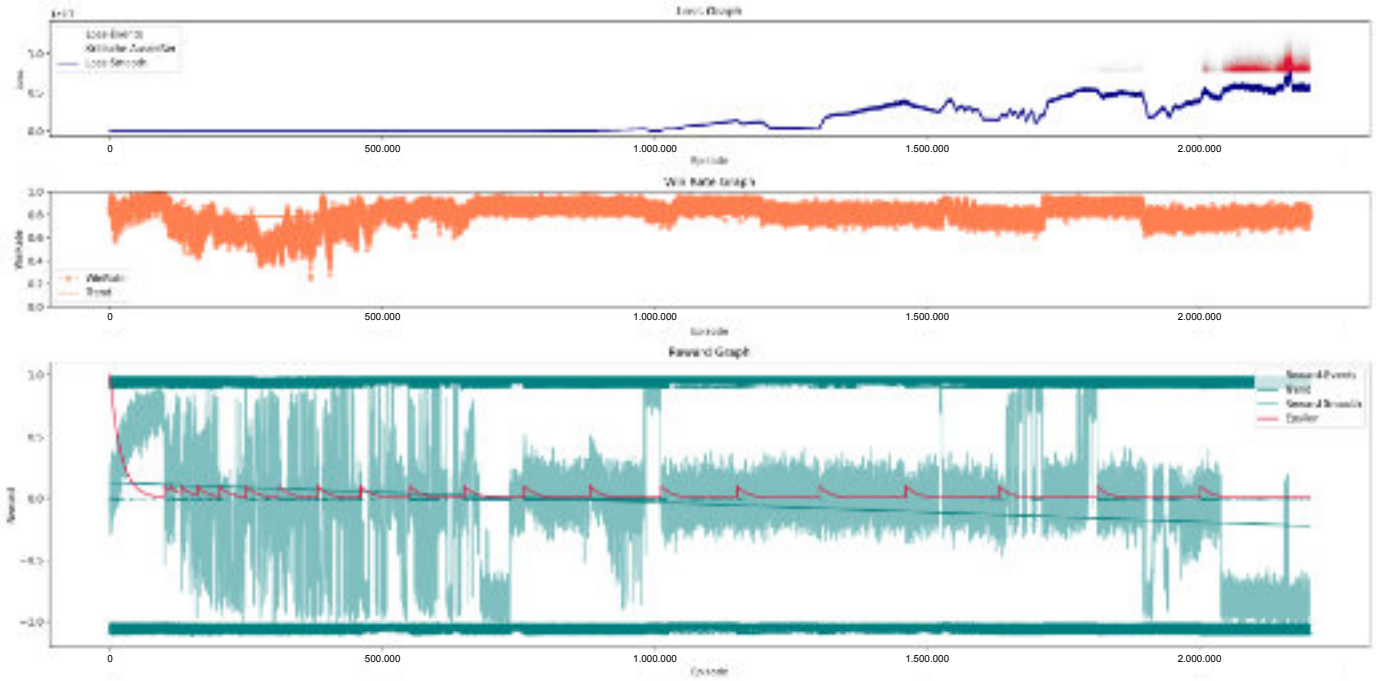


Trainer 13

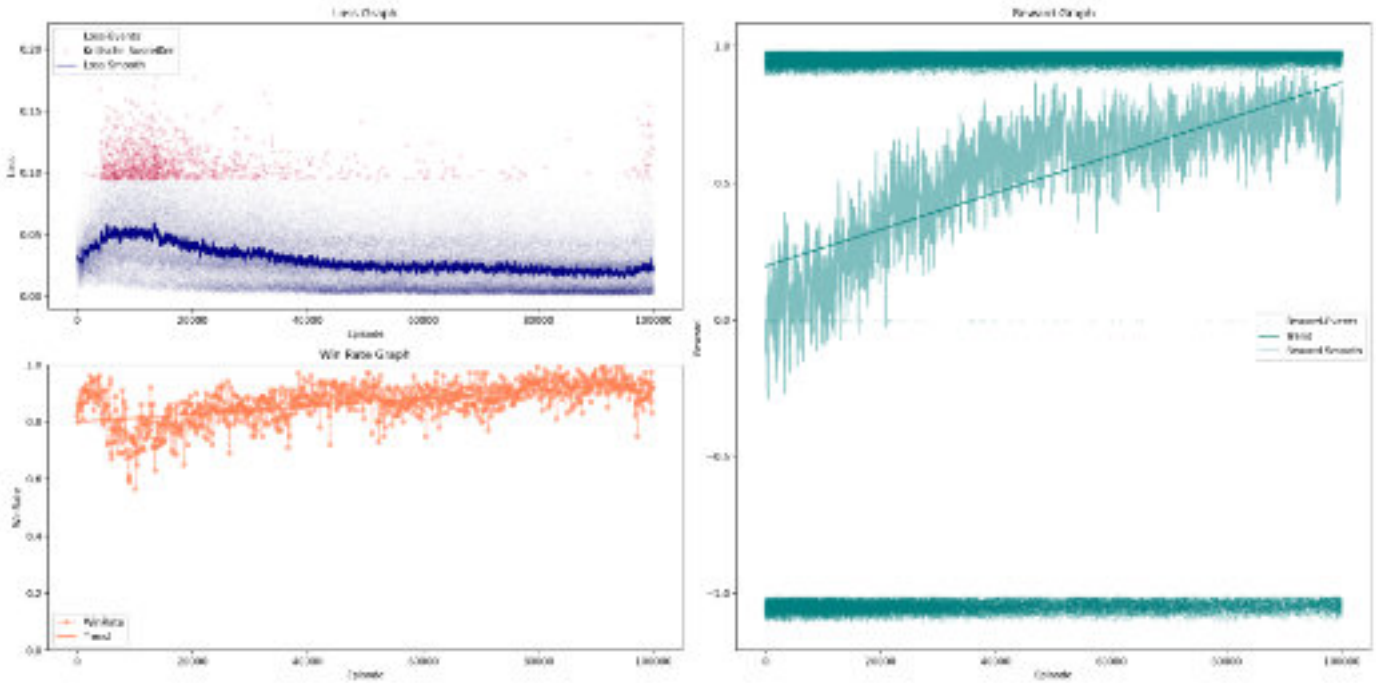
Haperparameter: hidden_size = 128, gamma = 0,99, batch_size = 64, survive_reward = -0,005

Erwartung: Durch den höheren Gamma-Wert in Kombination mit dem survive_reward und der hohen Netzwerkgröße, könnte der Agent vergleichbar mit Trainer 2 oder 4.

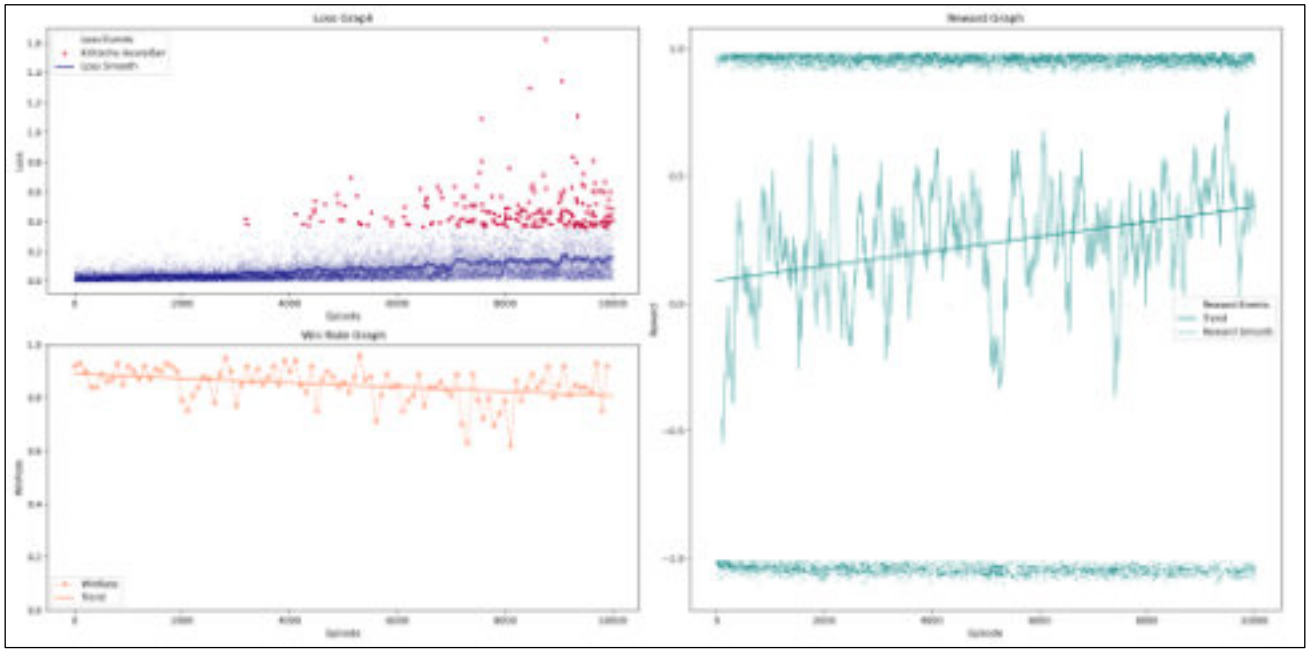
Überblick



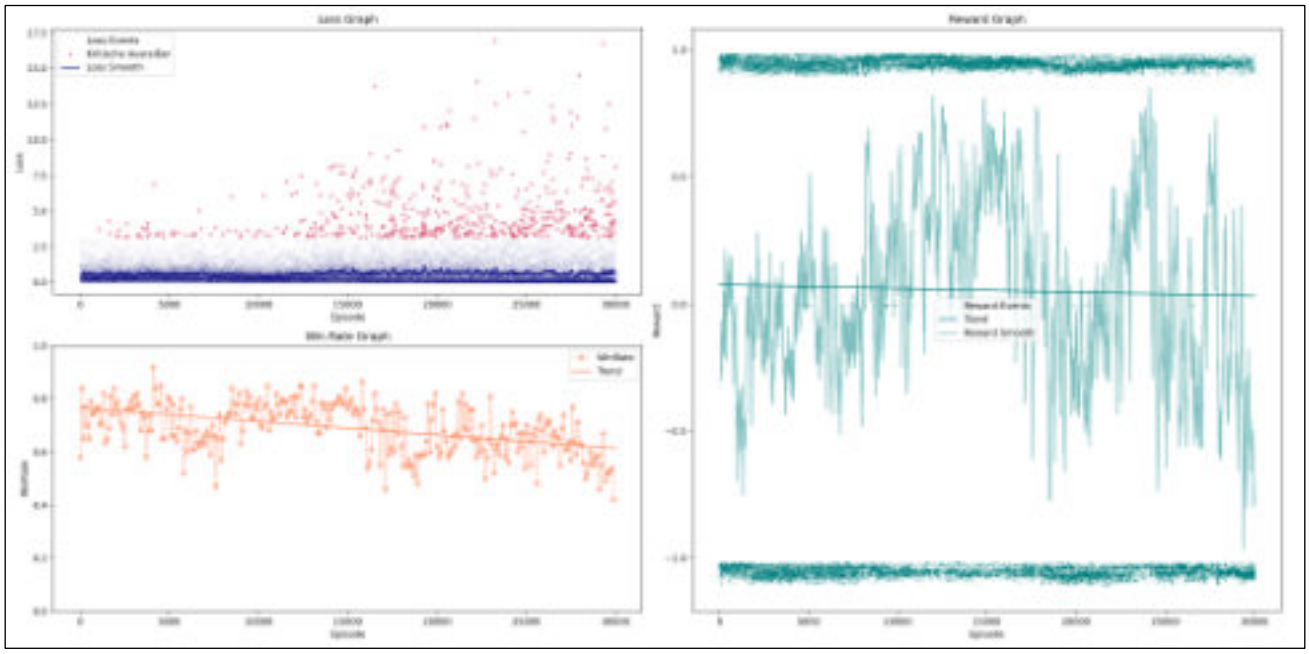
base_model



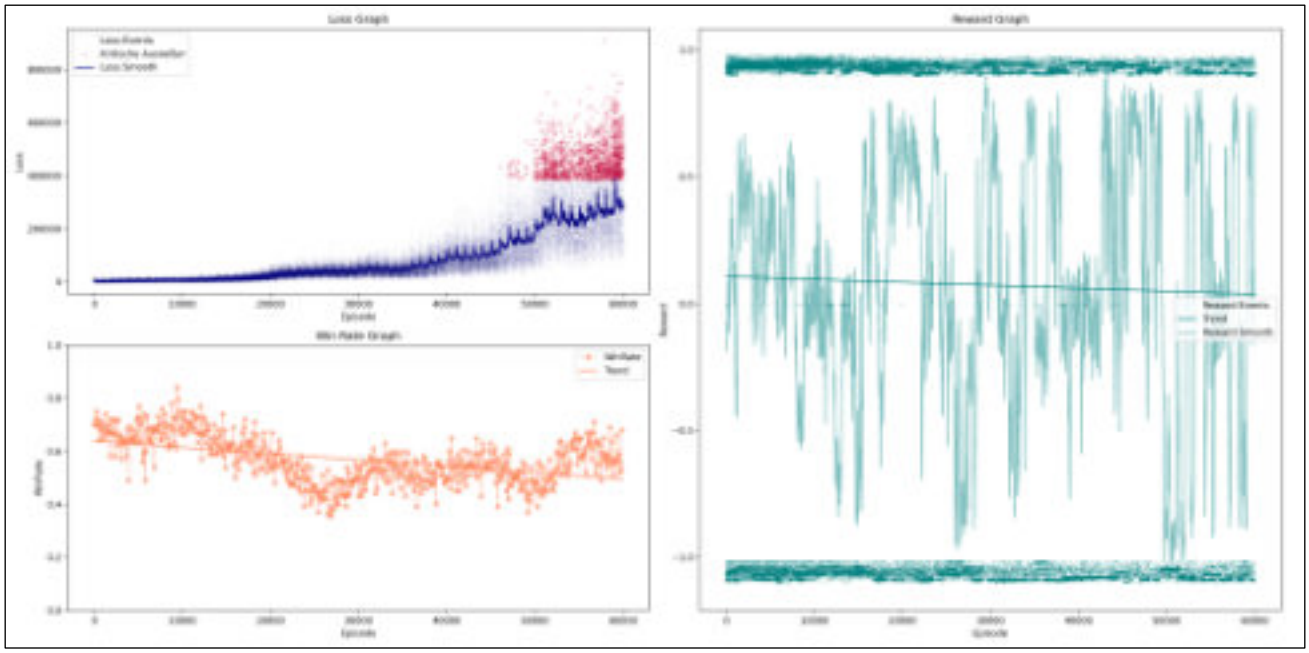
league_play_model_1



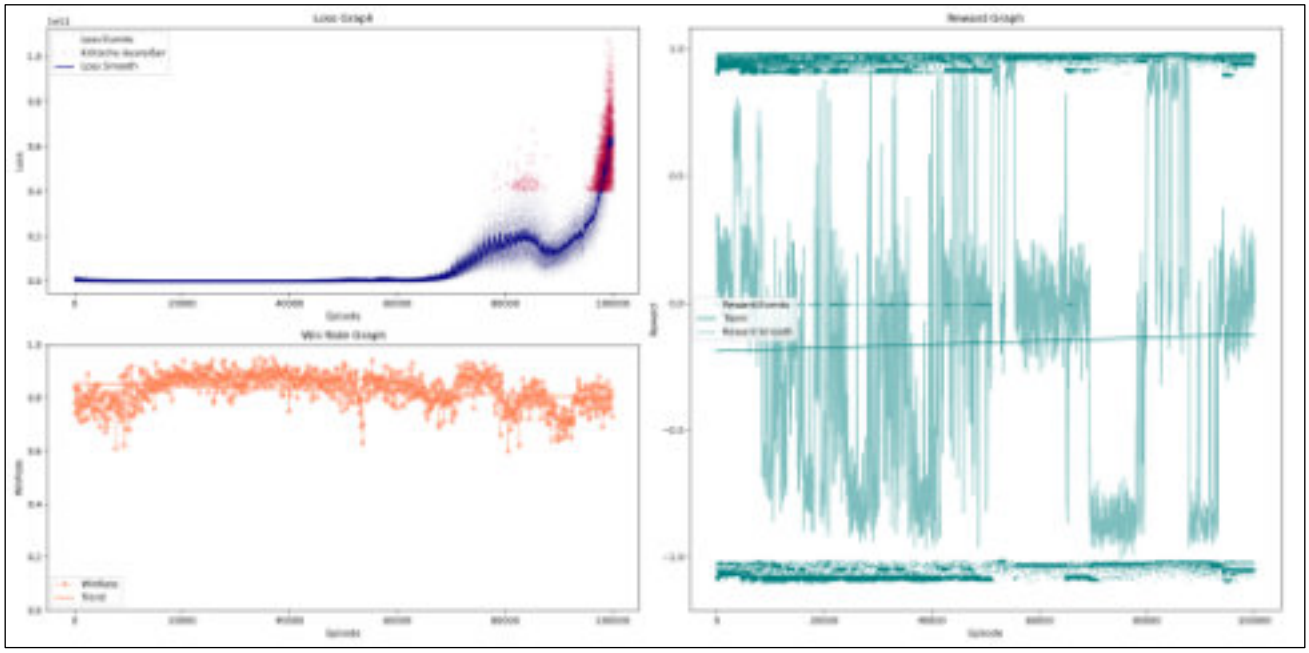
league_play_model_3



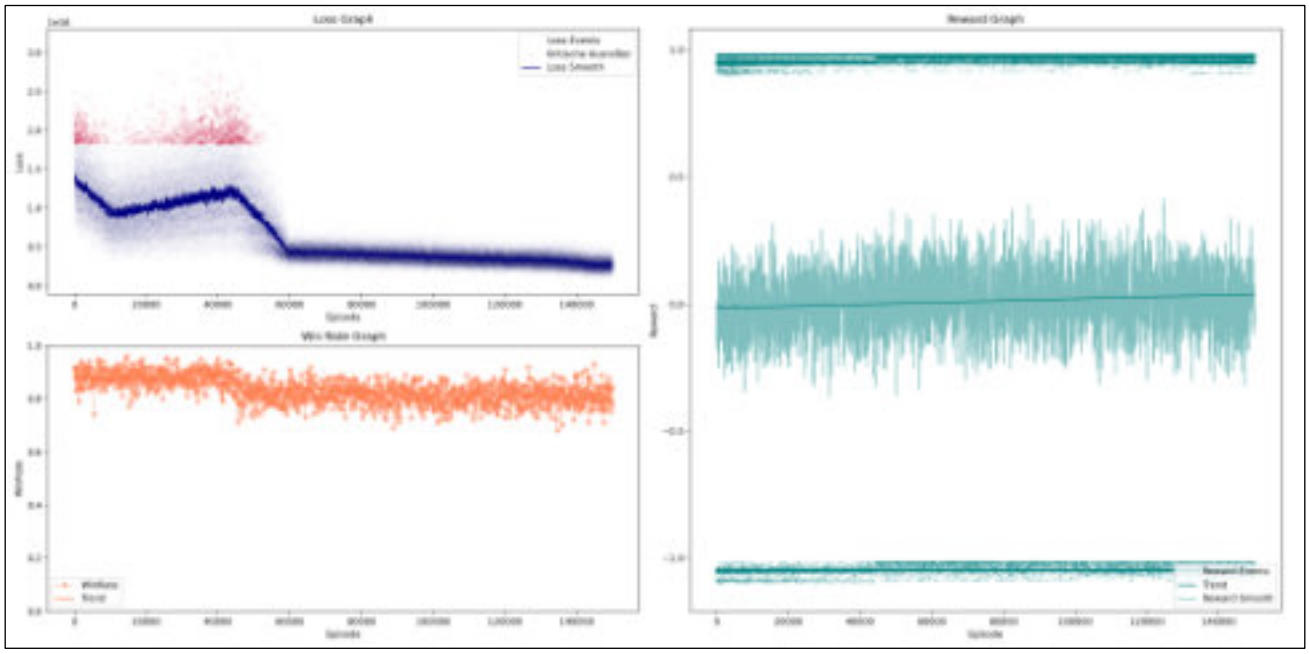
league_play_model_6



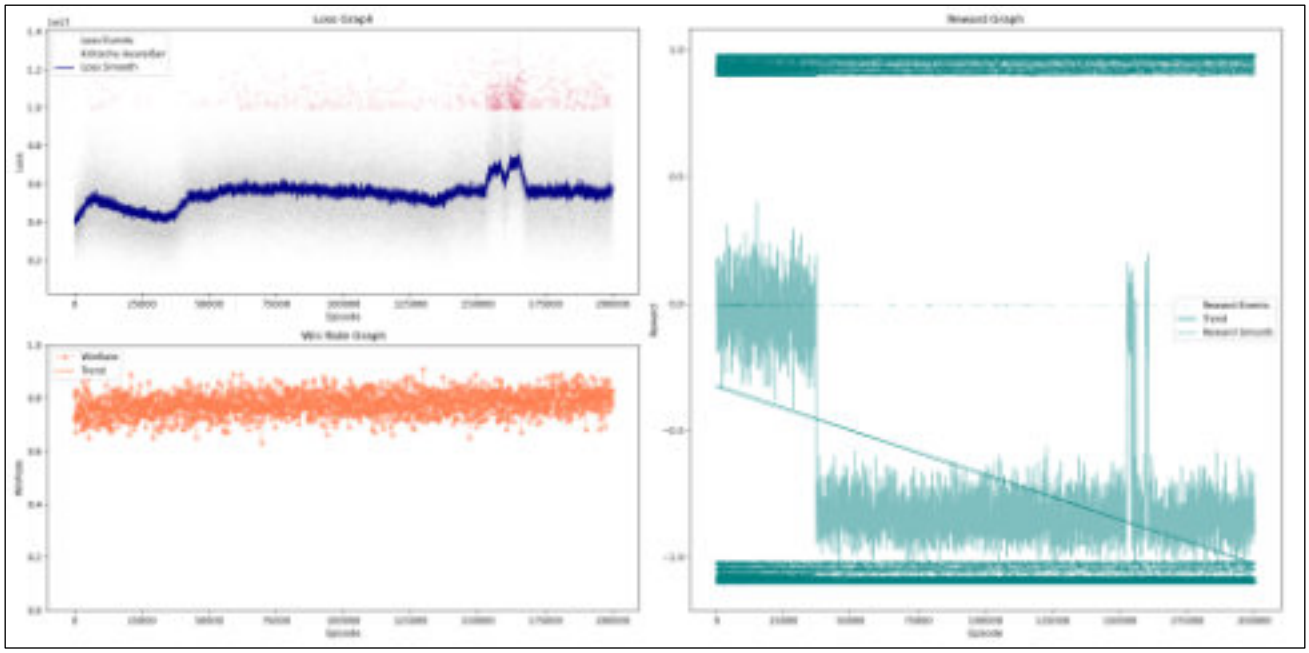
league_play_model_10



league_play_model_15



league_play_model_20

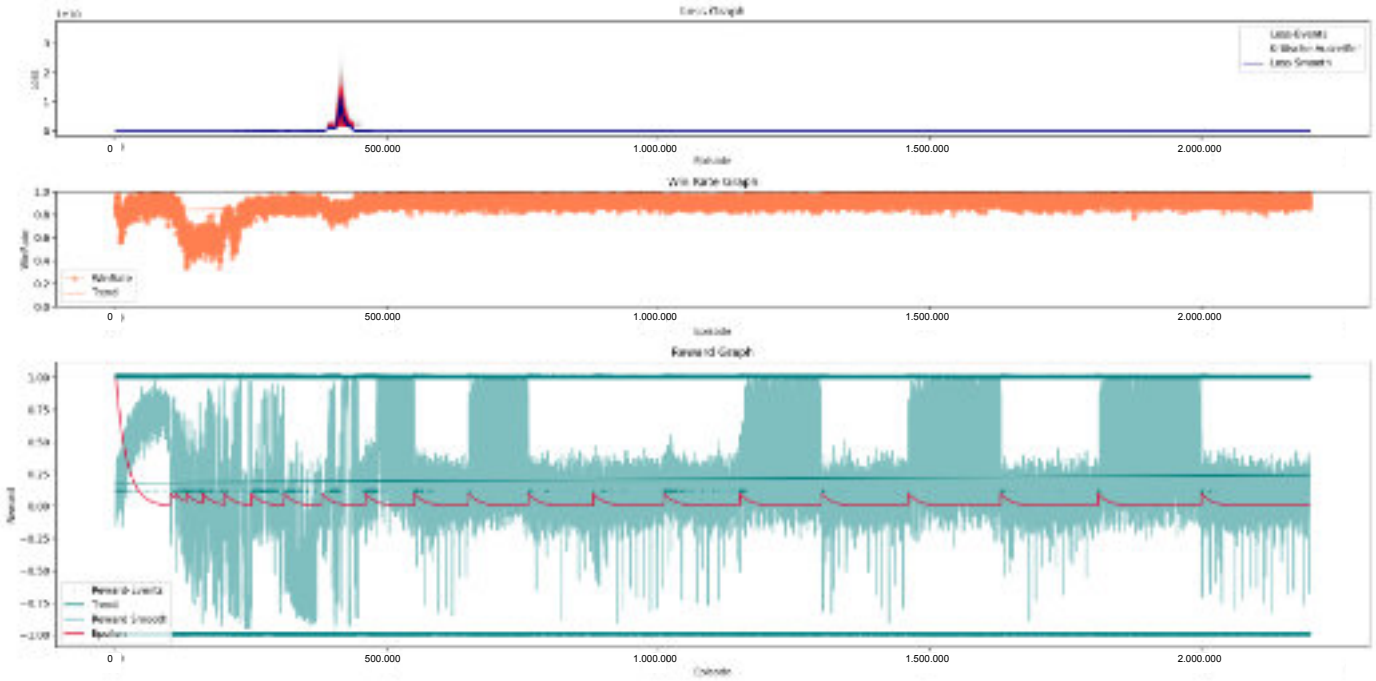


Trainer 14

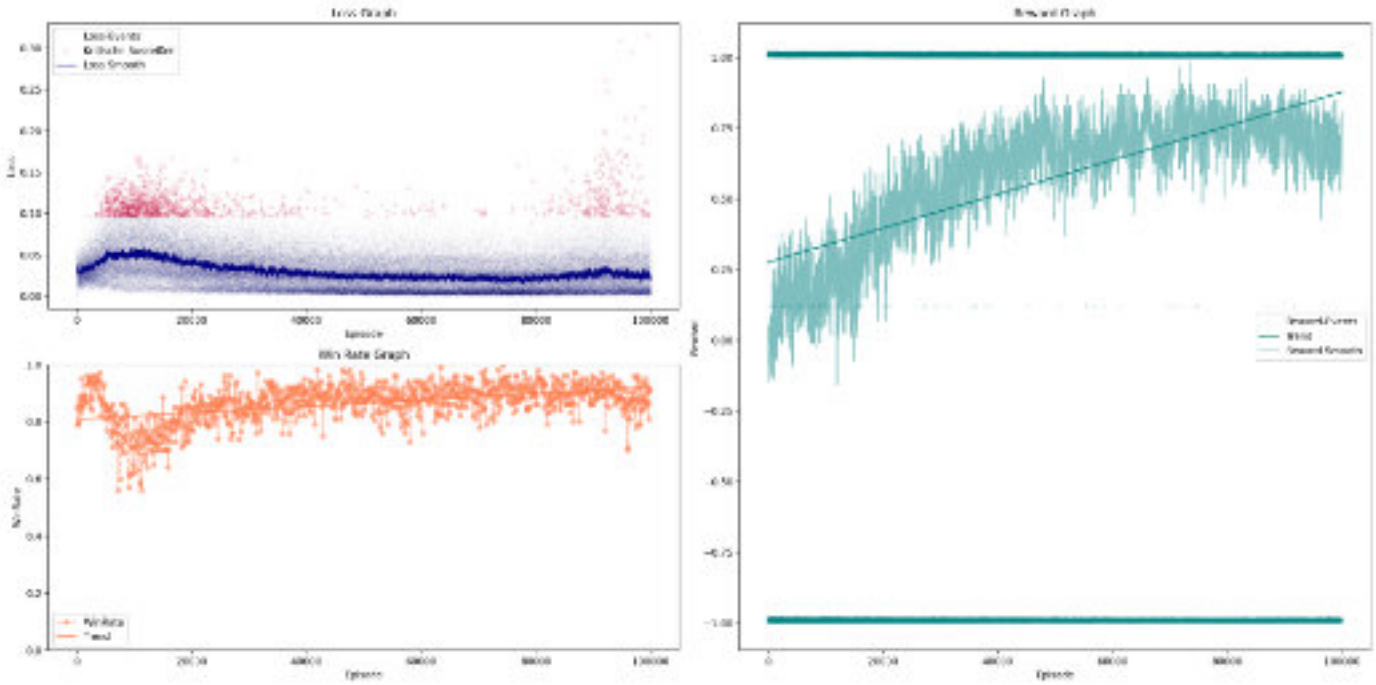
Hyperparameter: hidden_size = 128, gamma = 0,99, batch_size = 64, survive_reward = 0,001

Erwartung: Durch den höheren Gamma-Wert in Kombination mit dem positiven survive_reward und der hohen Netzwerkgröße, sollte der Agent am langfristigsten handeln.

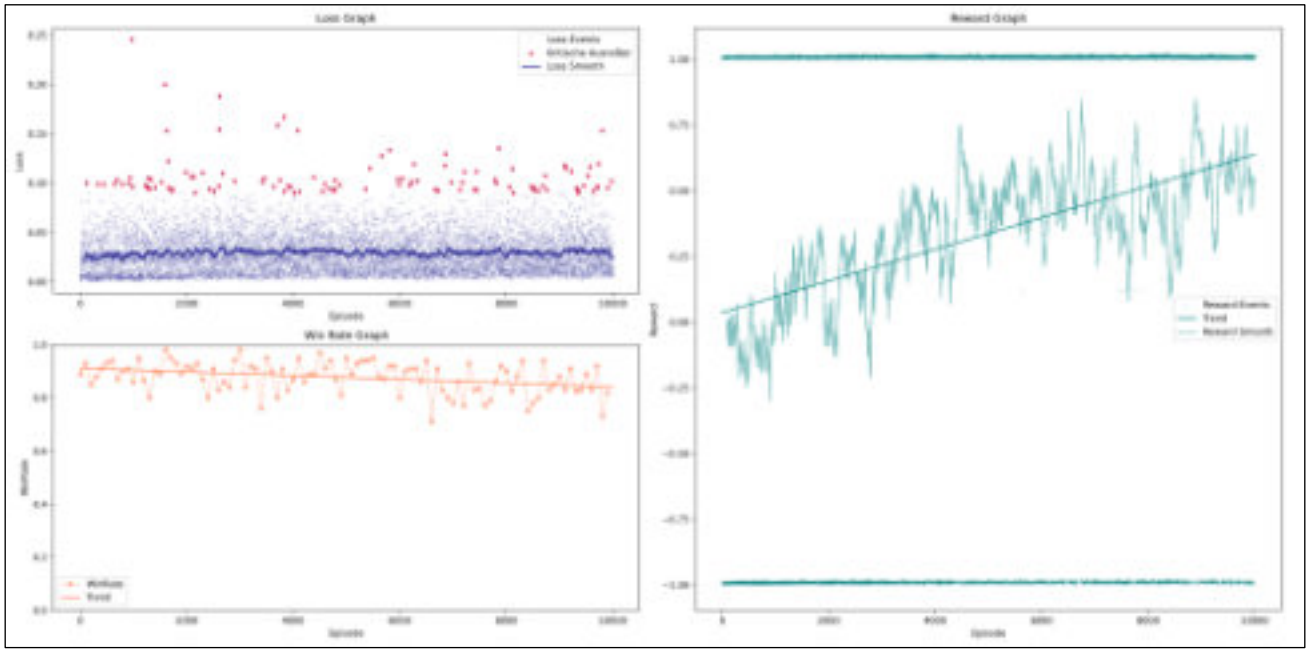
Überblick



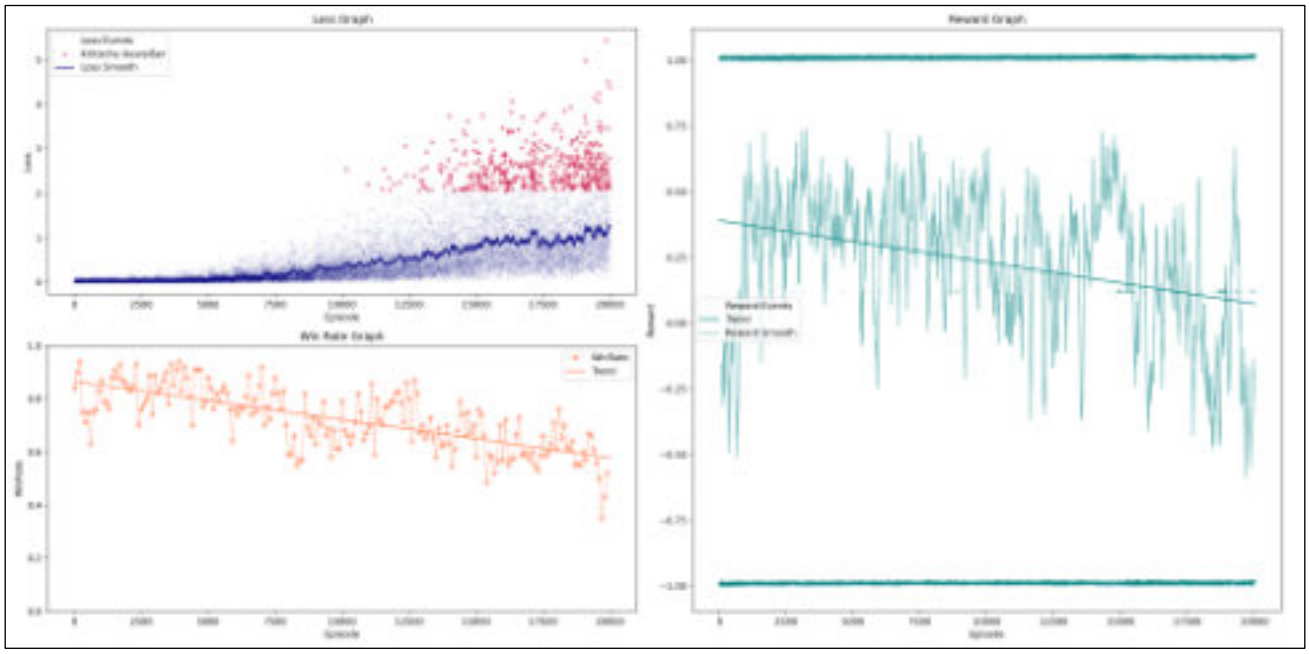
base_model



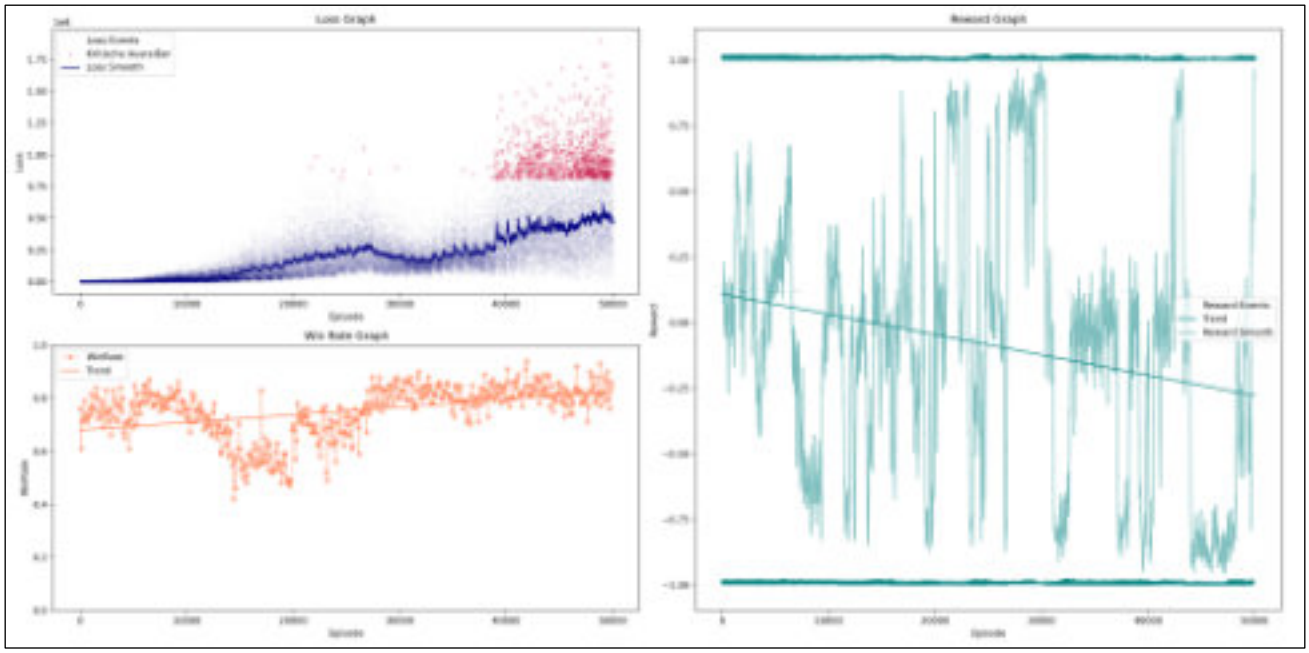
league_play_model_1



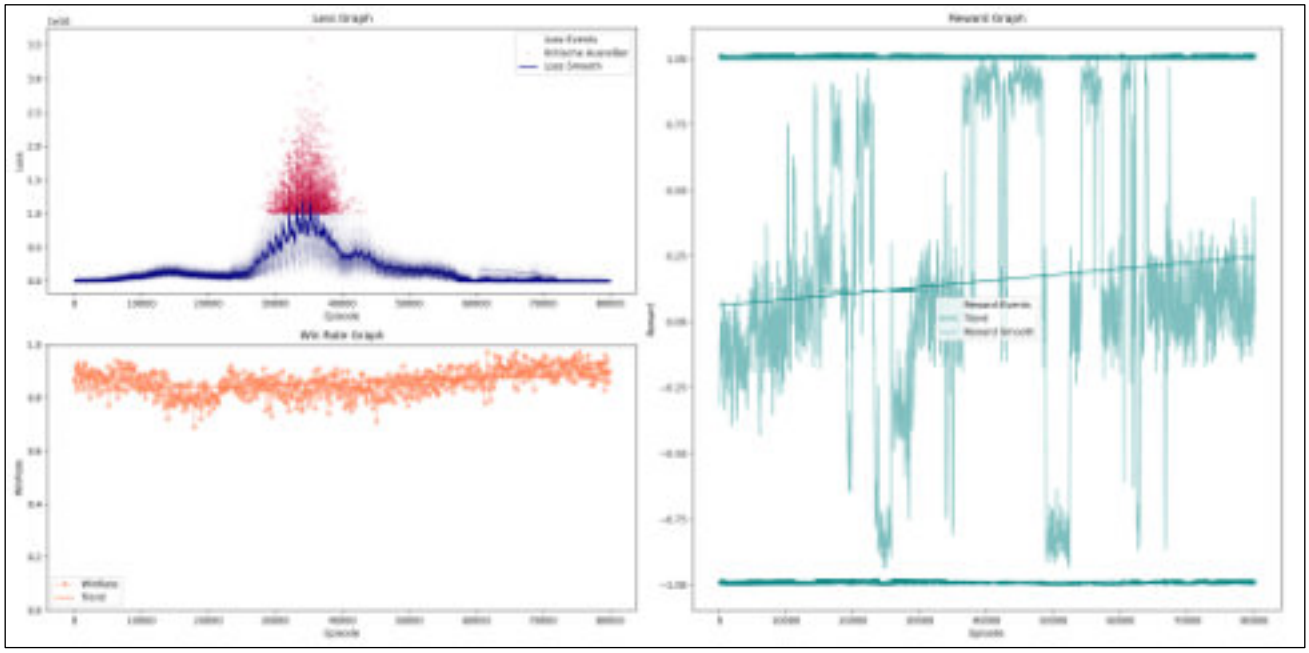
league_play_model_2



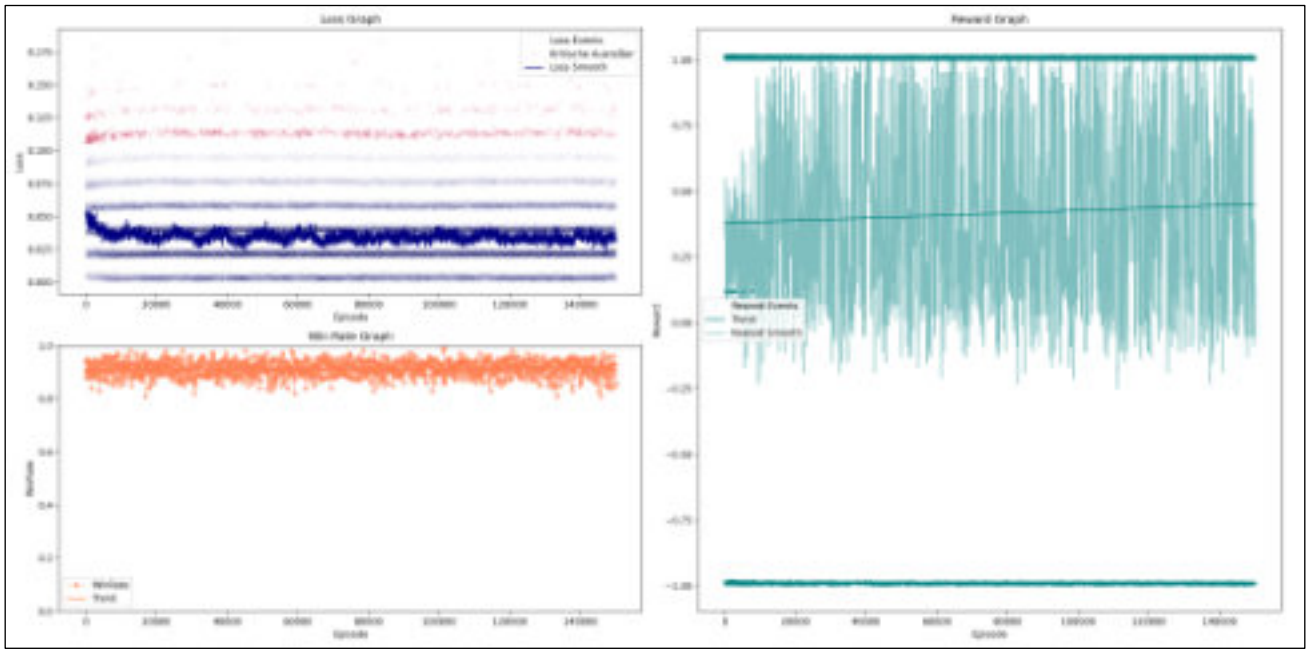
league_play_model_5



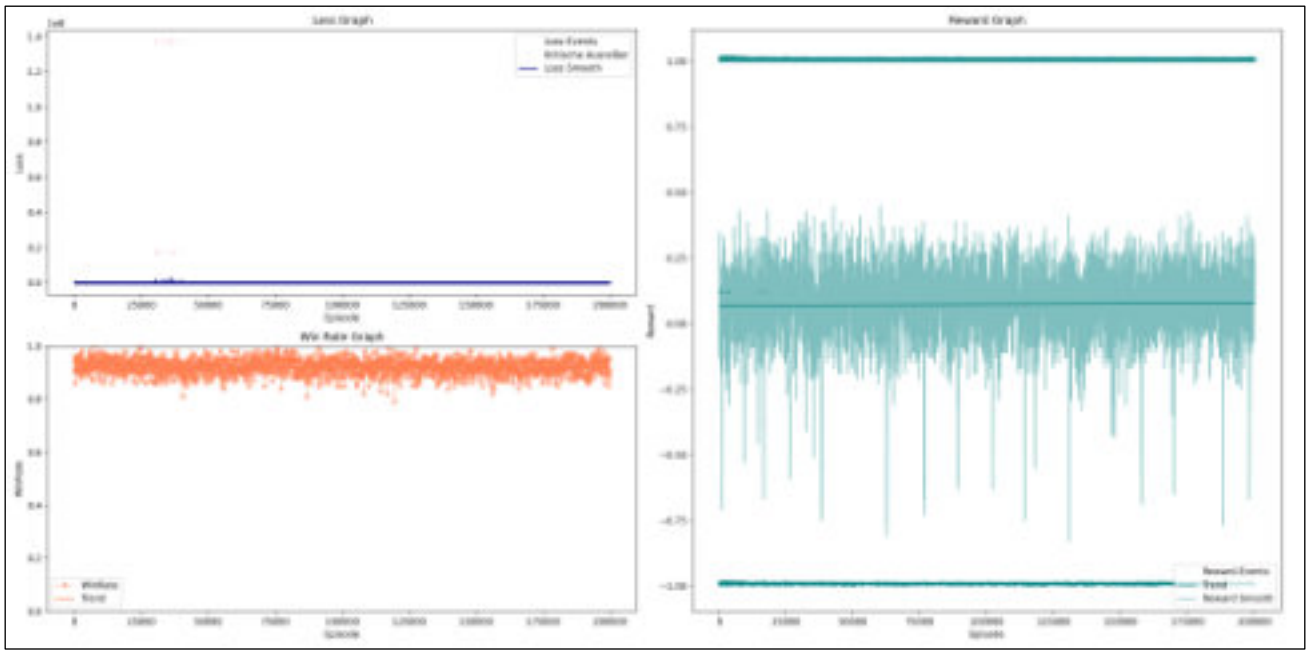
league_play_model_8



league_play_model_15



league_play_model_20

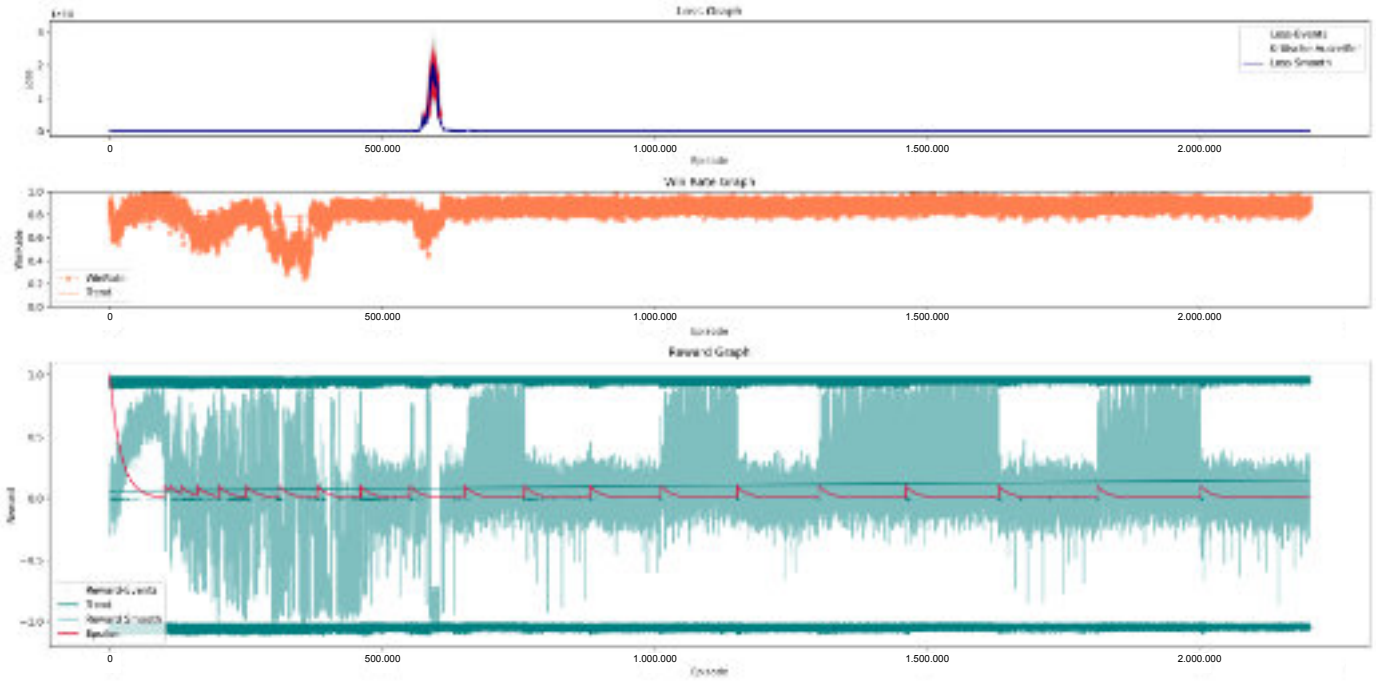


Trainer 15

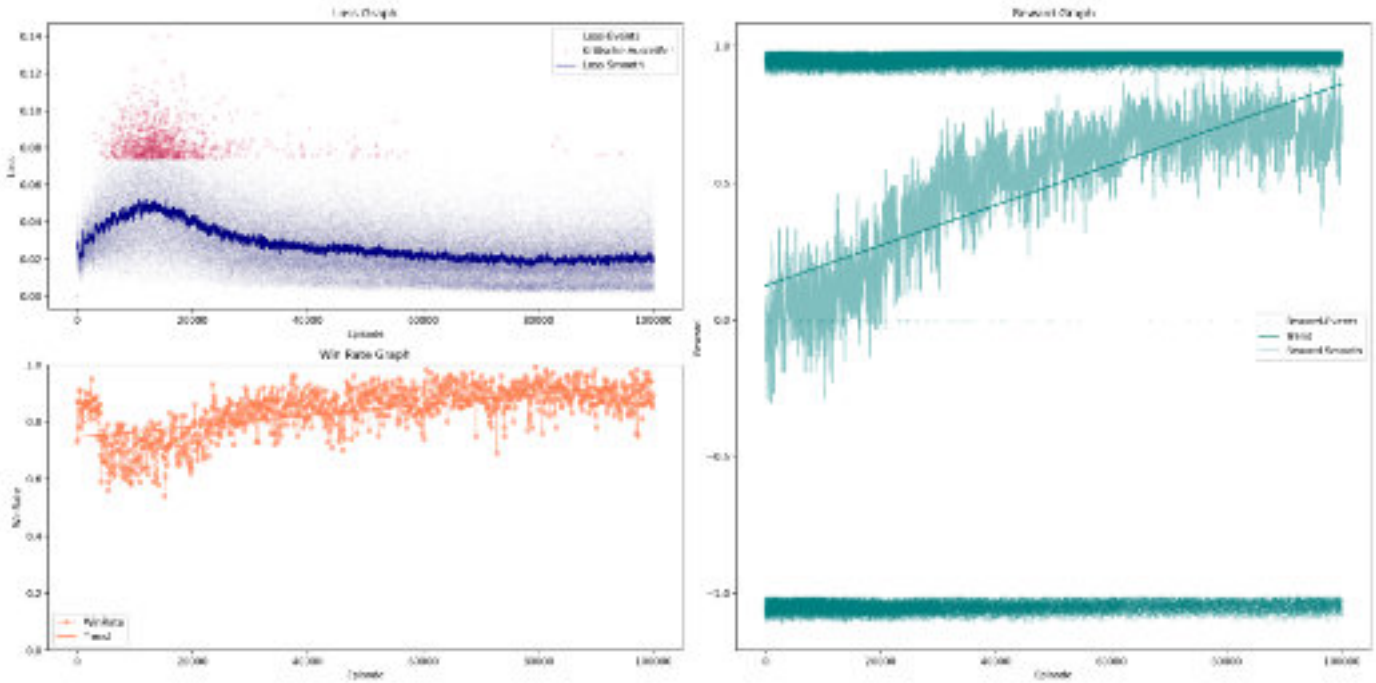
Hyperparameter: hidden_size = 128, gamma = 0,99, batch_size = 128, survive_reward = -0,005

Erwartung: Der Agent ähnelt Trainer 5, nur mit höherer batch_size.

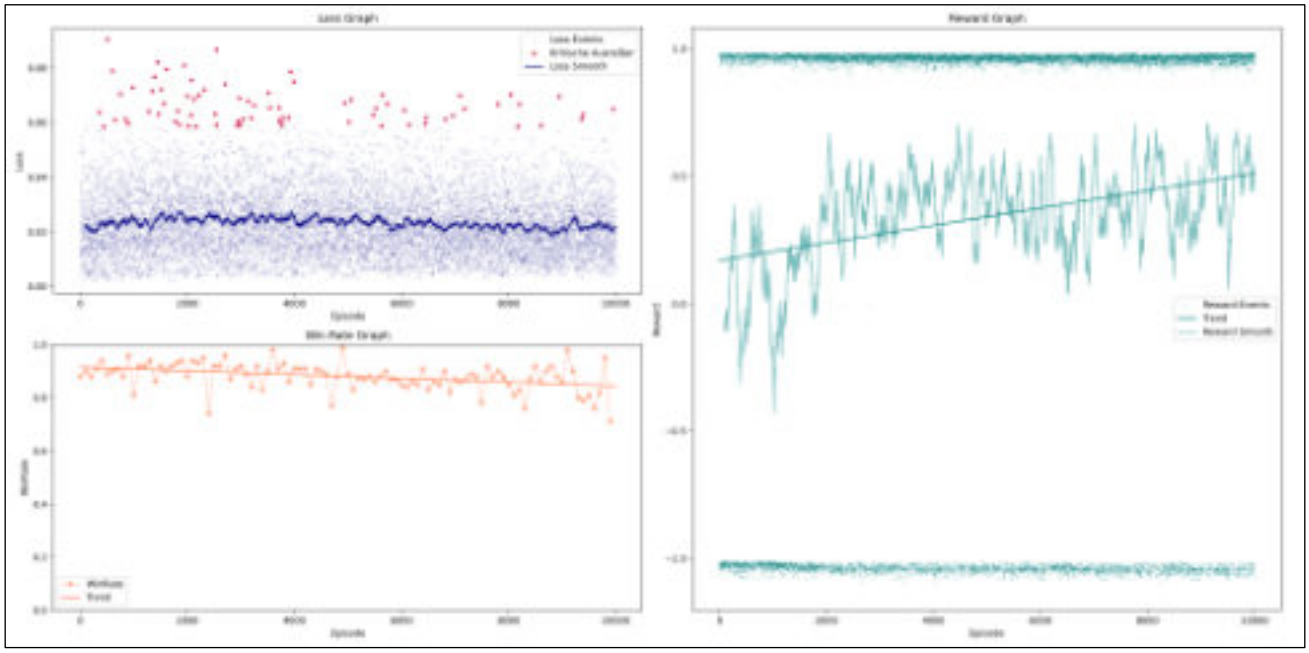
Überblick



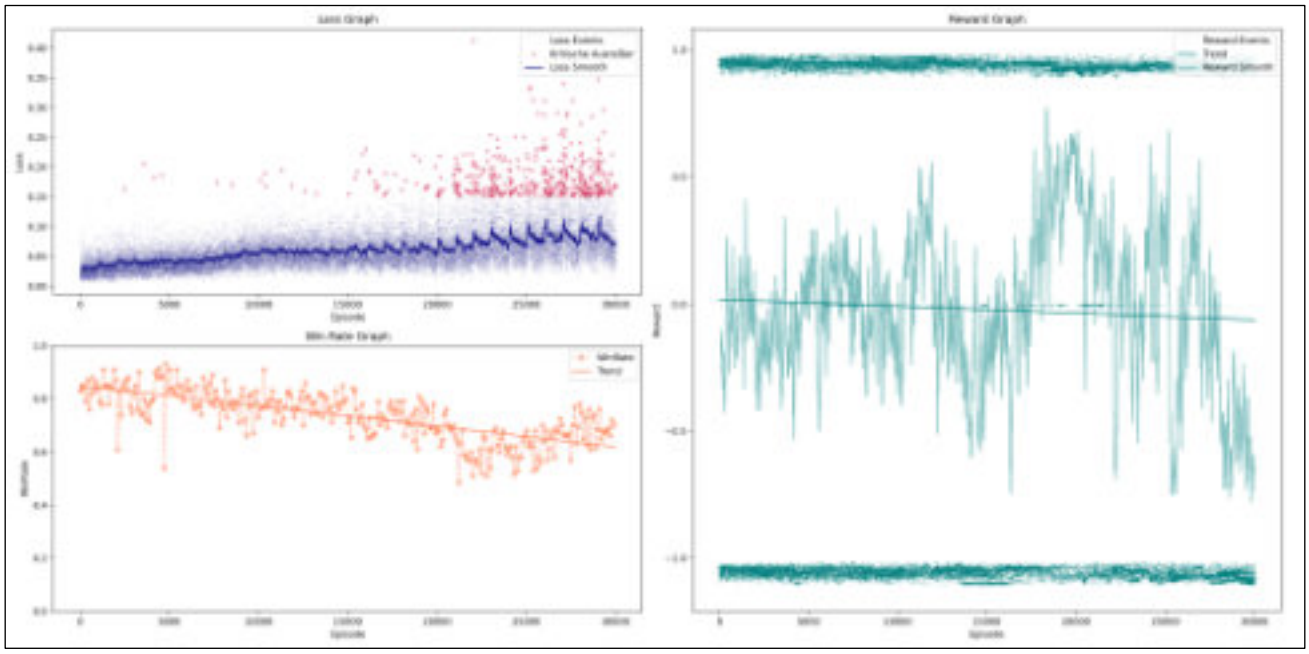
base_model



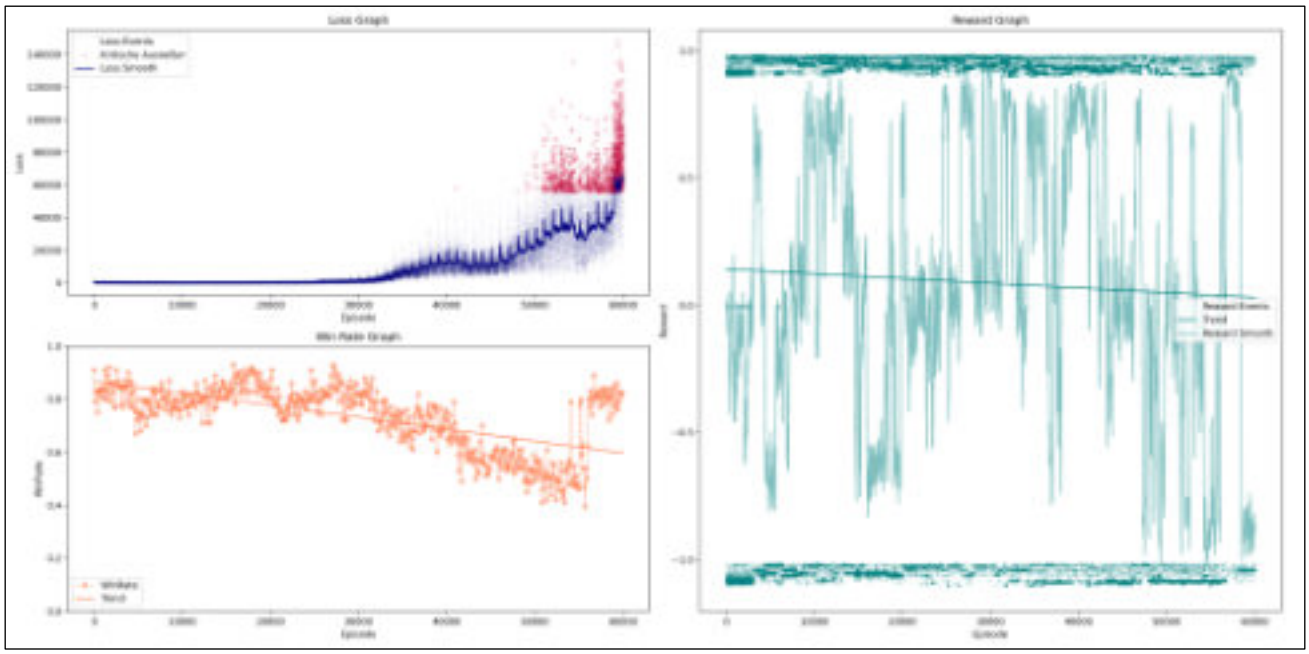
league_play_model_1



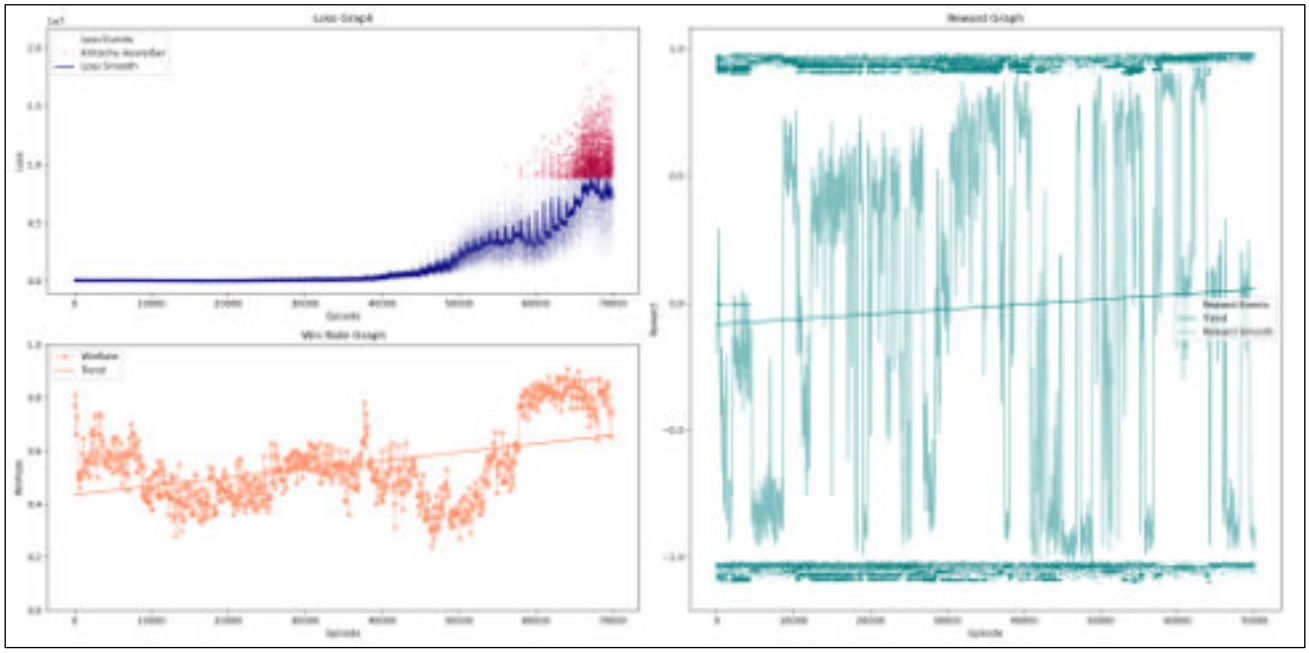
league_play_model_3



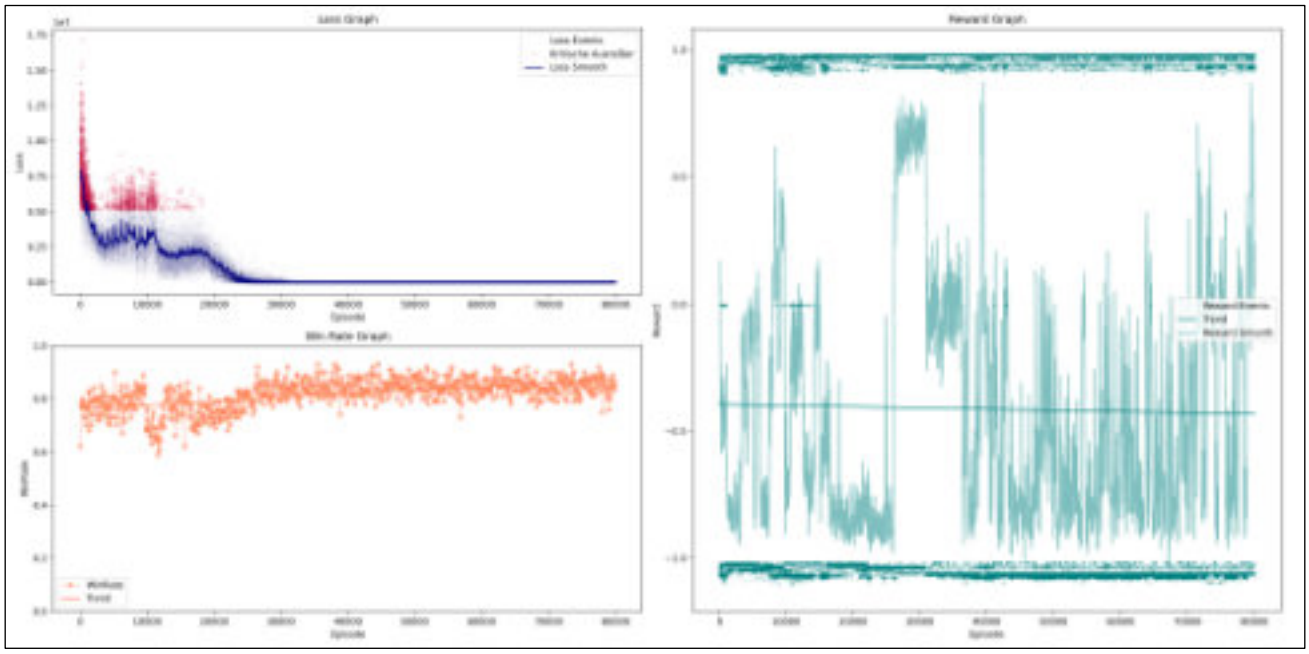
league_play_model_6



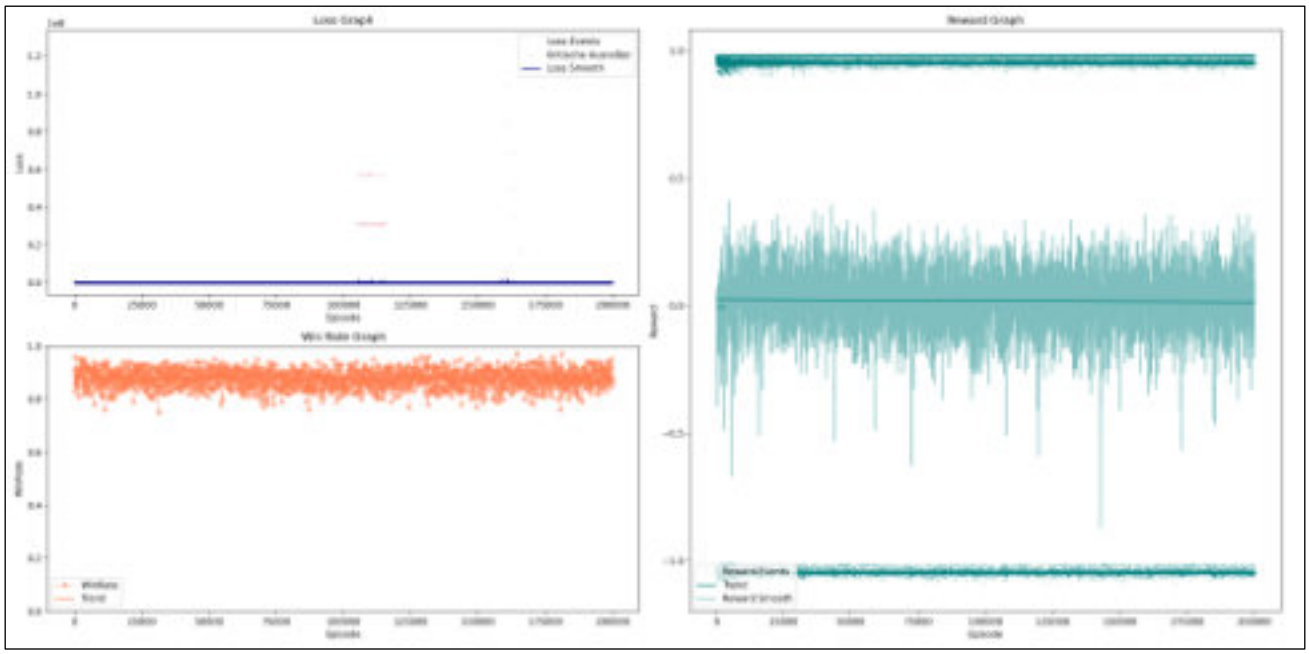
league_play_model_7



league_play_model_8



league_play_model_20

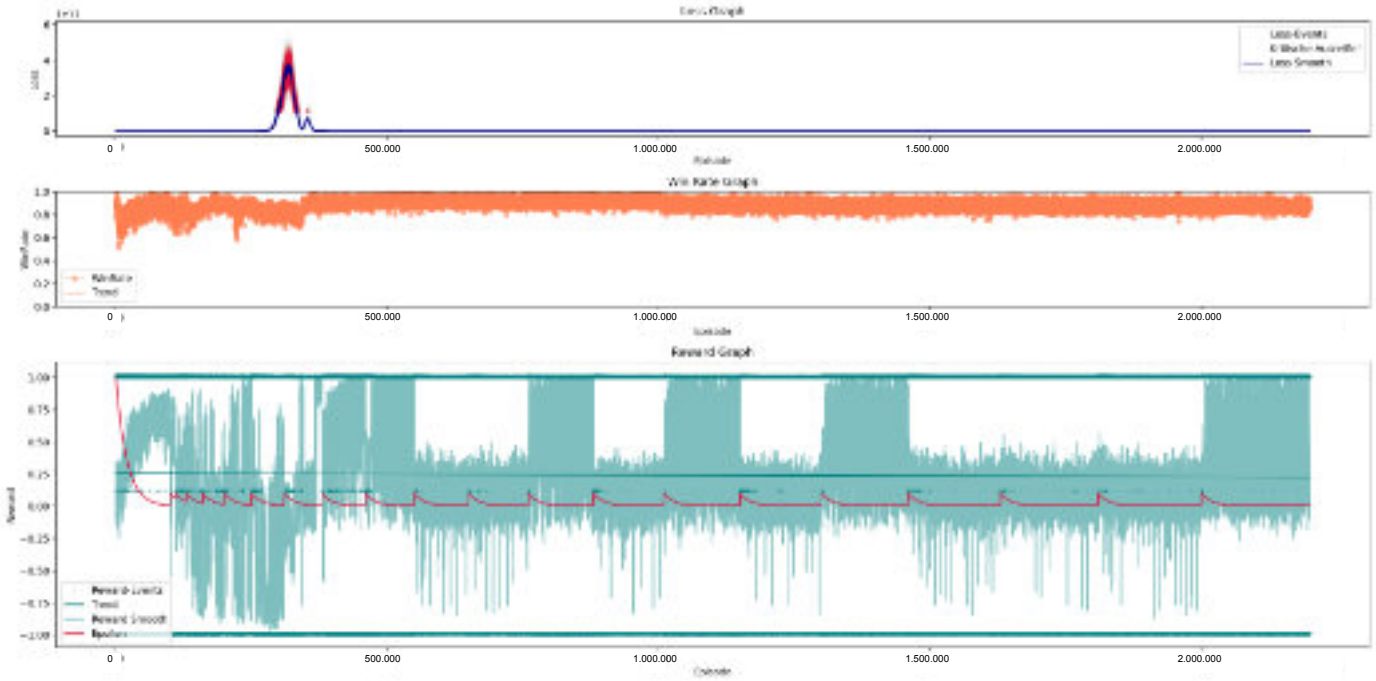


Trainer 16

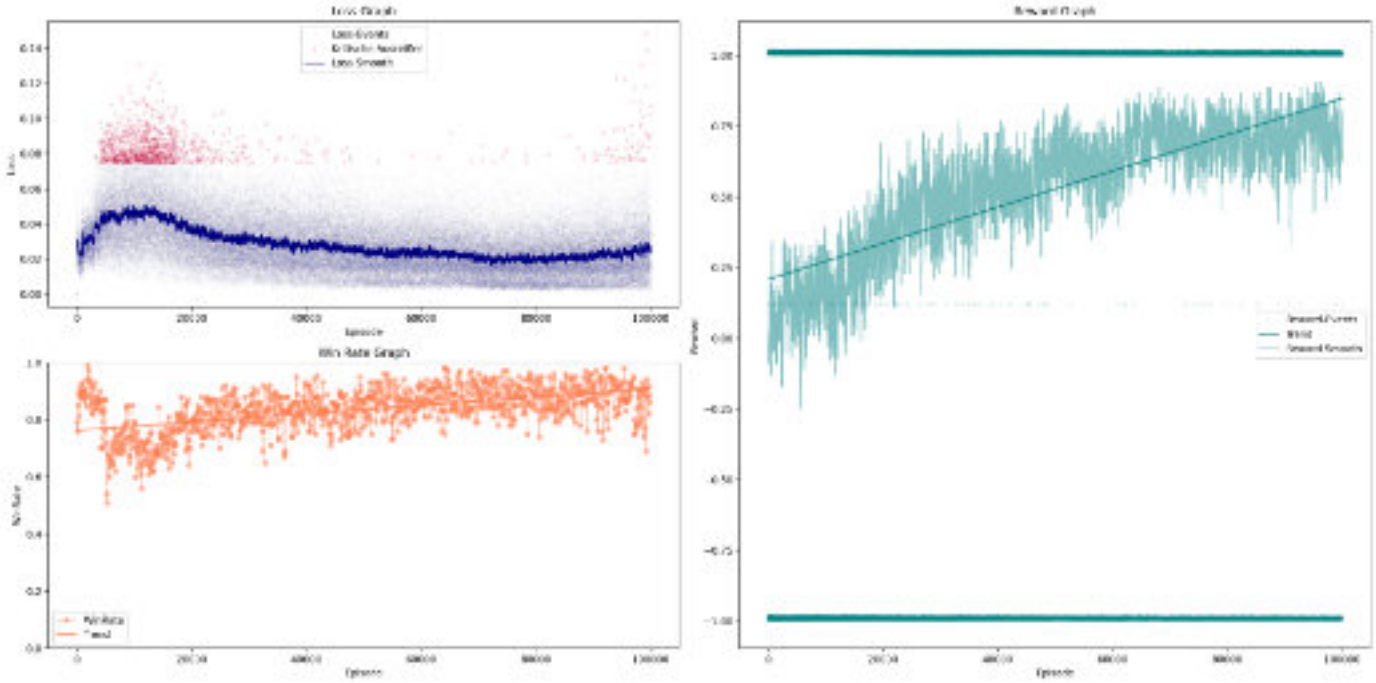
Hyperparameter: hidden_size = 128, gamma = 0,99, batch_size = 128, survive_reward = 0,001

Erwartung: Der Agent ähnelt Trainer 6, nur mit höherer batch_size. Dadurch könnte er noch langristiger handeln.

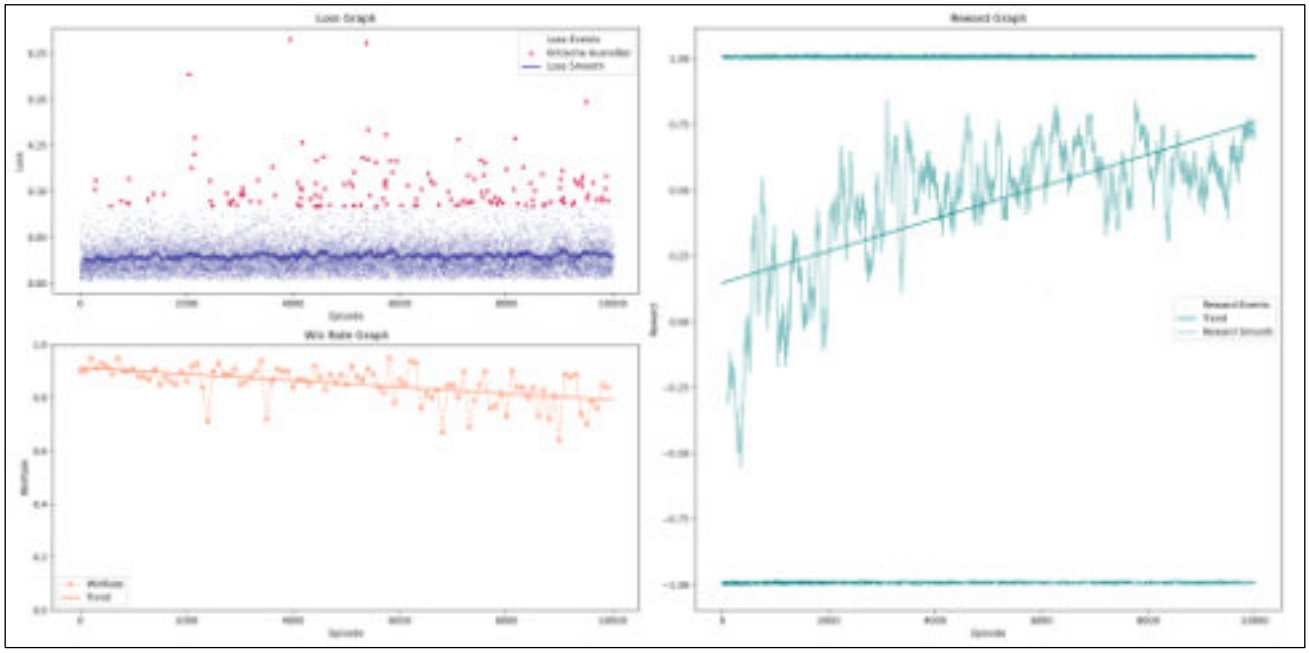
Überblick



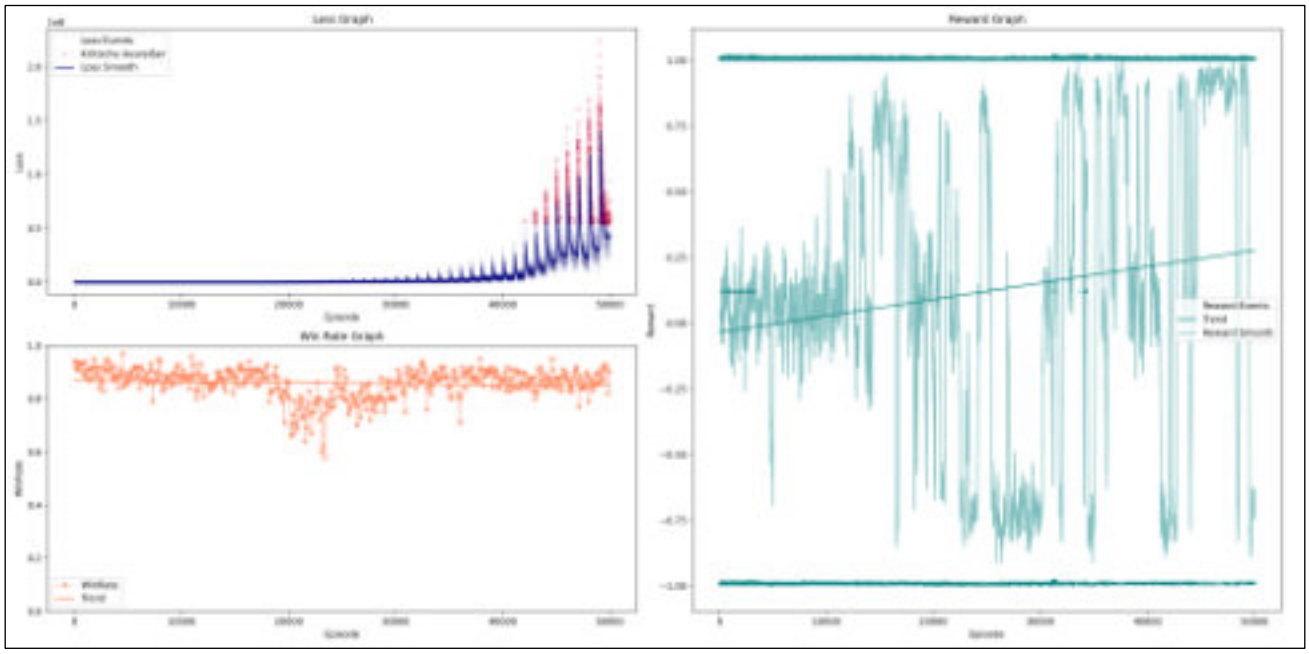
base_model



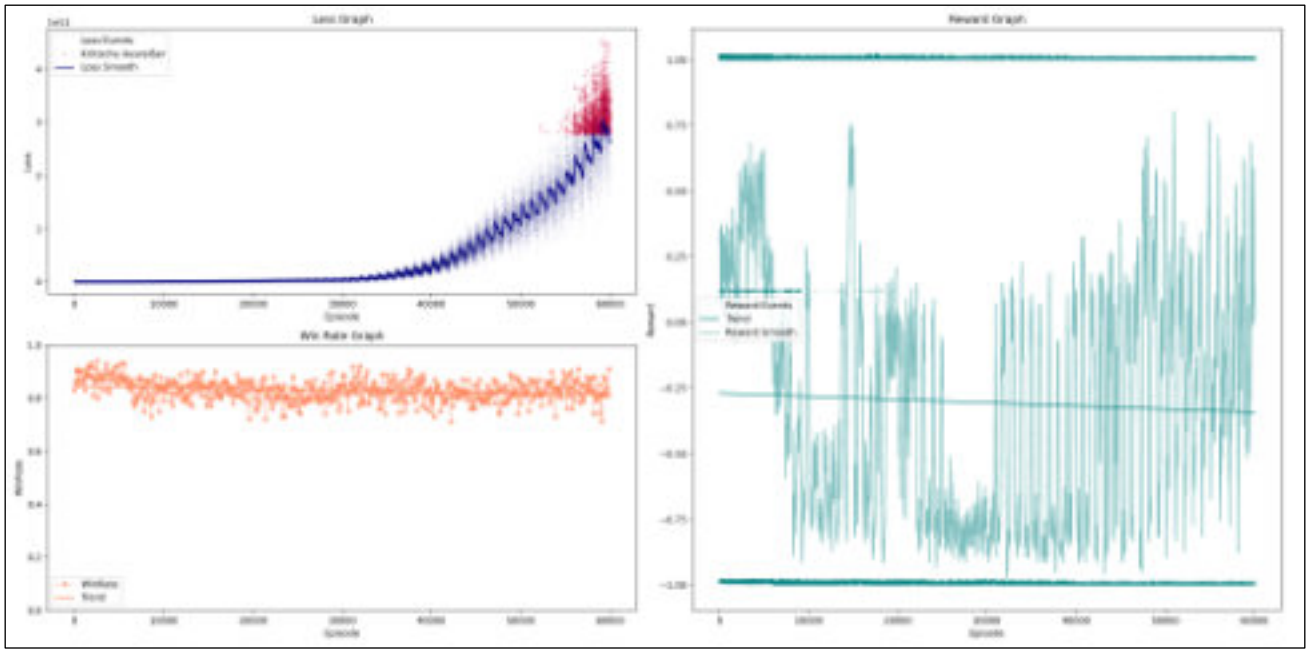
league_play_model_1



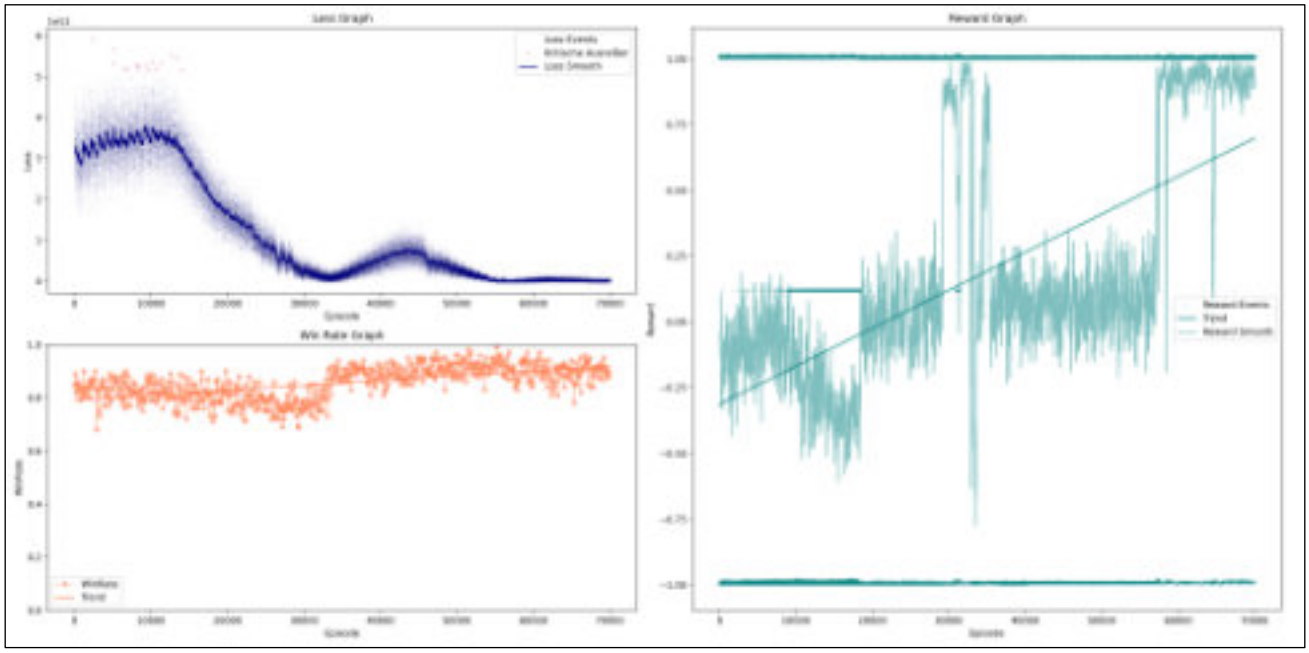
league_play_model_5



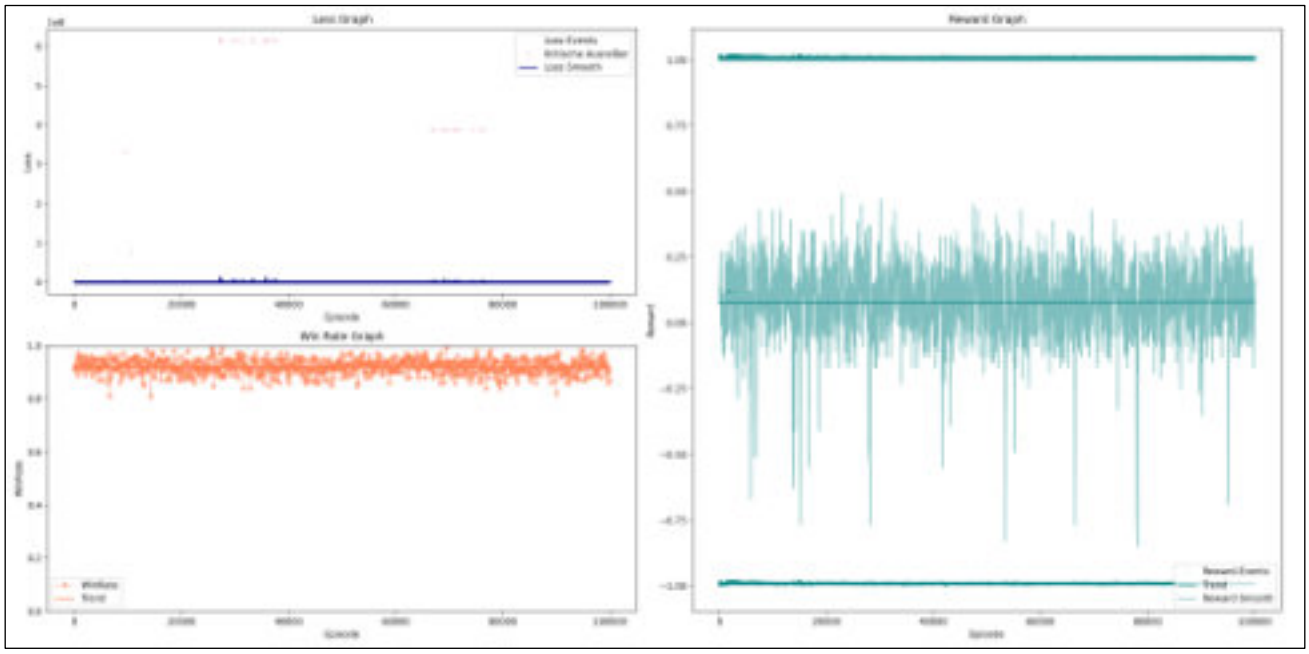
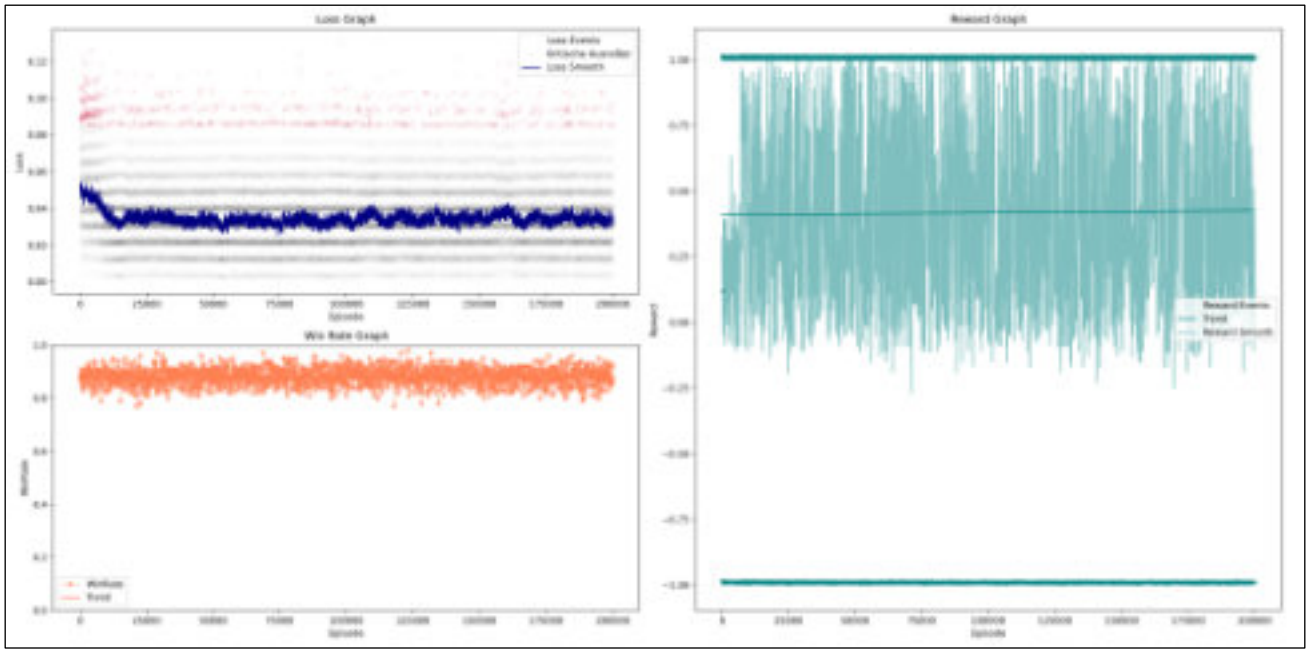
league_play_model_6



league_play_model_7



league_play_model_10



league_play_model_20